Time Series Analysis

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Some sections have been taken from previous notes by Guy Nason, Skevi Michael and Tobias Kley.

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Chapter 1

Introduction

A **time series** is a collection of observations of some process made sequentially in time.

Time series are special statistical data because:

- Observations are ordered in time.
- Observations at time t can influence those at time $t + u$, *i.e.* they aren't usually independent.

The crucial factor here is the **correlation** between data points. Many conventional statistical methods rely on the assumption that observations are independent and identically distributed, and so their applicability to time series is severely restricted.

There is a whole branch of statistics devoted exclusively to studying the way in which one goes about answering the mathematical and statistical questions posed by these time correlations, namely **time series analysis**.

Despite the challenges they pose, understanding time series is well worth the effort, with applications throughout all areas of society from rising global temperatures to stock markets and medical tests. Some more detailed examples are given below.

1.1 Typical Fields of Application

Economics and Financial Time Series

Examples

- Exchange rates, recorded as daily closing price.
- Stock indices, recorded as daily closing price (Figure 1.1).

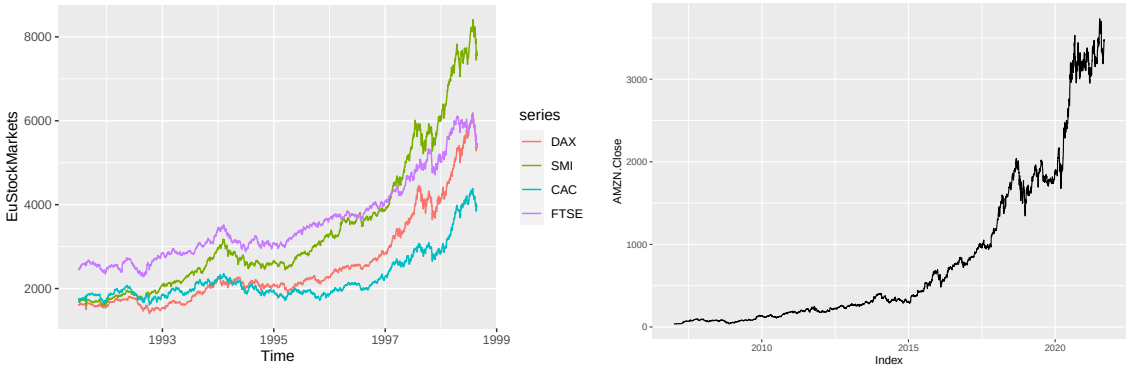


Figure 1.1: Daily Closing Prices of (a) Major European Stock Indices, 1991-1998, (b) Amazon, 2007-2021.

Questions one might ask:
What is the long term trend? What will tomorrow's closing price be? What caused the 'jumps'?

Physical Time Series

Examples

- Average monthly temperatures (Figure 1.2(a)).
- Riverflow (Figure 1.2(b)).

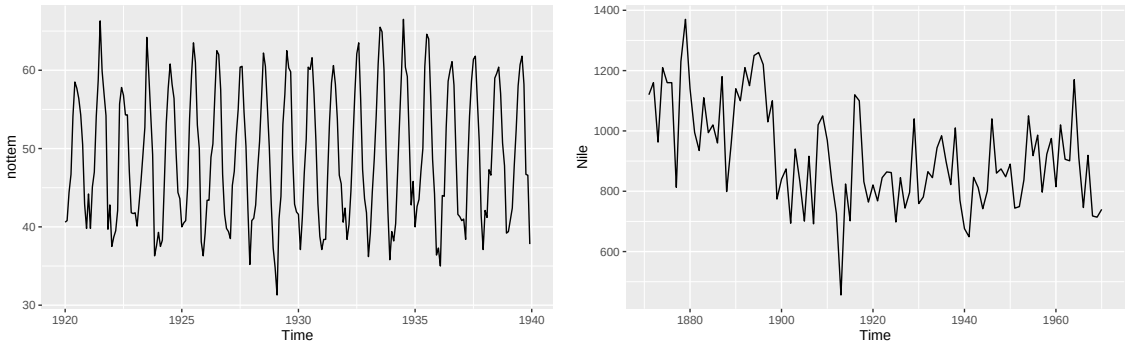


Figure 1.2: Examples of Physical Time Series: (a) Average Monthly Temperatures at Nottingham, 1920-1939, (b) Measurements of the annual flow of the river Nile at Aswan, 1871-1970.

Questions one might ask:
What is the long term trend? Can we predict monthly changes? What caused the change in mean?

Marketing Time Series

Examples

- Sales figures, recorded weekly/monthly etc.
- Monthly numbers of passengers (Figure 1.3).

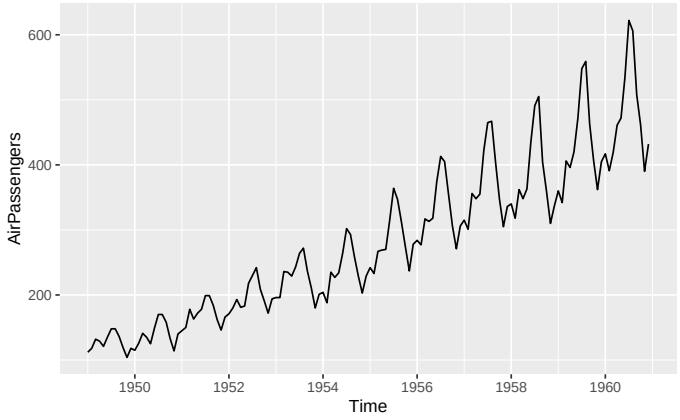


Figure 1.3: Monthly totals of international airline passengers (in thousands), 1949-1960.

Questions one might ask:
Can we predict the sales for the next period? Can we understand the series?

Demographic Time Series

Examples

- Population of England every year over 50 years.
- Population of the United States (Figure 1.4).

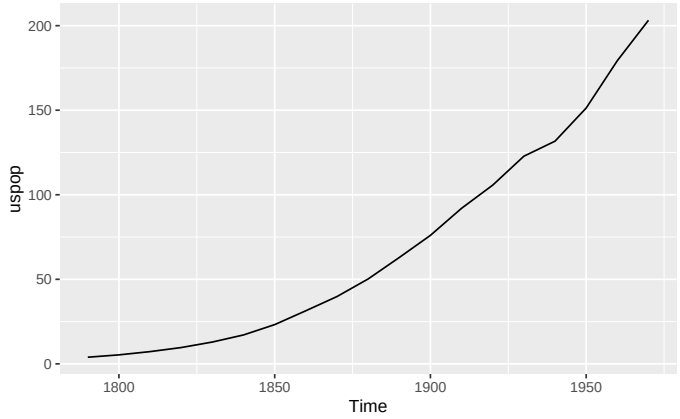


Figure 1.4: Population of the United States (in millions) as recorded by the decennial census.

Questions one might ask:
Can we plan ahead?

Process Control and Quality Management Time Series

Examples

- Controlling manufacturing tolerances (Figure 1.5).

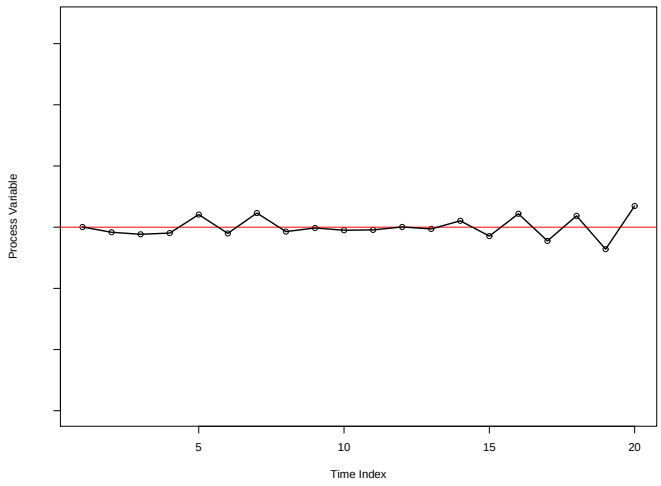


Figure 1.5: Variation of a process control variable over time.

Questions one might ask:
Can we predict when the process will exceed our limits?

Point Processes

Random events happening through time.

Examples

- Airplanes landing at BRS airport (Figure 1.6).
- Times of tries scored in a rugby match.
- Infections of children with measles.

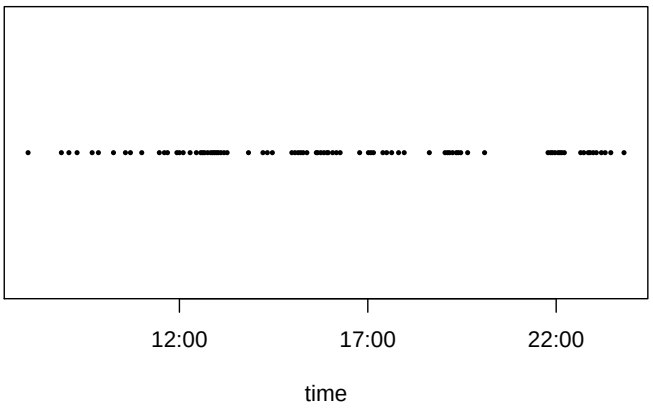


Figure 1.6: Arrival times at Bristol airport on a Friday.

Biomedical Time Series

Examples

- Heart rate of an infant sampled every 16 seconds (Figure 1.7).

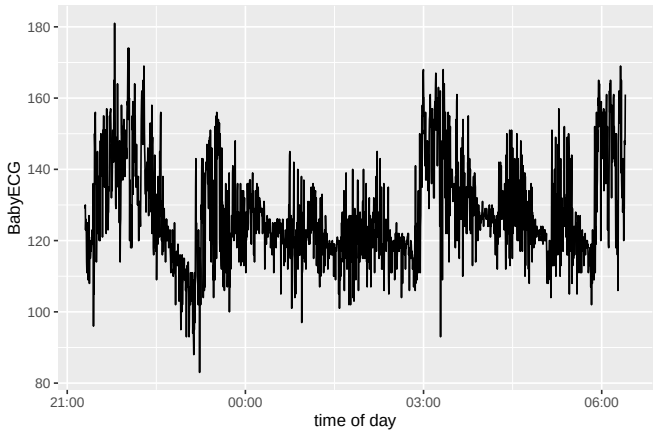


Figure 1.7: Infant's heart rate (in beats per minute) sampled every 16 seconds.

Questions one might ask:
What do the peaks and troughs correspond to?

Binary (or Categorical) Time Series

Take 0 or 1 (or discrete values).

Examples

- Communication networks.
- Biological examples (Figure 1.8)

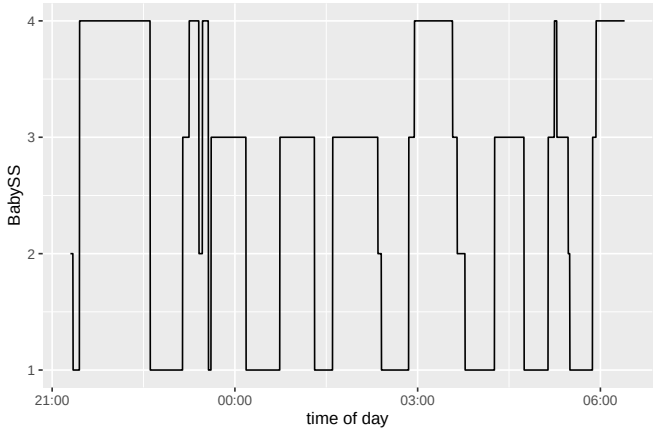


Figure 1.8: Infant's sleep state (1=quiet sleep, 2=between quiet and active sleep, 3=active sleep, 4=awake) sampled every 16 seconds.

Questions one might ask:
Does a baby go through sleep cycles?

1.2 Plotting a Time Series for Visual Inspection

There are dedicated classes available in **R** to handle time series data. Objects of type `ts` are common. Such `ts` object can be plotted by using the `plot` command. The following code can be used to create a plot of monthly mean relative sunspot numbers from 1749 to 1983. The plot is shown in Figure 1.9.

```
plot(sunspots)
```

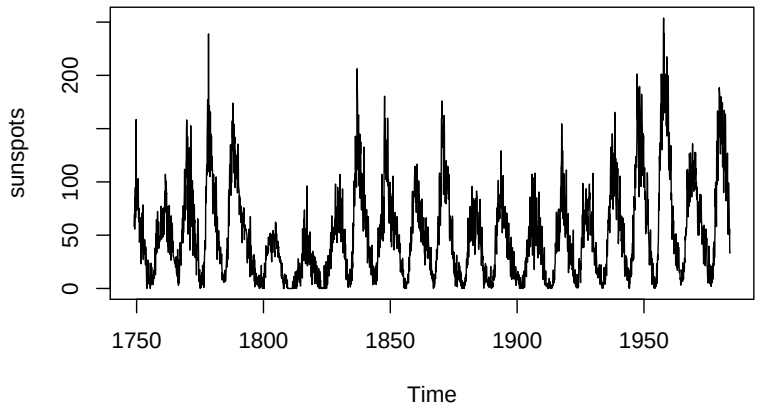


Figure 1.9: Yearly numbers of sunspots from 1700 to 1988 (rounded to one digit). Plotted with `plot()`.

The same data can also be plotted using the function `autoplot` from the package `forecast` that returns a `ggplot` object. The following code can be used to create the plot. The plot is shown in Figure 1.10.

```
library(forecast)
autoplot(sunspots)
```

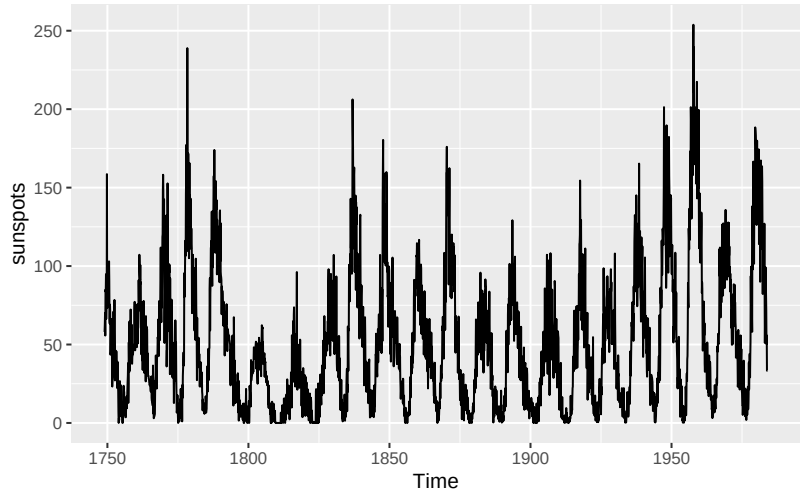


Figure 1.10: Yearly numbers of sunspots from 1700 to 1988 (rounded to one digit). Plotted with `forecast::autoplot()`.

1.3 Terminology

- A time series is said to be in **continuous** time when observations are made continuously in time, *i.e.* $X(t)$ exists for all $t \in \mathbb{R}$.

- A time series is in **discrete** time when observations are made at specific, discrete times. These times often equally spaced (but not always). In this case then we denote our observations by X_t, where X_t exists for all $t \in \mathbb{Z}$.

  Discrete time series can arise in several ways:

  - Sampling a continuous time series - this means we take the instantaneous value of some variable at specific, discrete, times e.g. temperature or heart rate.
  - Aggregation of values - this is more useful where we're interested in the number of events to occur over an interval of time e.g. monthly totals of air passengers.
  - Some time series are naturally discrete e.g. the interest paid monthly on a savings account.

- A **deterministic** time series is one that is governed by precise mathematical equations. If you know the equations and the starting reference exactly, then you can predict the future observations exactly.

- A time series is **stochastic** if past values only partly influence the future. In general, exact predictions are impossible and predictive values are distributed according to some probability distribution.

1.4 The Objectives of Time Series Analysis

a) **Description:** Our first aim is to describe a time series. The first tool we use to do this, which we have already seen in action, is to plot the series against time. This is a good way to get an overall impression of the data. It allows us to spot trends, seasonal effects, outliers and increases in variation. It is also a good way of conveying information about the data to other people, especially non-experts.

b) **Explanation:** We also want to understand the behaviour of a time series and be able to explain its behaviour. To achieve this we will employ statistical models.

c) **Prediction or forecasting:** This is perhaps our most important goal. We want to be able to use the information we have about a time series from its past behaviour in order to predict what will happen next. This is one of the key differences with other statistical analysis - we have to use *extrapolation* to predict future values (outside the range of our observed data) rather than being able to rely on *interpolation* (which only allows us to predict values within the range of our observed data).

d) **Control:** In some contexts we might want to go even further than predicting future values and actually be able to control the future behaviour. E.g. Industrial quality settings, environmental pollutants, the path of a space craft.

Chapter 2

Simple Descriptive Techniques

At the end of the previous chapter we introduced four key objectives of time series analysis. In this chapter our focus will be on the first of these: how to describe the key features of a time series. In later chapters we will develop more sophisticated inferential techniques for analysing time series, but the first step is always to produce a plot and to look for systematic features such as trend or seasonality. In addition to a straightforward time series plot showing how our series varies with time, we will also introduce the correlogram. The main difference between time series data and 'ordinary' iid data is that successive observations may be related. The correlogram is a visual representation of the correlation between observations at different time lags.

2.1 Decomposition According to Source of Variation

One approach to time series analysis is to decompose the variation in a series into components representing trend and seasonal variation. The aim is to estimate, record and remove these features and then examine what remains for any further pattern.

$$\boxed{\text{Time Series} = \text{Seasonality} + \text{Trend} + \text{Residuals}}$$

The different sources of variation will now be described in more detail:

- **Seasonal effects**

  Many time series, for example weather data, sales figures and unemployment rates, exhibit cyclic variation. In some texts the terms seasonal and cyclic are used to distinguish variation that follows an annual pattern and that which is associated with other periods. Here, we use the terms seasonal and cyclic interchangeably to refer to a period of any duration. This variation can be aligned with a variety of different periods from a classic annual period (for example average monthly air temperatures, see figure 1.2) to monthly effects (for example expenditure) or a combination of a number of seasonal effects of different periods (for example electricity consumption by a town, which is likely to have both a strong daily effect and a strong annual effect). Generally, one wishes first to estimate a seasonal effect and then one can remove it to form a deseasonalised series.

- **Trend**

  When we refer to trend then we are talking about the 'long-term change in mean'. The trouble with this definition is in deciding what we mean by 'long term'. It is possible to confuse a slowly varying cyclic effect with trend. For example the cyclic variation of a climatic variable with a period of 50 years may look like a trend if one had just 20 years of data. It depends on how often the series is sampled (sampling rate) and the length of the series with respect to the period of the cyclic effect. Once again, the trend of a time series is generally estimated, recorded and then removed.

- **Residuals**

  After seasonal effects and trend have been removed we are left with a series of residuals that may appear to vary randomly. In the next chapter we will see how the apparently irregular fluctuations of this data may be examined for any further pattern. Generally this involves fitting various time series models to the residuals and assessing the fit. A good model can usually help with explanation and prediction.

2.2 Transformations

Before analysing and removing features like seasonality and trend it is often sensible to consider transforming the data. Reasons for making a transformation include:

- **To stabilise the variance**: If there is a trend in the series and the standard deviation is proportional to the mean then a log (or square root) transform may be advisable. This is the case for the example of monthly international airline passengers (see figure 2.1).
- **To make a multiplicative seasonal effect additive**: Taking logarithms turns a model of the form $X_t = m_t S_t \varepsilon_t$ into the model $\log X_t = \log m_t + \log S_t + \log \varepsilon_t$ (m_t is the deseasonalised mean level, S_t is the seasonal effect and ε_t is the random error, all at time t).
- **To make the data normally distributed**: Often we carry out model building and forecasting based on the assumption that the data are normally distributed. In practice this may not be the case and a transformation may be necessary.

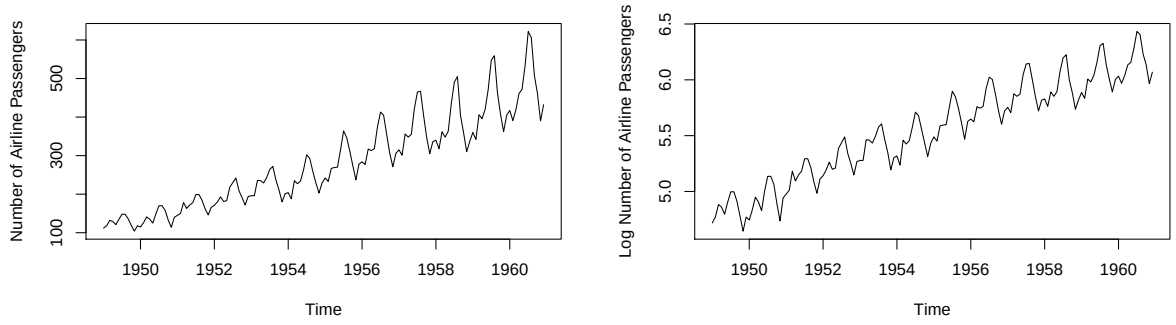


Figure 2.1: Demonstration of a log transformation to stabilise the variance applied to monthly totals of international airline passengers (in thousands), 1949-1960.

2.3 Analysing Trend

Once your data is suitably transformed, the next step is to estimate, record and remove any trend in the data. In this section we will discuss some general approaches to this process.

2.3.1 Curve Fitting

The first option is to simply fit a parametric curve (e.g. model the trend as a low order polynomial and estimate the coefficients by minimising ordinary least squares).

Possible models for x_t include:

- A simple linear fit

$$x_t = a + bt \tag{2.1}$$

- A polynomial regression up to order K

$$x_t = \sum_{j=0}^{K} b_j t^j \tag{2.2}$$

- A Gompertz curve

$$\log x_t = a - br^t \text{ where } r \in (0, 1) \tag{2.3}$$

- A Logistic curve

$$x_t = a/(1 + b\exp(-ct)) \tag{2.4}$$

both of the latter two are S-shaped and approach an asymptotic value as $t \to \infty$.

Note: The first two models are linear and can be fitted in **R** using the familiar function `lm()`. The second two are non-linear and can be fitted using the function `nls()`, which performs least squares regression for non-linear parametric trend curves.

Returning to our example of international airline passengers, figure 2.2 shows each model fitted to the log transformed data and figure 2.3 shows the detrended residuals corresponding to the cubic polynomial fit.

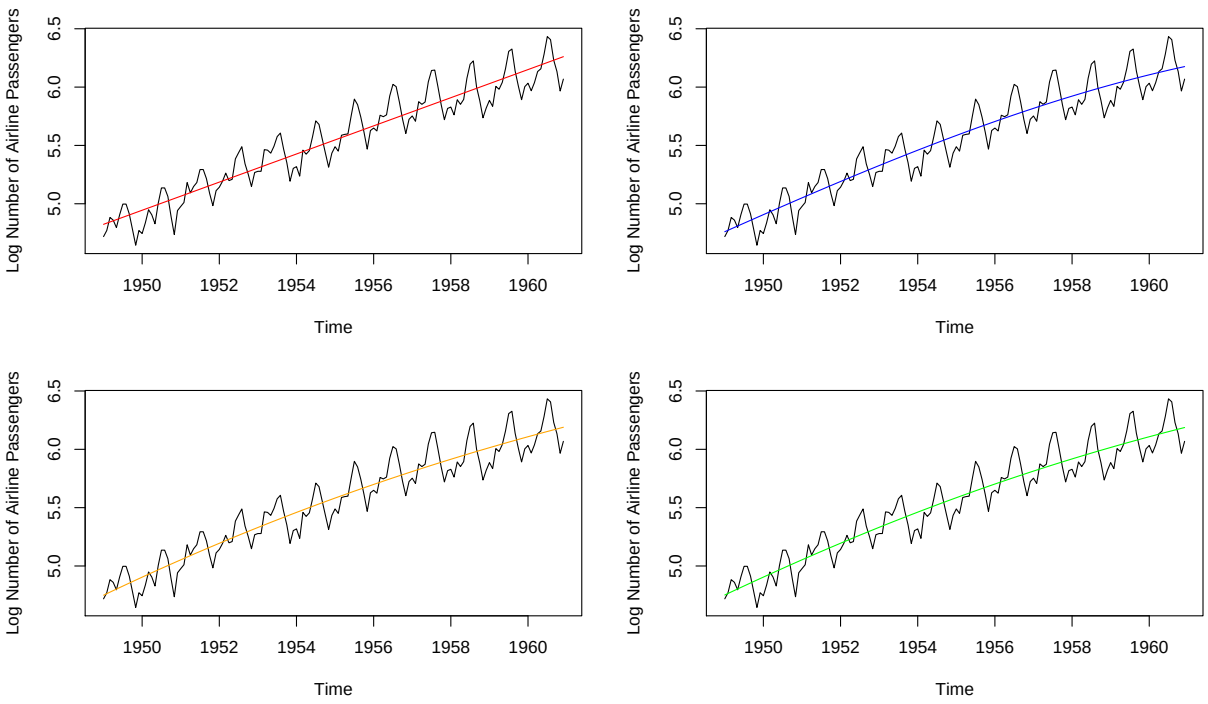


Figure 2.2: Demonstration of curve fitting using: a simple linear fit (red), polynomial regression with K=3 (blue), a Gompertz curve (orange) and a Logistic curve (green) applied to monthly totals of international airline passengers (in thousands), 1949-1960.

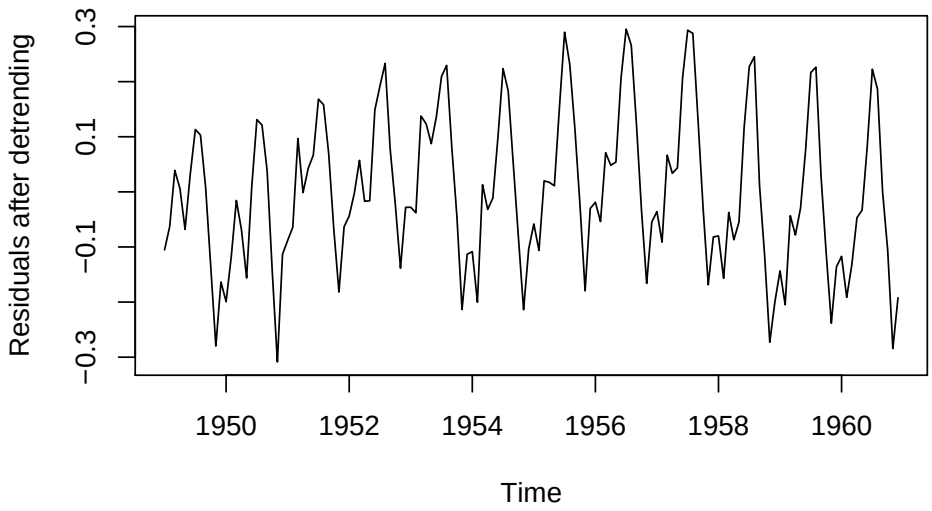


Figure 2.3: Residual monthly totals of international airline passengers after variance stabilisation and detrending using cubic polynomial regression.

We have now successfully stabilised the variance and both estimated and removed the trend. Looking at the residual data there is clearly still a seasonal effect. We will deal with this shortly, but first let's discuss some other ways of removing a trend.

2.3.2 Linear Filtering

A **linear filter** converts one time series $\boldsymbol{x} = \{x_t\}_{t\in\mathbb{Z}}$ into another time series $\boldsymbol{y} = \{y_t\}_{t\in\mathbb{Z}}$ by the linear operation

$$\boldsymbol{y} = \mathrm{Sm}(\boldsymbol{x}),$$

$$\text{where} \quad y_t = \sum_{r=-q}^{s} a_r x_{t+r} = a_{-q}x_{t-q} + a_{-q+1}x_{t-q+1} + \ldots a_0 x_t + \ldots + a_{s-1}x_{t+s-1} + a_s x_{t+s}$$

Notes:

- This is indeed a linear operation, because

$$\mathrm{Sm}(\lambda_1 \boldsymbol{x}_1 + \lambda_2 \boldsymbol{x}_2) = \lambda_1 \mathrm{Sm}(\boldsymbol{x}_1) + \lambda_2 \mathrm{Sm}(\boldsymbol{x}_2).$$

- The set $\mathrm{Sm} = \{a_r\}_{r=-q}^{s}$ are called the **weights** of the filter.

There are three main types of filter that we will consider:

a) **Moving Average Filter**

The filter is known as a moving average filter if

$$\sum_{r=-q}^{s} a_r = 1.$$

Moving average filters can be used to smooth out local fluctuations and estimate the local mean. They are often symmetric about 0 with $s = q$ and $a_j = a_{-j}$. The simplest moving average filter is associated with the weights $a_r = 1/(2q+1)$, for $r \in [-q, q]$, corresponding to

$$\mathrm{Sm}(x_t) = \frac{1}{2q+1} \sum_{r=-q}^{q} x_{t+r}.$$

b) **Asymmetric Filter**

An Asymmetric filter is given by

$$\mathrm{A}(x_t) = \sum_{j=0}^{q} a_j x_{t-j}.$$

Note that in this case the filtered value at time t only uses data x_s with $s \leq t$.

c) **Exponential Smoothing**

A very popular technique is to consider an asymmetric linear filter with $q = \infty$ and the weights set to $a_j = \alpha(1-\alpha)^j$, where α is a chosen constant and such that $\alpha \in (0, 1)$. Observations in the past do contribute to the smooth curve, but each previous generation receives geometrically less weight.

The **residuals** from a smooth curve can be written as:

$$\begin{aligned}\text{Res}(x_t) &= x_t - y_t \\ &= x_t - \text{Sm}(x_t) \\ &= \sum_{r=-q}^{s} b_t x_{t+r},\end{aligned}$$

where $b_0 = 1 - a_0$ and $b_r = -a_r$.

Note: if Sm is a moving average filter, then $\sum_{r=-q}^{s} b_r = 0$.

High and low pass filters

A smoothing filter Sm is known as a low-pass filter as high frequency oscillations within the series are smoothed out.

The residuals $\text{Res}(x_t)$ are what is left after removing the low frequencies and so is known as a high-pass filter.

An example of the application of a simple moving average filter with equal weighting and $q = 2$ is shown in figure 2.4.

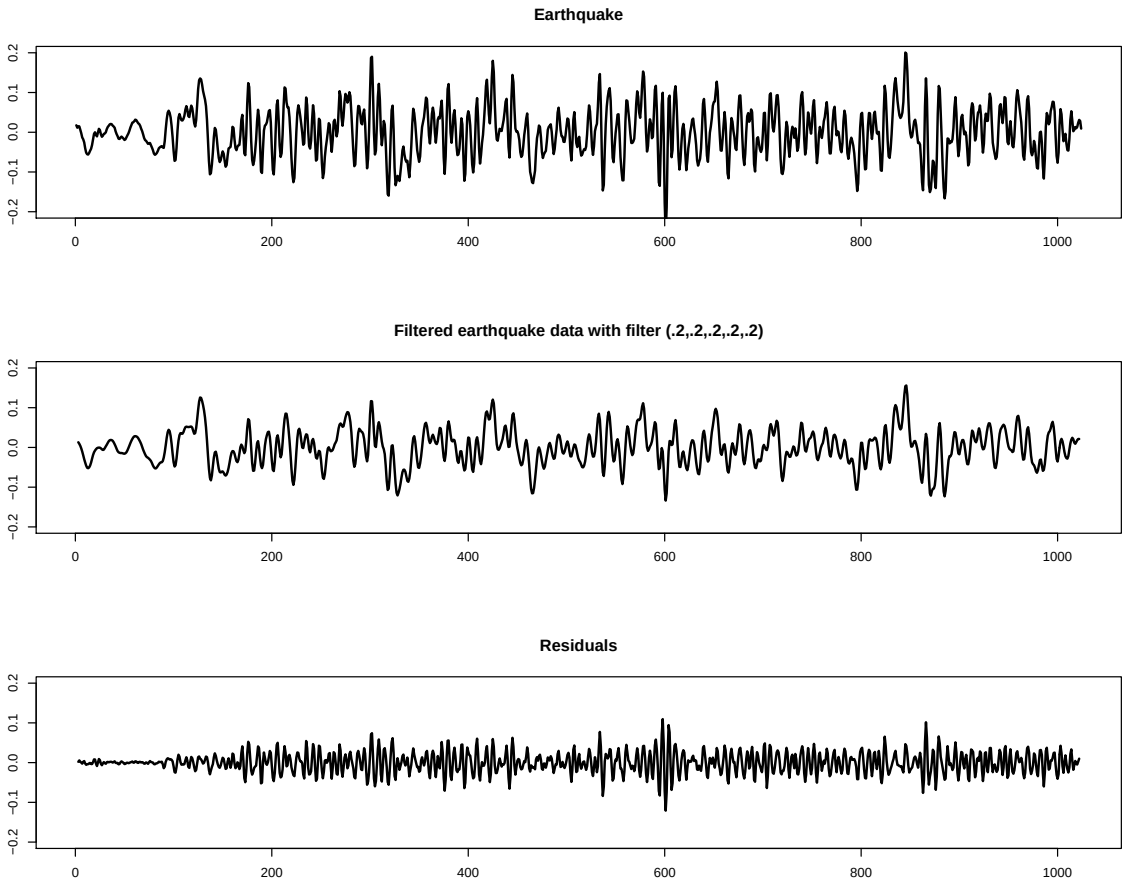


Figure 2.4: Seismic activity after an earthquake, filtered data and residuals.

2.3.3 Differencing

Differencing is a particularly powerful method for removing a trend. Suppose that $\{x_t\}_{t\in\mathbb{Z}}$ is a time series, then:

- First order differencing of lag 1 is written as

$$y_t = \nabla x_t = x_t - x_{t-1}$$

- First order differencing of lag k is written as

$$y_t = \nabla_k x_t = x_t - x_{t-k}$$

- Second order differencing (of lag 1) is achieved by the ∇^2 operator and is written as

$$y_t = \nabla^2 x_t = \nabla x_t - \nabla x_{t-1} = x_t - 2x_{t-1} + x_{t-2}$$

.

Note: The operation of differencing constitutes a linear filter.

2.4 Seasonal Fluctuations

Seasonal effects can be estimated and removed by:

- Filters designed for this purpose.
- Seasonal differencing (e.g. ∇_{12} to remove an annual cyclic effect from monthly data).
- Taking the mean value for a particular period.

Returning to our example of international airline passengers, we apply the third option. Thus, we estimate the seasonal effect by calculating the average for each month, from January to December. This is a simple, but effective option to estimate the seasonal effect for a time series that has already been detrended. To carry out this process in **R** you can make use of the function `smean()`. The resulting estimated seasonality is shown in figure 2.5 along with the residuals, which have now been log transformed to stabilise the variance, detrended using a cubic polynomial regression model and finally deseasonalised using a simple averaging model.

2.5 Analysing the Residuals

At this point we have stripped away all of the obvious patterns in the data and are left with a seemingly random series of residuals. The remaining series, which exhibits no systematic change in the mean (no trend), no systematic change in the variance and no periodic variations can be loosely referred to as **stationary**. If we were to be more precise then there is actually no such thing as a 'stationary time series' as the mathematical definition of stationarity (which will be introduced in the next chapter) is for a probability model. However, the phrase is often applied to time series data that could be sensibly modelled using a stationary probability model. In the next chapter we will see how the stationary residuals may be examined for further pattern using stationary probability models. It is worth noting that the non-stationary components such as trend and seasonality might be of more interest, but fitting a good model to the residuals can also be very helpful, in particular when predicting future values of the series.

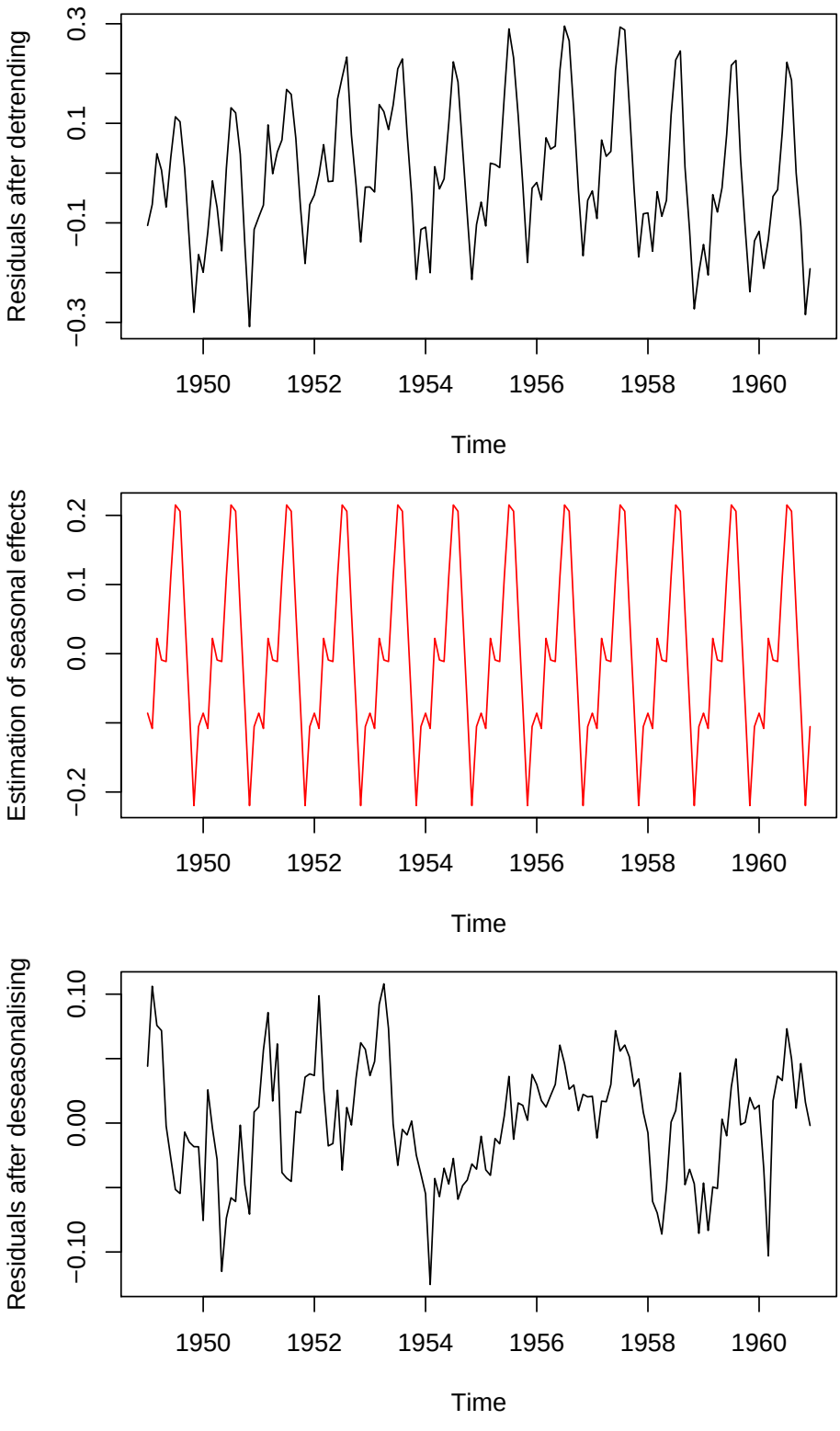


Figure 2.5: Demonstration of a simple technique for deseasonalisation applied to monthly totals of international airline passengers (in thousands), 1949-1960.

2.6 Autocorrelation

One of the key features of time series data is that successive observations in time can be correlated. Simply put, this means that what happens now (x_t) can influence what will happen next (x_{t+k} for $k > 0$). This means we cannot make the assumption that the data is iid. In order to study the relationship between what happens at different points in time we introduce the idea of **autocorrelation**.

The **sample correlation coefficient** you are familiar with measures the linear association between two samples $\{x_i\}_{i=1}^N$ and $\{y_i\}_{i=1}^N$ and is given by

$$r = \frac{\sum(x_i - \overline{x})(y_i - \overline{y})}{\sqrt{\sum(x_i - \overline{x})^2 \sum(y_i - \overline{y})^2}}$$

Meanwhile, the **sample autocorrelation coefficient** measures the linear association between observations in a time series that are separated by different distances or lags.

Given N time series observations $x_1, \ldots, x_N$ we can form $N - k$ pairs of observations (x_t, x_{t+k}) for $t = 1, ..., N - k$ and $0 \leq k < N$. For example, when $k = 1$ we have the $N - 1$ pairs

$$(x_1, x_2), \quad (x_2, x_3), \ldots, (x_{N-1}, x_N).$$

Regarding the first of each pair as one variable and the other as another variable we can form the sample autocorrelation at lag 1 by

$$r_1 = \frac{\sum_{t=1}^{N-1}(x_t - \overline{x}_{(1)})(x_{t+1} - \overline{x}_{(2)})}{\sqrt{\sum_{t=1}^{N-1}(x_t - \overline{x}_{(1)})^2 \sum_{t=1}^{N-1}(x_{t+1} - \overline{x}_{(2)})^2}}$$

where

$$\overline{x}_{(1)} = \frac{\sum_{t=1}^{N-1} x_t}{N-1} \quad \text{and} \quad \overline{x}_{(2)} = \frac{\sum_{t=2}^{N} x_t}{N-1}$$

are the means of the first and last $N - 1$ observations respectively.

Usually, for large N we have that the mean estimates and the variance estimates (in the denominator) computed from the $N - 1$ values are close to the estimates computed from all N values, *i.e.*

$$\overline{x}_{(1)} \simeq \overline{x}_{(2)} \simeq \overline{x} = \frac{\sum_{t=1}^{N} x_t}{N}.$$

As a result the **sample autocorrelation coefficient** at lag 1 is given by

$$r_1 = \frac{\sum_{t=1}^{N-1}(x_t - \bar{x})(x_{t+1} - \bar{x})}{\sum_{t=1}^{N}(x_t - \bar{x})^2}. \tag{2.5}$$

In a similar way we can find the correlation between observations that are k time-steps apart.

Thus, we define the **sample autocorrelation coefficient** at lag k by

$$r_k = \frac{\sum_{t=1}^{N-k}(x_t - \bar{x})(x_{t+k} - \bar{x})}{\sum_{t=1}^{N}(x_t - \bar{x})^2} \tag{2.6}$$

for $0 \leq k < N$.

Note: In practice, the autocorrelation coefficients are usually computed by computing a series of autocovariance coefficients $\{c_k\}$, where

$$c_k := \frac{1}{N} \sum_{t=1}^{N-k} (x_t - \bar{x})(x_{t+k} - \bar{x}).$$

This is known as the **autocovariance** at lag k and we have $r_k = c_k/c_0$.

2.6.1 The correlogram

The correlogram is a graphical representation of the autocorrelations. For each lag, k, a vertical line corresponding to the sample autocorrelation coefficient r_k is plotted. The correlogram of a given observed time series may be plotted in **R** using the command `acf()`.

Note: The first sample autocorrelation coefficient is always given by $r_0 \equiv 1$! This is because it describes the correlation of an observation in a time series with itself.

How do we tell what is significant correlation and what is just random noise?

If $x_1, x_2, ..., x_N$ are completely random then, provided N is large, $r_k \approx 0$ for all $k \neq 0$. More precisely, it can be shown that for a random series

$$r_k \sim \mathcal{N}(0, 1/N)$$

and thus we can form a 95% confidence interval for the "true" autocorrelations by $\pm 2/\sqrt{N}$ and any autocorrelations outside of this confidence interval indicate that the series is not completely random.

Note: You would still expect 1 in 20 (5%) of the r_k to be outside of this confidence interval.

An example random series and its correlogram is shown in figure 2.6. The blue dashed line shows the tolerance limits of $\pm 2/\sqrt{N}$.

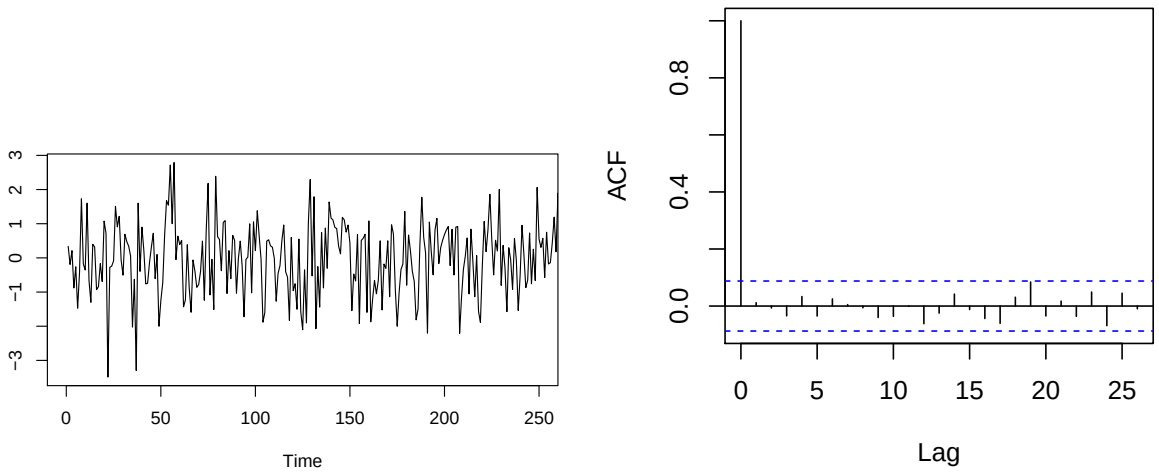


Figure 2.6: A random series and the corresponding correlogram.

Typical Features of the Correlogram

Now that we have determined how to recognise when the autocorrelation coefficient is significantly different from 0, let's consider some typical features of the correlogram:

- A time series with **short term autocorrelation** is characterised by a fairly large value of $|r_1|$ followed by successively smaller values, with $|r_k| \to 0$ as $k \to \infty$.

  The body temperature of a beaver exhibits short term correlation as shown in figure 2.7 (the dots show when the beaver is awake/asleep).

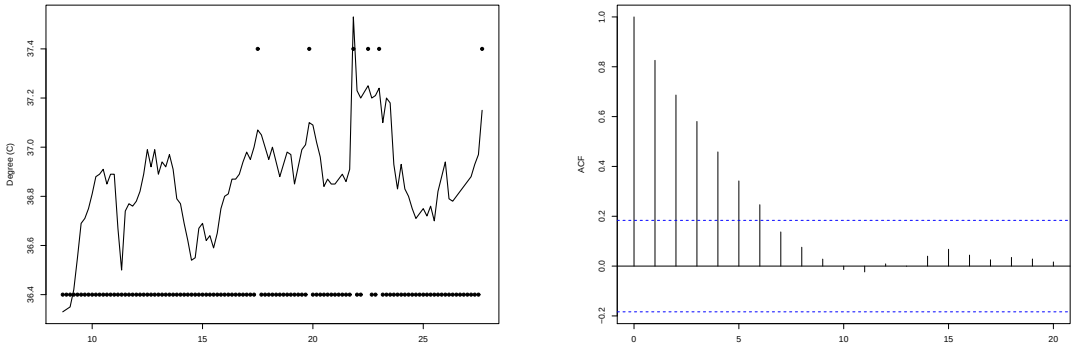


Figure 2.7: The body temperature of a beaver as a function of time (in hours) and the corresponding correlogram.

- An **alternating time series** is characterised by alternating positive and negative values of r_k, e.g. $r_1 < 0$, $r_2 > 0$, $r_3 < 0$ etc.

  Figure 2.8 shows an example of an alternating series and its correlogram.

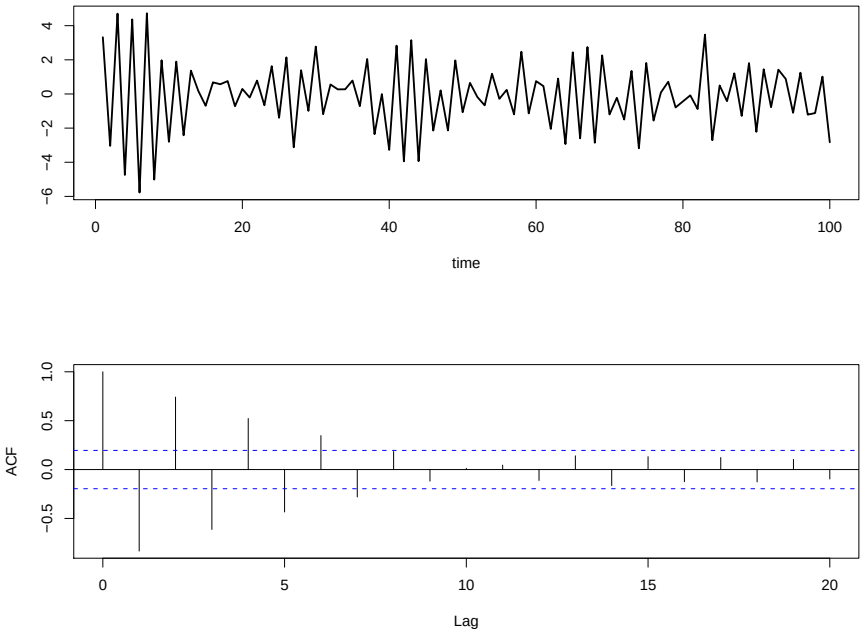


Figure 2.8: An alternating series and the corresponding correlogram

- If the time series contains **seasonal variation**, then the correlogram will contain an oscillation at the same frequency. In particular, if $x_t = acos(\omega t)$, where a is a constant and $0 < \omega < \pi$, then $r_k = cos(\omega k)$ for large N.

  Average air temperatures exhibit a seasonal pattern as shown in figure 2.9.

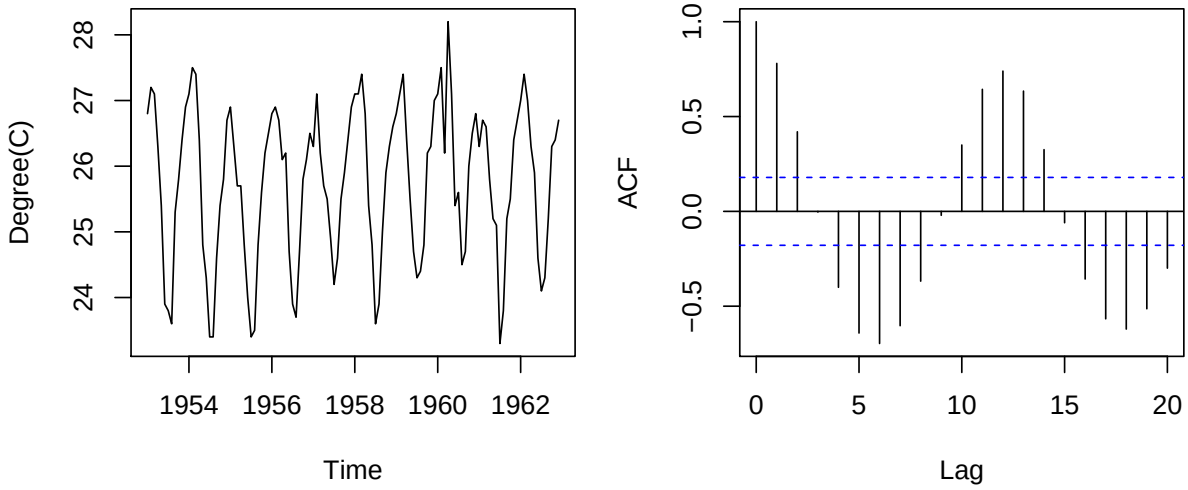


Figure 2.9: Average monthly air temperatures at Recife, Brazil, and the corresponding correlogram

We tend to deseasonalise the time series before taking autocorrelation.

Figure 2.10 shows the deseasonalised air temperatures and corresponding correlogram, which now shows some slight short term correlation.

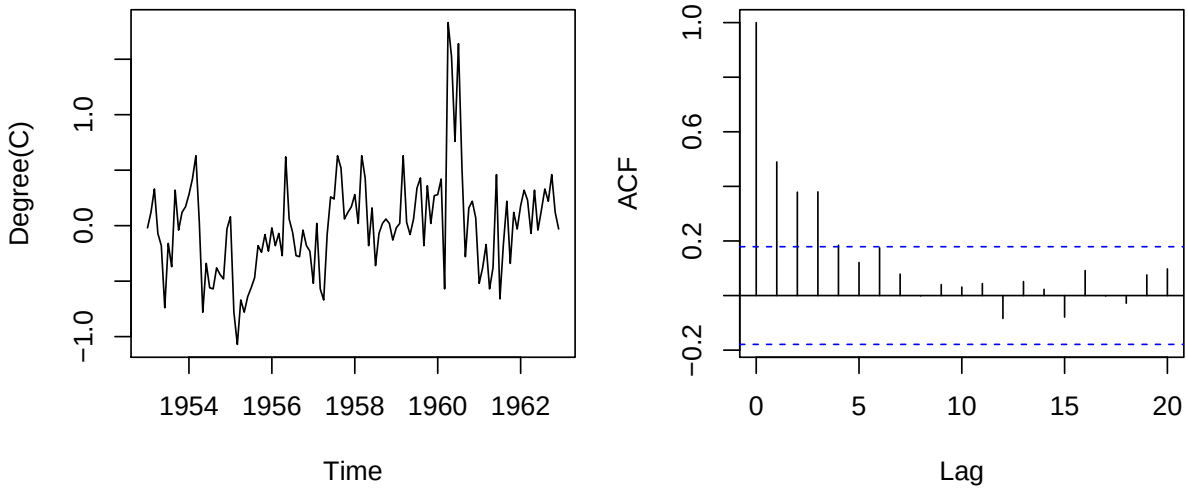


Figure 2.10: Deseasonalised average monthly air temperatures at Recife, Brazil, and the corresponding correlogram

- If there is a **trend**, then r_k will not tend to zero except for very large k. Typically, we detrend the series before taking the autocorrelation.

  Global temperature deviations from a baseline are, unfortunately, an example of a time series with an increasing trend as shown in figure 2.11.

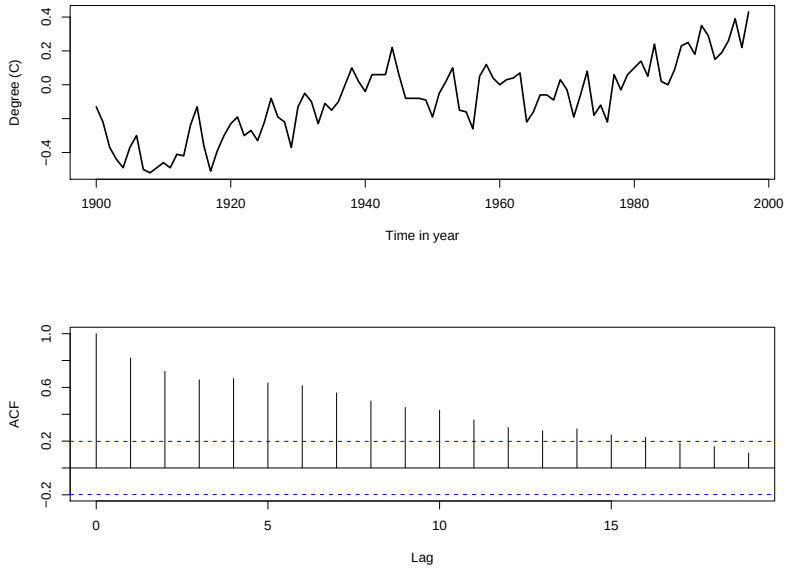


Figure 2.11: A time series plot of global temperature deviations and the corresponding correlogram.

- **Outliers** usually seriously affect correlogram as they can cause spurious correlations at a lag of equal distance to that between the outliers.

  Figure 2.12 demonstrates this effect, showing a random time series with two outliers at a separation of three time units and the corresponding correlogram exhibiting a spurious correlation at lag three. To avoid this we remove the outliers before taking the autocorrelation.

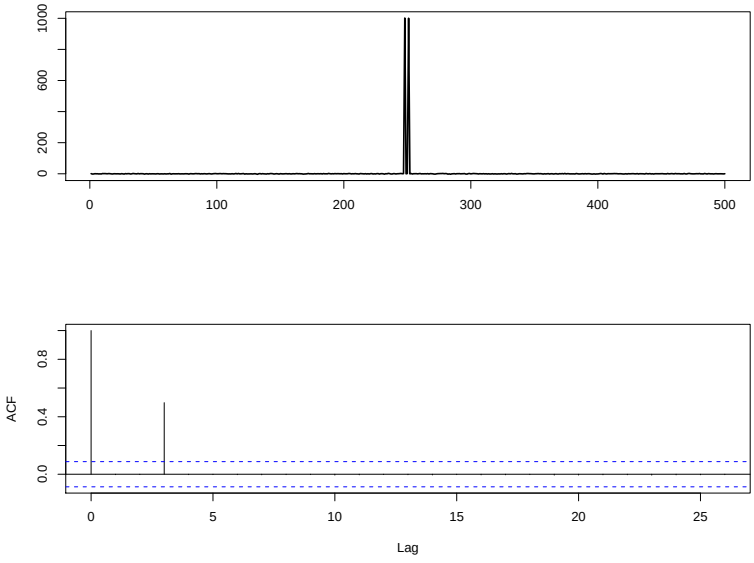


Figure 2.12: A random series with outliers and the corresponding correlogram.

Chapter 3

Linear Models for Time Series

In the previous chapter we looked at some simple descriptive techniques for time series. In particular, we discussed how systematic features of the time series, such as seasonality and trend, could be estimated, recorded and removed leaving a set of 'stationary' residuals. From an intuitive point of view this means that the properties of one section of the data are much like those of any other. This process is useful not only in terms of analysing the systematic features of the data, but also because much of the probability theory of time series is concerned with stationary models. In this chapter we will give a mathematical definition of stationarity as well as introducing several different types of stochastic processes that may be appropriate when setting up a model for a time series.

3.1 Probabilistic Behaviour of a Time Series

The probabilistic behaviour of a time series is fully specified by the joint probability distribution of $X_{t_1}, \ldots, X_{t_n}$ for any set of times $t_1, \ldots, t_n$ and any value of n. However, this is rather complicated and is usually not attempted in practice. A more useful way of describing a time series is through its moments. Particularly the first and second moments, which are given by the mean, variance and autocovariance as follows:

- **Mean**: $\mu(t) = \mathbb{E}[X_t]$,
- **Variance**: $\sigma^2(t) = \mathrm{Var}[X_t]$,
- **Autocovariance**: $\gamma(t_1, t_2) = \mathrm{Cov}(X_{t_1}, X_{t_2}) = \mathbb{E}[(X_{t_1} - \mu_X(t_1))(X_{t_2} - \mu_X(t_2))]$.

3.2 Stationary Processes

A **strictly stationary** time series is one where the joint distribution of $(X_{t_1}, \ldots, X_{t_n})$ is the same as the joint distribution of $(X_{\tau+t_1}, \ldots, X_{\tau+t_n})$ for all $t_1, \ldots, t_n$, τ and n.

In plain English: Shifting the time origin by a set amount has no effect on the joint distribution.

Properties:

- The mean and variance do not depend on t, that is

$$\mu(t) := \mathbb{E}[X_t] = \mu \quad \text{and} \quad \sigma^2(t) := \mathrm{Var}[X_t] = \sigma^2.$$

- The autocovariance only depends on the lag and can be written as

$$\gamma(k) = \mathrm{Cov}(X_t, X_{t+k}) = \mathbb{E}[(X_t - \mu)(X_{t+k} - \mu)].$$

  This is called the **autocovariance function** (a.c.v.f.) at lag k.

In practice it is useful to define stationarity in a less restrictive way than strict stationarity.

A process is called **second-order stationary** (or weakly stationary) if the first and second moments are time invariant, that is:

- its mean is constant, $\mathbb{E}[X_t] = \mu$
- its variance is constant and finite, $\mathrm{Var}[X_t] = \gamma(0) < \infty$
- its a.c.v.f depends only on the lag, k,

$$\mathrm{Cov}(X_t, X_{t+k}) = \mathrm{Cov}(X_0, X_k) =: \gamma(k).$$

Properties:

- The second condition is implied by the third since

$$\mathrm{Var}[X_t] = \mathrm{Cov}(X_t, X_t) = \gamma(0) = \text{constant}.$$

- If $\{X_t\}$ is strictly stationary then $\{X_t\}$ is second-order stationary.
- If $\{X_t\}$ is second-order stationary and the joint distribution of $X_{t_1}, \ldots, X_{t_n}$ is multivariate normal for all $t_1, \ldots, t_n$ and n, then $\{X_t\}$ is strictly stationary.

From now on whenever we use the term stationary we shall mean second-order stationary unless we specifically indicate otherwise.

3.3 The Autocorrelation Function

The **autocorrelation function** (a.c.f.) at lag k, $\rho(k)$, of a stationary stochastic process $\{X_t\}$ is given by

$$\rho(k) = \gamma(k)/\gamma(0) = \gamma(k)/\sigma^2.$$

Properties:

1. $\boxed{\rho(0) = \gamma(0)/\gamma(0) = 1.}$
2. The autocorrelation function is an even function of the lag, *i.e.* $\boxed{\rho(k) = \rho(-k)}$.

   Proof:

   $$\begin{aligned} \gamma(k) &= \mathrm{Cov}(X_t, X_{t+k}) \\ &= \mathrm{Cov}(X_{t+k}, X_t) && \text{by symmetry of the covariance function} \\ &= \mathrm{Cov}(X_t, X_{t-k}) && \text{by stationarity} \\ &= \gamma(-k) \end{aligned}$$

3. $\boxed{|\rho(k)| \leq 1}$ This is the 'usual' property of correlation.

   Proof:

   $$\begin{aligned} 0 &\leq \mathrm{Var}(\lambda_1 X_t + \lambda_2 X_{t+k}) \\ &= \lambda_1^2 \mathrm{Var}(X_t) + \lambda_2^2 \mathrm{Var}(X_{t+k}) + 2\lambda_1\lambda_2 \mathrm{Cov}(X_t, X_{t+k}) \\ &= (\lambda_1^2 + \lambda_2^2)\gamma(0) + 2\lambda_1\lambda_2\gamma(k) && \text{by stationarity} \end{aligned}$$

   for any lag k and any λ_1, $\lambda_2 \in \Re$. The two bounds can then be determined as follows:

   - Setting $\lambda_1 = \lambda_2 = 1$ yields $\rho(k) \geq -1$ by a simple rearrangement.
   - Similarly, setting $\lambda_1 = -\lambda_2 = 1$ yields $\rho(k) \leq 1$.

3.4 Purely Random Processes

A discrete process $\{Z_t\}$ is called a **purely random process** if the random variables Z_t are a sequence of mutually independent, identically distributed variables.

Notes:

- If the mean and variance exist then they are constant for all t and

$$\gamma(k) = \mathrm{Cov}(Z_t, Z_{t+k}) = \begin{cases} \sigma_Z^2 & \text{for } k = 0 \\ 0 & \text{for } k \neq 0 \end{cases}$$

- It follows from the previous point that purely random processes are **second-order stationary**.
- Purely random processes are also **strictly stationary**, this follows directly from the fact that the Z_t are iid.
- Purely random processes are also known as white noise.

3.5 Random Walk

Suppose $\{Z_t\}$ is a purely random process with mean μ_Z and variance σ_Z^2. A process $\{X_t\}$ is said to be a **random walk** if

$$X_t = X_{t-1} + Z_t.$$

Notes:

- The process is usually started at zero when $t = 0$ so that

$$X_1 = Z_1 \quad \text{and} \quad X_t = \sum_{i=1}^{t} Z_i.$$

  It follows that the mean and variance are given by $\mathbb{E}[X_t] = \mu_Z t$ and $\text{Var}[X_t] = \sigma_Z^2 t$

- Since the mean and variance depend on time, random walks are **not second-order stationary**.

3.6 Moving Average Processes

Suppose $\{Z_t\}$ is a purely random process with mean 0 and variance σ_Z^2. Then a process $\{X_t\}$ is said to be a **moving average process of order** q, and is denoted by MA(q), if

$$X_t = \beta_0 Z_t + \beta_1 Z_{t-1} + \cdots + \beta_q Z_{t-q} = \sum_{i=0}^{q} \beta_i Z_{t-i} \tag{3.1}$$

where the $\{\beta_i\}$ are constants.

Thus, a moving average process is one in which what we observe now, X_t, is influenced by a set of purely random events that happened in the past, Z_t, Z_{t-1}, ..., Z_{t-q}. The order, q, of the process defines how far reaching the influence of these past events is.

Notes:

- The $\{Z_t\}$ are usually scaled so that $\beta_0 = 1$.
- It is immediate that the mean and variance are given by

$$\mathbb{E}[X_t] = \sum_{i=0}^{q} \mathbb{E}[\beta_i Z_{t-i}] = 0 \quad \text{and} \quad \text{Var}[X_t] = \sum_{i=0}^{q} \text{Var}[\beta_i Z_{t-i}] = \sum_{i=0}^{q} \beta_i^2 \text{Var}[Z_{t-i}] = \sigma_Z^2 \sum_{i=0}^{q} \beta_i^2.$$

3.6.1 The Autocovariance Function, $\gamma(k)$

The autocovariance function is given by

$$\gamma(k) = \begin{cases} \sigma_Z^2 \sum_{j=|k|}^{q} \beta_{j-|k|}\beta_j & \text{for } |k| \leq q \\ 0 & \text{for } |k| > q \end{cases} = \begin{cases} \sigma_Z^2 \sum_{i=0}^{q-|k|} \beta_i\beta_{i+|k|} & \text{for } |k| \leq q \\ 0 & \text{for } |k| > q. \end{cases} \tag{3.2}$$

Proof: Suppose $k \geq 0$. Then

$$\begin{aligned} \text{Cov}(X_t, X_{t+k}) &= \text{Cov}\left(\sum_{i=0}^{q} \beta_i Z_{t-i}, \sum_{j=0}^{q} \beta_j Z_{t+k-j}\right) \\ &= \sum_{i=0}^{q}\sum_{j=0}^{q} \beta_i\beta_j \text{Cov}(Z_{t-i}, Z_{t+k-j}) && \text{by linearity of the Covariance} \\ &= \sum_{i=0}^{q}\sum_{j=0}^{q} \beta_i\beta_j \delta_{i,j-k}\sigma_Z^2 && \text{by independence of the } Z_t \\ &= \sigma_Z^2 \sum_{j=k}^{q} \beta_{j-k}\beta_j && \text{for } 0 \leq k \leq q. \end{aligned}$$

where $\delta_{i,j} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$ is the Kronecker delta function. Note that we must be careful to enforce the conditions $0 \leq i, j \leq q$ when determining the non-zero values in the sum:

- Since we considering $k \geq 0$ then we have non-zero values when $i = j - k$ where $k \leq j <\leq q$.
- Meanwhile, if we were considering $k \leq 0$ then we would have $j = i + k$ where $-k \leq i \leq q$.

Consequences:

- It follows that MA processes are **second-order stationary** for all values $\{\beta_i\}$.
- Furthermore if the $\{Z_i\}$ are normally distributed then the process is also strictly stationary.
- The autocorrelation function is given by

$$\rho(k) = \frac{\gamma(k)}{\gamma(0)} = \begin{cases} 1 & \text{for } k = 0 \\ \dfrac{\sum_{i=0}^{q-|k|} \beta_i\beta_{i+|k|}}{\sum_{i=0}^{q} \beta_i^2} & \text{for } |k| = 1, 2, ..., q \\ 0 & \text{for } |k| > q \end{cases} \tag{3.3}$$

- Notice that the a.c.f. cuts off at lag q, which is a special feature of MA processes and can be used to identify the order by inspection of the correlogram.

- Note that we could have instead written

$$X_t = \tilde{\mu} + \beta_0 Z_t + \beta_1 Z_{t-1} + \cdots + \beta_q Z_{t-q} = \tilde{\mu} + \sum_{i=0}^{q} \beta_i Z_{t-i}$$

so that the moving average process oscillates instead around some non-zero mean, $\mathbb{E}(X_t) = \tilde{\mu}$, for constant $\tilde{\mu}$. This addition does not affect the a.c.f. and so has been omitted for simplicity.

3.6.2 Examples

Example 3.1. Consider the MA(1) process with $\beta_0 = 1$

$$X_t = Z_t + \beta Z_{t-1}.$$

Compute the a.c.f.

(a) By using the formula.

Answer:

In this example then we have $q = 1$, $\beta_0 = 1$ and $\beta_1 = \beta$. Plugging these values into the formula for $\rho(k)$ we find that

- $\rho(0) = 1$ as always
- when $k = \pm 1$ we evaluate the sums for $|k| = q = 1$ to give

$$\rho(1) = \rho(-1) = \frac{\beta_0 \beta_1}{\beta_0^2 + \beta_1^2} = \frac{\beta}{1 + \beta^2}$$

- $\rho(k) = 0$ for $|k| > 1$.

Combining these cases we find that

$$\rho(k) = \begin{cases} 1 & \text{for } k = 0 \\ \dfrac{\beta}{1+\beta^2} & \text{for } k = \pm 1 \\ 0 & \text{for } |k| > 1 \end{cases}$$

(b) By first calculating $\gamma(k) = \mathrm{Cov}(X_t, X_{t+k})$ from first principles and then using $\rho(k) = \gamma(k)/\gamma(0)$.

Answer:

In this case then we begin by determining the autocovariance, $\gamma(k)$, for general k

$$\begin{aligned} \gamma(k) &= \mathrm{Cov}(X_t, X_{t+k}) \\ &= \mathrm{Cov}(Z_t + \beta Z_{t-1}, Z_{t+k} + \beta Z_{t+k-1}) \\ &= \mathrm{Cov}(Z_t, Z_{t+k}) + \beta \mathrm{Cov}(Z_t, Z_{t+k-1}) + \beta \mathrm{Cov}(Z_{t-1}, Z_{t+k}) + \beta^2 \mathrm{Cov}(Z_{t-1}, Z_{t+k-1}) \end{aligned}$$

having expanded using properties of the covariance. Now we consider the different cases (noting that we need only consider $k \geq 0$ since the cases where $k < 0$ may then be obtained by symmetry):

– If $k = 0$, then

$$\begin{aligned}\gamma(0) &= \text{Cov}(Z_t, Z_t) + \beta\text{Cov}(Z_t, Z_{t-1}) + \beta\text{Cov}(Z_{t-1}, Z_t) + \beta^2(Z_{t-1}, Z_{t-1}) \\ &= (1 + \beta^2)\sigma_Z^2\end{aligned}$$

since $\text{Cov}(Z_i, Z_j) = 0$ if $i \neq j$ by independence.

– Similarly, if $k = 1$ then

$$\begin{aligned}\gamma(1) &= \text{Cov}(Z_t, Z_{t+1}) + \beta\text{Cov}(Z_t, Z_t) + \beta\text{Cov}(Z_{t-1}, Z_{t+1}) + \beta^2(Z_{t-1}, Z_t) \\ &= \beta\sigma_Z^2.\end{aligned}$$

– If $k > 1$ then it is clear that none of the indices will match and so $\text{Cov}(X_t, X_{t+k}) = 0$ in this case.

Combining we find that

$$\rho(k) = \frac{\gamma(k)}{\gamma(0)} = \begin{cases} 1 & \text{for } k = 0 \\ \dfrac{\beta}{1+\beta^2} & \text{for } k = \pm 1 \\ 0 & \text{for } |k| > 1 \end{cases}$$

which agrees with our answer obtained using the formula.

Exercise 3.1. Consider the MA(q) process with equal weights, $\beta_0 = \beta_1 = ... = \beta_q = \frac{1}{q+1}$,

$$X_t = \frac{1}{q+1}\sum_{i=0}^{q} Z_{t-i}.$$

Show that

$$\rho(k) = \begin{cases} 1 - \dfrac{|k|}{q+1} & \text{for } |k| \leq q \\ 0 & \text{for } |k| > q. \end{cases}$$

3.7 Operator Notation

The backward shift operator, B, is defined by

$$B^j X_t = X_{t-j}, \qquad \forall j \in \mathbb{Z}$$

i.e. B shifts a discrete series 'back' one unit of time.

Using this notation we can now write the MA(q) process as

$$\begin{aligned} X_t &= \beta_0 Z_t + \beta_1 Z_{t-1} + \cdots + \beta_q Z_{t-q} \\ &= (\beta_0 + \beta_1 B + \beta_2 B^2 + \cdots + \beta_q B^q) Z_t \\ &= \theta_\beta(B) Z_t \end{aligned}$$

where $\theta_\beta(z) = \beta_0 + \beta_1 z + \beta_2 z^2 + \cdots + \beta_q z^q$ is a polynomial of degree q. This is particularly useful when considering the *invertibility* of a process *i.e.* whether we can write our process in the form

$$Z_t = \theta_\beta^{-1}(B) X_t = \sum_{i=0}^{\infty} \pi_i X_{t-i}$$

where $\{\pi_i\}$ are some new set of constants. If we can, and if this infinite sum is convergent ($\sum_{i=0}^{\infty} |\pi_i| < \infty$), then we call the process **invertible**. It turns out that this condition for invertibility can be written more simply as follows:

An MA(q) process is called **invertible** if the roots of the polynomial

$$\theta_\beta(z) = \beta_0 + \beta_1 z + \beta_2 z^2 + \cdots + \beta_q z^q$$

lie outside the unit circle where we regard z as a complex variable, not as an operator.

Proof for an MA(1) process
For the MA process

$$X_t = Z_t + \theta Z_{t-1}$$

then we find (by successive substitution) that

$$\begin{aligned} Z_t &= X_t - \theta Z_{t-1} \\ &= X_t - \theta(X_{t-1} - \theta Z_{t-2}) = X_t - \theta X_{t-1} + \theta^2 Z_{t-2} \\ &= ... \\ &= X_t - \theta X_{t-1} + \theta^2 X_{t-2} - ... \end{aligned}$$

If $|\theta| < 1$ then the series of coefficients of X_{t-j} converges. Comparing this result to the corresponding polynomial $\theta_\beta(z) = 1 + \theta z$ we find that the process is invertible if the root $z = -1/\theta$ is outside the unit circle, *i.e.* $|z| = \frac{1}{|\theta|} > 1$.

Note: The imposition of the invertibility condition ensures that there is a unique MA process for a given a.c.f. To see how this works one can consider a second MA(1) process given by $X_t = Z_t + \frac{1}{\theta} Z_{t-1}$. It has the same a.c.f. as the above example, but is not invertible.

3.8 Autoregressive Processes

Suppose $\{Z_t\}$ is a purely random process with mean 0 and variance σ_Z^2. Then a process $\{X_t\}$ is said to be an **autoregressive process of order** p, and is denoted by AR(p), if

$$X_t = \alpha_1 X_{t-1} + \alpha_2 X_{t-2} + \cdots + \alpha_p X_{t-p} + Z_t = \sum_{i=1}^{p} \alpha_i X_{t-i} + Z_t \tag{3.4}$$

where the $\{\alpha_i\}$ are constants.

Thus, an autoregressive process is one in which what we observe now, X_t, may be described as a function of our past observations, X_{t-1}, X_{t-2}, ..., X_{t-p} plus some random noise, Z_t. The order of the process, p, determines the number of past observations needed to predict what's happening now. This is reminiscent of a multiple regression (hence *regressive*), but X_t is regressed on past values of itself (hence *auto*) rather than on other independent processes.

Since an autoregressive process depends on past values of itself, it becomes much more challenging to determine the mean, variance and a.c.f. and hence to establish whether the process is stationary. So, it would be convenient if we could rewrite our AR process as an MA process instead. Thus, we turn to an idea closely related to *invertibility* of an MA process, namely **causality**.

We can use operator notation to represent an AR(p) process as follows,

$$\begin{aligned} X_t - \alpha_1 X_{t-1} - \alpha_2 X_{t-2} - \cdots - \alpha_p X_{t-p} &= Z_t \\ (1 - \alpha_1 B - \alpha_2 B^2 - \cdots - \alpha_p B^p) X_t &= Z_t \\ \phi_\alpha(B) X_t &= Z_t \end{aligned}$$

where $\phi_\alpha(z) = 1 - \alpha_1 z - \alpha_2 z^2 - \cdots - \alpha_p z^p$ is a polynomial of degree p.

Then, we may rewrite our process as

$$X_t = \phi_\alpha^{-1}(B) Z_t = \sum_{i=0}^{\infty} \beta_i Z_{t-i}$$

for some new set of constants, $\{\beta_i\}$. Thus, we have rewritten our finite order AR(p) process as an MA process of infinite order. If we can do this, and if the infinite sum is convergent ($\sum_{i=0}^{\infty} |\beta_i| < \infty$) then we call the process **causal**. Analagously to invertibility it turns out that there is a simpler condition for causality.

An AR(p) process is called **causal** if the roots of the polynomial

$$\phi_\alpha(z) = 1 - \alpha_1 z - \alpha_2 z^2 - \cdots - \alpha_p z^p$$

lie outside the unit circle where we regard z as a complex variable.

Proof for an AR(1) process
For the AR process

$$X_t = \alpha X_{t-1} + Z_t$$

then we find (by successive substitution) that

$$\begin{aligned} X_t &= Z_t + \alpha X_{t-1} \\ &= Z_t + \alpha(Z_{t-1} + \alpha X_{t-2}) = Z_t + \alpha Z_{t-1} + \alpha^2 X_{t-2} \\ &= \dots \\ &= Z_t + \alpha Z_{t-1} + \alpha^2 Z_{t-2} + \dots \end{aligned} \tag{3.5}$$

If $|\alpha| < 1$ then the series of coefficients of Z_{t-j} converges. Comparing this result to the corresponding polynomial $\phi_\alpha(z) = 1 - \alpha z$ we find that the process is causal if the root $z = 1/\alpha$ is outside the unit circle, *i.e.* $|z| = \frac{1}{|\alpha|} > 1$.

An alternative (and equivalent) condition for causality that is sometimes more convenient is as follows.

An AR(p) process is called **causal** if the roots of the polynomial

$$f(z) = z^p - \alpha_1 z^{p-1} - \alpha_2 z^{p-2} - \cdots - \alpha_p$$

lie inside the unit circle where we regard z as a complex variable.

Proof:
The two polynomials are related through $\phi_\alpha(z) = z^p f(1/z)$ and so, if the roots of $f(z)$ are given by $\{\mu_i\}$, and we write $f(z)$ in terms of its roots as $f(z) = \prod_{i=1}^{p}(z - \mu_i)$, we may easily see that

$$\phi_\alpha(z) = z^p \prod_{i=1}^{p}(\frac{1}{z} - \mu_i) = \prod_{i=1}^{p}(1 - \mu_i z).$$

So, the roots $\{\nu_i\}$ of $\phi_\alpha(z)$ are reciprocals of those of $f(z)$, *i.e.* $\nu_i = 1/\mu_i$ and it follows that $|\nu_i| > 1 \Leftrightarrow |\mu_i| < 1$.

So far, so good, but how does this help us with determining stationarity? Well, it turns out that

Causality $\implies$ **Stationarity**.

Proof
Let's compute our mean, variance and a.c.f. so we can check the conditions for stationarity

$$\mathbb{E}[X_t] = 0, \qquad \text{Var}[X_t] = \sigma_Z^2 \sum_{i=0}^{\infty} \beta_i^2 \qquad \text{and} \qquad \gamma(k) = \sigma_Z^2 \sum_{j=|k|}^{\infty} \beta_{j-|k|}\beta_j = \sigma_Z^2 \sum_{i=0}^{\infty} \beta_i \beta_{i+|k|}. \tag{3.6}$$

Thus,

1. The mean is constant, $\mathbb{E}[X_t] = 0$.
2. The variance is constant and finite provided that the sum $\sum_{i=0}^{\infty} \beta_i^2$ converges. A sufficient condition for this sum to converge is that $\sum_{i=0}^{\infty} |\beta_i| < \infty$.
3. The autocovariance depends only on the lag, k.

So, if a process is causal it will also be stationary.

Now that we have established a condition for stationarity we turn out attention to calculating the autocorrelation function of an AR process. We can, in principle find the a.c.f. for a general order AR(p) process using (3.6). But the constants $\{\beta_i\}$ may be hard to find so we will also discuss an alternative option for computing the a.c.f.

3.8.1 The Autocorrelation of a First Order Autogressive Process, AR(1)

First, let's consider an AR(1) process defined by

$$X_t = \alpha X_{t-1} + Z_t.$$

In this case there are two main methods for computing $\rho(k)$:

Method 1:
In this simple case then we *can* readily determine the constants $\{\beta_i\}$ by using successive substitution to rewrite the process as an MA process of infinite order as in (3.5). Thus we have

$$X_t = Z_t + \alpha Z_{t-1} + \alpha^2 Z_{t-2} + \cdots = \sum_{i=0}^{\infty} \alpha^i Z_{t-i}$$

which corresponds to a MA(∞) process with $\beta_i = \alpha^i$. Substituting directly into our formula for the a.c.v.f. (3.6) we find that

$$\gamma(k) = \sigma_Z^2 \sum_{i=0}^{\infty} \beta_i \beta_{i+|k|} = \sigma_Z^2 \sum_{i=0}^{\infty} \alpha^i \alpha^{(i+|k|)} = \sigma_Z^2 \alpha^{|k|} \sum_{i=0}^{\infty} \alpha^{2i} = \frac{\sigma_Z^2 \alpha^{|k|}}{1-\alpha^2}$$

where the geometric series converges provided $|\alpha| < 1$. So the autocorrelation function is given by

$$\rho(k) = \frac{\gamma(k)}{\gamma(0)} = \frac{\sigma_Z^2 \alpha^{|k|}/(1-\alpha^2)}{\sigma_Z^2/(1-\alpha^2)} = \alpha^{|k|} \qquad \text{for } |\alpha| < 1.$$

Method 2:
Alternatively, you can *first check the process is stationary* (*i.e.* $|\alpha| < 1$) then multiply the equation defining X_t by X_{t-k} for $k \geq 1$ and take expectations to give

$$\mathbb{E}[X_t X_{t-k}] = \alpha \mathbb{E}[X_{t-1} X_{t-k}] + \mathbb{E}[Z_t X_{t-k}].$$

Now, dealing with the terms one by one we have

- Using our assumption of stationarity along with the fact that $\mathbb{E}[X_t] = 0$ then

$$\gamma(k) = \text{Cov}(X_t, X_{t+k}) = \text{Cov}(X_t, X_{t-k}) = \mathbb{E}[X_t X_{t-k}].$$

- Similarly, $\gamma(k-1) = \mathbb{E}[X_{t-1} X_{t-k}]$.
- Finally, we use the fact that Z_t is independent of past values of X_t (*i.e.* Z_t is independent of X_{t-k} for $k > 0$) to evaluate

$$\mathbb{E}[Z_t X_{t-k}] = \mathbb{E}[Z_t]\mathbb{E}[X_{t-k}] = 0.$$

Combining we find that

$$\gamma(k) = \alpha\gamma(k-1) = \alpha^2\gamma(k-2) = \cdots = \alpha^k \gamma(0)$$

and so, using $\rho(-k) = \rho(k)$, we ultimately gain the result

$$\rho(k) = \frac{\gamma(k)}{\gamma(0)} = \alpha^{|k|}$$

as before.

3.8.2 The Autocorrelation of General Order Autoregressive Processes, AR(p)

Next, we consider a general order AR(p) process defined by

$$X_t = \alpha_1 X_{t-1} + \alpha_2 X_{t-2} + \cdots + \alpha_p X_{t-p} + Z_t.$$

The coefficients, $\{\beta_i\}$, in the MA(∞) representation of higher order AR(p) processes are increasingly algebraically complex. Thus, we generally compute the a.c.f. using a method method analogous to 'method 2' above. But first, to give an idea of how this process continues, we consider the MA(∞) representation of an AR(2) process.

Using the backward shift operator we write

$$Z_t = \phi_\alpha(B) X_t$$

where $\phi_\alpha(z) = 1 - \alpha_1 z - \alpha_2 z^2 \equiv (1 - \mu_1 z)(1 - \mu_2 z)$ is the characteristic polynomial, whilst μ_1 and μ_2 are the roots of the alternative version $f(z) = z^2 - \alpha_1 z - \alpha_2$.

Using partial fraction decomposition we find that

$$\phi_\alpha^{-1}(z) = \frac{1}{(1 - \mu_1 z)(1 - \mu_2 z)} = \frac{A}{1 - \mu_1 z} + \frac{B}{1 - \mu_2 z}$$

where

$$A = \frac{\mu_1}{\mu_1 - \mu_2} \qquad \text{and} \qquad B = \frac{-\mu_2}{\mu_1 - \mu_2}.$$

If $|\mu_1| < 1$ and $|\mu_2| < 1$ (hence if the process is causal) then we may use the formula for the sum of an infinite geometric series to write

$$\phi_\alpha^{-1}(z) = \sum_{i=0}^{\infty} \frac{\mu_1^{i+1} - \mu_2^{i+1}}{\mu_1 - \mu_2} z^i$$

and so

$$X_t = \sum_{i=0}^{\infty} \beta_i Z_{t-i} \qquad \text{where} \qquad \beta_i = \frac{\mu_1^{i+1} - \mu_2^{i+1}}{\mu_1 - \mu_2} \tag{3.7}$$

is the MA(∞) representation of our AR(p) process. In principle we could use the coefficients $\{\beta_i\}$ in (3.6) to determine $\rho(k)$. However, this process is long and laborious even for $p = 2$ and will only become more so for higher order processes. So, we instead turn our attention to a different method.

3.8.3 The Yule-Walker Equations

This method is analogous to 'method 2' for an AR(1) process. First we check stationarity (*i.e.* $|\mu_i| < 1$ for $i = 1, ..., p$).

Then, we multiply the equation defining X_t

$$X_t - \alpha_1 X_{t-1} - \alpha_2 X_{t-2} - \cdots - \alpha_p X_{t-p} = Z_t$$

by X_{t-k} for $k \geq 0$ and take expectations to give

$$\mathbb{E}[X_t X_{t-k}] - \alpha_1 \mathbb{E}[X_{t-1} X_{t-k}] - \alpha_2 \mathbb{E}[X_{t-2} X_{t-k}] - \cdots - \alpha_p \mathbb{E}[X_{t-p} X_{t-k}] = \mathbb{E}[Z_t X_{t-k}].$$

Recall that $\gamma(k) = \mathbb{E}[X_t X_{t-k}]$ and so we have

$$\gamma(k) - \alpha_1 \gamma(k-1) - \alpha_2 \gamma(k-2) - \cdots - \alpha_p \gamma(k-p) = \begin{cases} \sigma_Z^2 & \text{if } k = 0 \\ 0 & \text{if } k > 0 \end{cases}$$

where

$$\mathbb{E}[Z_t X_{t-k}] = \mathbb{E}\left[Z_t \left(\sum_{i=0}^{\infty} \beta_i Z_{t-k-i} \right) \right] = \mathbb{E}\left[Z_t \left(Z_{t-k} + \beta_1 Z_{t-k-1} + \beta_2 Z_{t-k-2} + \dots \right) \right] = \begin{cases} \sigma_Z^2 & \text{if } k = 0 \\ 0 & \text{if } k > 0 \end{cases}$$

in which we have used the MA(∞) representation for a causal AR(p) from (3.7) along with the fact that $\beta_0 = 1$.

One approach to determining $\rho(k)$ is to first consider the equations corresponding to $k = 0, ..., p$. This provides $p+1$ equations for $\gamma(0)$, ..., $\gamma(p)$. Solving them and then using the recurrence relation

$$\gamma(k) = \alpha_1 \gamma(k-1) + \alpha_2 \gamma(k-2) + \cdots + \alpha_p \gamma(k-p)$$

for $k > p$ allows us to determine the a.c.v.f. (and hence the a.c.f.) for all lags, k.

Alternatively, we can deal with the a.c.f. directly. Dividing through by $\gamma(0)$ yields the **Yule-Walker equations**

$$\rho(k) = \alpha_1 \rho(k-1) + \alpha_2 \rho(k-2) + \cdots + \alpha_p \rho(k-p) \qquad \text{for } k > 0. \tag{3.8}$$

They have the general solution

$$\rho(k) = A_1 \mu_1^{|k|} + A_2 \mu_2^{|k|} + \cdots + A_p \mu_p^{|k|} \tag{3.9}$$

where $\{\mu_i\}$ are the roots of the alternative polynomial

$$f(z) = z^p - \alpha_1 z^{p-1} - \alpha_2 z^{p-2} - \cdots - \alpha_p.$$

The coefficients A_1, A_2, ..., A_p may then be determined by enforcing the condition $\rho(0) = 1$ along with $p-1$ further conditions obtained by solving the first $p-1$ Yule-Walker equations for $\rho(k)$ with $k = 1, ..., p-1$. Substitution into the general solution then yields a system of p linear equations for A_1, A_2, ..., A_p.

This is perhaps more easily understood in the context of an example.

Example 3.2. Consider the AR(2) process

$$X_t = X_{t-1} - \frac{1}{2}X_{t-2} + Z_t$$

where $\{Z_t\}$ is a purely random process mean 0 and variance σ_Z^2. Determine its autocorrelation function.

Our approach to answering questions of this type is somewhat algorithmic as you will see below:

Step 1: Find the roots μ_1 and μ_2.
In this case the alternative polynomial is given by

$$f(z) = z^2 - \alpha_1 z - \alpha_2 = z^2 - z + \frac{1}{2}$$

and has roots

$$\mu_{1,2} = \frac{1 \pm \sqrt{1-2}}{2} = \frac{1 \pm i}{2}.$$

Step 2: Check Stationarity.

$$|\mu_1| = |\mu_2| = \frac{1}{\sqrt{2}} < 1$$

so the roots of the polynomial $f(z)$ lie within the unit circle and hence the process is stationary.

Step 3: Use the Yule-Walker equations to find $\rho(1)$.
We have two remaining unknowns, A_1 and A_2, in the general solution so we need to enforce two conditions. Namely, the values of $\rho(0)$ and $\rho(1)$. $\rho(0) = 1$ as always. For $k > 0$ we need to use the Yule-Walker equations,

$$\rho(k) = \rho(k-1) - \frac{1}{2}\rho(k-2).$$

When $k = 1$ this becomes $\rho(1) = \rho(0) - \frac{1}{2}\rho(-1)$. Using $\rho(k) = \rho(-k)$ and $\rho(0) = 1$ we thus find $\rho(1) = 2/3$.

Step 4: Determine the unknown coefficients, A_1 and A_2.
The general solution is given by

$$\rho(k) = A_1\mu_1^{|k|} + A_2\mu_2^{|k|} = A_1\left(\frac{1+i}{2}\right)^{|k|} + A_2\left(\frac{1-i}{2}\right)^{|k|}.$$

We have two unknowns, so we need to enforce two conditions. Namely, the values of $\rho(0)$ and $\rho(1)$ found in the previous step.

- When $k = 0$ we have $\rho(0) = 1$. Substitution into the general solution yields

$$A_1 + A_2 = 1. \tag{3.10}$$

- When $k = 1$ we have $\rho(1) = 2/3$. Substitution into the general solution yields

$$\frac{2}{3} = A_1\left(\frac{1+i}{2}\right) + A_2\left(\frac{1-i}{2}\right). \tag{3.11}$$

Solving (3.10) and (3.11) simultaneously we ultimately find

$$A_1 = \frac{1}{2} - \frac{1}{6}i \qquad \text{and} \qquad A_2 = \frac{1}{2} + \frac{1}{6}i$$

Step 5: Substitute A_1 and A_2 back into the general solution.
Substitution yields

$$\rho(k) = \left(\frac{1}{2} - \frac{1}{6}i\right)\left(\frac{1}{2} + \frac{1}{2}i\right)^{|k|} + \left(\frac{1}{2} + \frac{1}{6}i\right)\left(\frac{1}{2} - \frac{1}{2}i\right)^{|k|}$$

which on first inspection appears complex. However, when we have complex roots, μ_1 and μ_2, it is typically helpful to express them in the form $r\,e^{i\theta}$. Thus, we find

$$\begin{aligned}\rho(k) &= \left(\frac{1}{\sqrt{2}}\right)^{|k|}\left[\left(\frac{1}{2} - \frac{1}{6}i\right)e^{i\pi|k|/4} + \left(\frac{1}{2} + \frac{1}{6}i\right)e^{-i\pi|k|/4}\right] \\ &= \left(\frac{1}{\sqrt{2}}\right)^{|k|}\left[\cos(\pi|k|/4) + \frac{1}{3}\sin(\pi|k|/4)\right].\end{aligned}$$

Note: We have only considered autoregressive processes with zero mean. Non-zero means can be dealt with by rewriting (3.4) as

$$X_t - \tilde{\mu} = \alpha_1(X_{t-1} - \tilde{\mu}) + \alpha_2(X_{t-2} - \tilde{\mu}) + \cdots + \alpha_p(X_{t-p} - \tilde{\mu}) + Z_t$$

for constant $\tilde{\mu}$. This addition does not affect the a.c.f. and so has been omitted for simplicity.

3.9 The Partial Autocorrelation Function

Note that, unlike the a.c.f. for an MA(q) process, there is no cut off in the a.c.f. for an AR(p) process. This might seem surprising since X_t only explicitly depends upon the previous p observations, X_{t-i} for $i = 1, ..., p$. But each of these past observations in turn depends on the observations before them. How this iterates is most easily explained for an AR(1) process. In this case then X_t only explicitly depends on X_{t-1}, but X_{t-1} in turn depends on X_{t-2} and X_{t-2} depends on X_{t-3} and so on... So, X_t in fact depends on *all* past values X_t, but this dependence is increasingly small as it is only felt through other intermediate observations and not directly.

We can see the consequence of this intuition in the general solution for the a.c.f. of an AR(p) process of arbitrary order (3.9). Indeed, $\rho(k)$ decays exponentially as $k \to \infty$ if the roots $\{\mu_i\}$ are all real. If some of the roots are complex, then they will be in conjugate pairs and $\rho(k)$ will dampen, in a sinusoidal fashion, exponentially fast as $k \to \infty$. So, a correlogram with rapid decay but no sharp cut off at any particular lag is indicative of an AR(p) process.

In order to eliminate the effect of intermediate values and consider only the direct effect of past observations we introduce the idea of partial correlation. The **partial correlation function**, $\rho_{X,Y \cdot Z_1,...,Z_n}$, is defined to be the correlation between the residuals e_X and e_Y resulting from the linear regression of X with $Z_1, ..., Z_n$ and of Y with $Z_1, ..., Z_n$, respectively. It is designed to measure dependence between X and Y with the influence of $Z_1, ..., Z_n$ removed. In time series analysis we consider the partial autocorrelation function.

The **partial autocorrelation function** at lag k, $\pi(k)$, of a stationary process $\{X_t\}$ with a.c.f. ρ_X is defined as

$$\pi(k) = \rho_{X_t, X_t - k \cdot X_{t-1}, \ldots, X_{t-k+1}}$$

and thus measures the dependence between observations separated by lag k with the influence of intermediate observations removed.

It can be computed recursively using $\rho_X(0), ..., \rho_X(k)$, but the computational complexity of this process increases exponentially with the lag and is beyond the scope of the course. We are predominantly interested in its properties:

- $\pi(1) \equiv \rho_X(1)$ since there are no intermediate observations at lag 1.
- For an AR(p) process $\boxed{\pi(p) = \alpha_p}$ since it measures the excess correlation after the other observations X_{t-1}, ..., X_{t-p+1} have been taken into account.
- It can be shown that the p.a.c.f. of an AR(p) process **cuts off at lag** p, *i.e.* $\pi(k) = 0$ for $k > p$. This is a special feature of the AR process and can be used to identify the order by inspection of the correlogram.
- For an invertible MA process we can write

  $$X_t = \sum_{i=1}^{\infty} \alpha_i X_{t-i} + Z_t$$

  for some constants $\{\alpha_i\}$ satisfying $\sum_{i=1}^{\infty} |\alpha_i| < \infty$. Since there is direct correlation between X_t and its past values X_{t-i} for all i it is clear that there is ***no* cut off in the p.a.c.f.** This is another key point when fitting models, as we will see in the next chapter.

3.10 Mixed models

A useful class of time series is formed by combining MA and AR processes.

A **mixed autoregressive moving average (ARMA) process** containing p AR terms and q MA terms is said to be of order (p, q) and is defined as

$$X_t = \alpha_1 X_{t-1} + \cdots + \alpha_p X_{t-p} + Z_t + \beta_1 Z_{t-1} + \cdots + \beta_q Z_{t-q}.$$

Or, using the backward shift operator,

$$\phi_\alpha(B) X_t = \theta_\beta(B) Z_t$$

where $\phi_\alpha(z)$ and $\theta_\beta(z)$ are polynomials of degree p and q respectively where

$$\phi_\alpha(z) = 1 - \alpha_1 z - \cdots - \alpha_p z^p \qquad \text{and} \qquad \theta_\beta(z) = 1 + \beta_1 z + \cdots + \beta_q z^q.$$

Note: The importance of ARMA processes lies in the fact that a stationary time series may often be adequately modelled by an ARMA model involving fewer parameters than a pure MA or AR process by itself.

If the polynomials $\phi_\alpha(z)$ and $\theta_\beta(z)$ have no common zeroes, then the previously seen conditions for invertibility and causality still apply. So:

- An ARMA(p,q) model is **invertible** if the roots of the polynomial $\theta_\beta(z)$ lie outside the unit circle. In this case then we can write

  $$Z_t = \theta_\beta^{-1}(B)\phi_\alpha(B)X_t =: \psi(B)X_t$$

  where

  $$\psi(z) = 1 - \sum_{i=1}^{\infty} \psi_i z^i$$

  for some new set of constants $\{\psi_i\}$ in which the infinite sum is convergent. This allows us to rewrite our ARMA(p,q) process as an infinite order autoregressive process,

  $$X_t = \sum_{i=1}^{\infty} \psi_i X_{t-i} + Z_t.$$

- An ARMA(p,q) model is **causal** if the roots of the polynomial $\phi_\alpha(z)$ lie outside the unit circle. In this case then we can write

  $$X_t = \phi_\alpha^{-1}(B)\theta_\beta(B)Z_t =: \nu(B)Z_t$$

  where

  $$\nu(z) = \sum_{i=0}^{\infty} \nu_i z^i$$

  for some new set of constants $\{\nu_i\}$ and this infinite sum will be convergent. This allows us to rewrite our ARMA(p,q) process as an infinite order moving average process,

  $$X_t = \sum_{i=0}^{\infty} \nu_i Z_{t-i}.$$

Note: Since $Z_t = \psi(B)X_t$ and $X_t = \nu(B)Z_t$ we have that $\psi(B)\nu(B) \equiv 1$.

The coefficients $\{\psi_i\}$ and $\{\nu_i\}$ may be found either directly by evaluating

$$\psi(z) = \theta_\beta^{-1}(z)\phi_\alpha(z) \quad \text{and} \quad \nu(z) = \phi_\alpha^{-1}(z)\theta_\beta(z)$$

or by equating powers in

$$\theta_\beta(z)\psi(z) = \phi_\alpha(z) \quad \text{and} \quad \phi_\alpha(z)\nu(z) = \theta_\beta(z).$$

Exercise 3.2. Consider the ARMA(1,1) process defined by

$$X_t = 0.5X_{t-1} + Z_t - 0.3Z_{t-1}$$

(a) Show that the process is causal and invertible.

(b) Find the weights $\{\psi_i\}$ in the AR(∞) representation

$$X_t = \sum_{i=1}^{\infty} \psi_i X_{t-i} + Z_t.$$

(c) Find the weights $\{\nu_i\}$ in the MA(∞) representation

$$X_t = \sum_{i=0}^{\infty} \nu_i Z_{t-i}.$$

3.10.1 Autoregressive Integrated Moving Average Processes

In chapter 2 we saw that time series which are non-stationary in the mean can be detrended by differencing the data. This can be achieved by replacing X_t by

$$W_t = \bigtriangledown^d X_t.$$

The resulting, differenced, time series may then be modelled using an ARMA(p, q) process.

The general **autoregressive integrated moving average (ARIMA) process** is of the form

$$\begin{aligned} & W_t = \bigtriangledown^d X_t \\ \text{where} \quad & W_t = \alpha_1 W_{t-1} + \cdots + \alpha_p W_{t-p} + Z_t + \beta_1 Z_{t-1} + \cdots + \beta_q Z_{t-q}, \end{aligned}$$

and is written as ARIMA(p, d, q).

Notes:

- The ARIMA process is called *integrated* because the stationary model which is fitted to the differenced data has to be summed or integrated to provide a model for the non-stationary series.
- A random walk is just an ARIMA(0,1,0) process, indeed in this case

  $$W_t = Z_t \qquad \text{and} \qquad Z_t = \bigtriangledown X_t = X_t - X_{t-1}$$

  from which we regain a random walk since rearranging gives $X_t = X_{t-1} + Z_t$.

Chapter 4

Fitting Linear Time Series Models

In the previous chapter we introduced several different probability models that may be used to describe time series and we discussed their properties. In particular, we discussed the behaviour of the autocorrelation function of each process. In this chapter we will consider how to go about fitting a suitable model to an observed time series. The key tool we use to fit models to our sample data is the sample autocorrelation coefficient since the behaviour we would expect to see for each of our theoretical models is quite distinctive.

All of the processes introduced in the previous chapter are subsumed within the general autoregressive integrated moving average (ARIMA) process, so in this chapter our aim will be to fit an ARIMA process. To fit an ARIMA process to an observed time series one often follows these three steps:

1. Identify the model order (p,d,q).
2. Estimate the model parameters $\{\alpha_i, \beta_i, \mu, \sigma_Z^2\}$.
3. Perform diagnostic checks of the fitted model (e.g. analysis of the residuals).

4.1 Modelling an Autoregressive Process

If the sample autocorrelation coefficient of a given time series $x_1, \ldots, x_N$ indicates that an autoregressive process would be a good fit then there are two things we need to do:

1. Identify the order, p.
2. Estimate the parameters $\{\alpha_1, \ldots, \alpha_p\}$ and μ.

We will begin by considering the second point.

4.1.1 Estimation of Parameters when p is Known

Suppose we have an AR(p) process with constant mean μ, given by

$$X_t - \mu = \alpha_1(X_{t-1} - \mu) + \cdots + \alpha_p(X_{t-p} - \mu) + Z_t.$$

Given N observations $x_1, \ldots, x_N$ the parameters μ, α_1, $\ldots$, α_p may be estimated by **least squares** by minimizing

$$S(\mu, \alpha_1, \ldots, \alpha_p) = \sum_{t=p+1}^{N} [(x_t - \mu) - \alpha_1(x_{t-1} - \mu) - \cdots - \alpha_p(x_{t-p} - \mu)]^2 \tag{4.1}$$

over μ, α_1, $\ldots$, α_p. For an $AR(1)$ process we find that

$$\hat{\mu} = \frac{\overline{x}_{(2)} - \hat{\alpha}\overline{x}_{(1)}}{1 - \hat{\alpha}} \qquad \text{and} \qquad \hat{\alpha} = \frac{\sum_{t=1}^{N-1}(x_t - \hat{\mu})(x_{t+1} - \hat{\mu})}{\sum_{t=1}^{N-1}(x_t - \hat{\mu})^2}$$

where $\overline{x}_{(1)}$ and $\overline{x}_{(2)}$ are the means of the first and last $N-1$ observations respectively.

Proof
Substituting $p = 1$ into (4.1) we have

$$S(\mu, \alpha) = \sum_{t=2}^{N} [(x_t - \mu) - \alpha(x_{t-1} - \mu)]^2.$$

Taking partial derivatives gives

$$\begin{aligned} \frac{\partial S}{\partial \mu}(\mu, \alpha) &= 2(\alpha - 1) \sum_{t=2}^{N} [(x_t - \mu) - \alpha(x_{t-1} - \mu)] \\ &= 2(\alpha - 1)(N-1)[(\overline{x}_{(2)} - \mu) - \alpha(\overline{x}_{(1)} - \mu)] \end{aligned} \tag{4.2}$$

and

$$\begin{aligned} \frac{\partial S}{\partial \alpha}(\mu, \alpha) &= -2 \sum_{t=2}^{N} (x_{t-1} - \mu)[(x_t - \mu) - \alpha(x_{t-1} - \mu)] \\ &= -2 \sum_{t=2}^{N} (x_{t-1} - \mu)(x_t - \mu) + 2\alpha \sum_{t=1}^{N-1} (x_t - \mu)^2. \end{aligned} \tag{4.3}$$

Then, equating (4.2) to zero yields $\hat{\mu}$ whilst equating (4.3) to zero yields $\hat{\alpha}$.

Notes:

- If N is large then $\overline{x}_{(1)} \simeq \overline{x}_{(2)} \simeq \overline{x}$.

  Using this approximation in our estimate for μ and α we find that

  $$\hat{\mu} \simeq \overline{x} \qquad \text{and} \qquad \hat{\alpha} \simeq \frac{\sum_{t=1}^{N-1}(x_t - \overline{x})(x_{t+1} - \overline{x})}{\sum_{t=1}^{N-1}(x_t - \overline{x})^2} \simeq \frac{c_1}{c_0} = r_1 \tag{4.4}$$

  where we have also made the approximation $\sum_{t=1}^{N-1}(x_t - \overline{x})^2 \simeq \sum_{t=1}^{N}(x_t - \overline{x})^2$ in the denominator.

- For AR(2) process, similar approximations can be made and we have

  $$\begin{aligned} \hat{\mu} &\simeq \overline{x}, \\ \hat{\alpha}_1 &\simeq r_1(1 - r_2)/(1 - r_1^2), \\ \hat{\alpha}_2 &\simeq (r_2 - r_1^2)/(1 - r_1^2). \end{aligned}$$

- Both of the above approximations for large N correspond to the **Yule-Walker estimates**. These are gained by taking $\hat{\mu} \simeq \overline{x}$ and substituting the sample autocorrelation coefficients into the first p Yule-Walker equations in place of the corresponding theoretical autocorrelation functions in order to determine the remaining p estimates, $\{\hat{\alpha}_i\}_{i=1}^{p}$. That is, we solve the p linear equations given by

$$r_k = \hat{\alpha}_1 r_{k-1} + \cdots + \hat{\alpha}_p r_{k-p} \qquad \text{for } k = 1, ..., p.$$

  These approximations are very close to the 'exact' least squares estimates for large N, but for smaller observed time series (e.g. N less than about 50) they are not so good and this approximation should not be made.

4.1.2 Determining the Order of the Process, p

It is rarely possible to determine the order of an AR process using only the sample autocorrelation coefficient. In general it is necessary to fit candidate models AR(p) for successively higher orders, $p = 1, 2, ...$, and select which one is most appropriate. Below we will outline three commonly used approaches for this selection process.

- **The sample partial autocorrelation coefficient**

  The pth **sample partial autocorrelation coefficient**, denoted π_p, is given by the last coefficient, α_p, when an AR(p) model is fitted to the data.

  It is estimated by fitting models of successively higher orders, $p = 1, 2, ...$, and taking $\hat{\pi}_p = \hat{\alpha}_p$ in each case (*i.e.* $\hat{\pi}_1 = \hat{\alpha}_1$ when an AR(1) process is fitted, $\hat{\pi}_2 = \hat{\alpha}_2$ when an AR(2) process is fitted, and so on).

  When fitting an AR(p) model the last coefficient, α_p, measures the excess correlation at lag p which is not accounted for by an AR($p-1$) model. So, we would expect the sample partial autocorrelation coefficient to drop off at lag p as we saw for its theoretical counterpart in Chapter 3. In order to to determine p we plot $\hat{\pi}_p$ against p and look for a sharp cut-off to near zero (in practice, $|\hat{\pi}_p| \gg 2/\sqrt{N}$ whilst $|\hat{\pi}_k| \simeq 0$ for $k > p$), indicating that there isn't any significant excess correlation unaccounted for by an AR(p) process.

- **The residual sum of squares**
  This approach again involves fitting AR(p) processes of successively higher orders. We then plot the residual sum of squares $S(\hat{\mu}, \hat{\alpha}_1, ..., \hat{\alpha}_p)$ against p and look for a sharp drop-off followed by no significant further decrease. This drop-off point identifies p.

- **Akaike's Information Criterion (AIC)**
  This balances the reduction of estimated error variance with the number of parameters being fitted. We choose the order, p^*, that minimises

$$\text{AIC}(p) = N \log\left(\hat{\sigma}^2_{Z,p}\right) + 2p.$$

  Note: The estimated error variance is proportional to the residual sum of squares $S(\hat{\mu}, \hat{\alpha}_1, ..., \hat{\alpha}_p)$.

4.2 Modelling a Moving Average Process

If we think a moving average process would be a good fit for an observed time series then there are two things we need to do:

1. Identify the order, q.
2. Estimate the parameters $\{\beta_1, \ldots, \beta_q\}$ and μ.

4.2.1 Determining the Order of the Process, q

The autocorrelation function of an MA(q) process cuts off at lag q. So, in order to determine q we plot r_k against k (the correlogram) and look off for a sharp cut-off to near zero (in practice $|r_p| \gg 2/\sqrt{N}$ whilst $|r_k| \simeq 0$ for $k > p$).

4.2.2 Estimation of Parameters when q is Known

This is not easy. Consider the MA(1) process

$$X_t = \mu + Z_t + \beta Z_{t-1}$$

where μ and β are our unknown parameters and $\{Z_t\}$ is a purely random process. We would like to use least squares to estimate our parameters as we did before for an AR(p) process. The residual sum of squares is given by

$$\sum Z_t^2 = \sum (X_t - \mu - \beta Z_{t-1})^2,$$

but this contains Z_{t-1} in the sum on the right hand side. The $\{Z_t\}$ aren't observed, and so we cannot compute this. One simple solution is to set $Z_0 = 0$, then we have

$$\begin{aligned}
Z_1 &= X_1 - \mu \\
Z_2 &= X_2 - \mu - \beta Z_1 = X_2 - \mu - \beta(X_1 - \mu) \\
Z_3 &= X_3 - \mu - \beta Z_2 = X_3 - \mu - \beta(X_2 - \mu - \beta(X_1 - \mu))
\end{aligned}$$

etc.

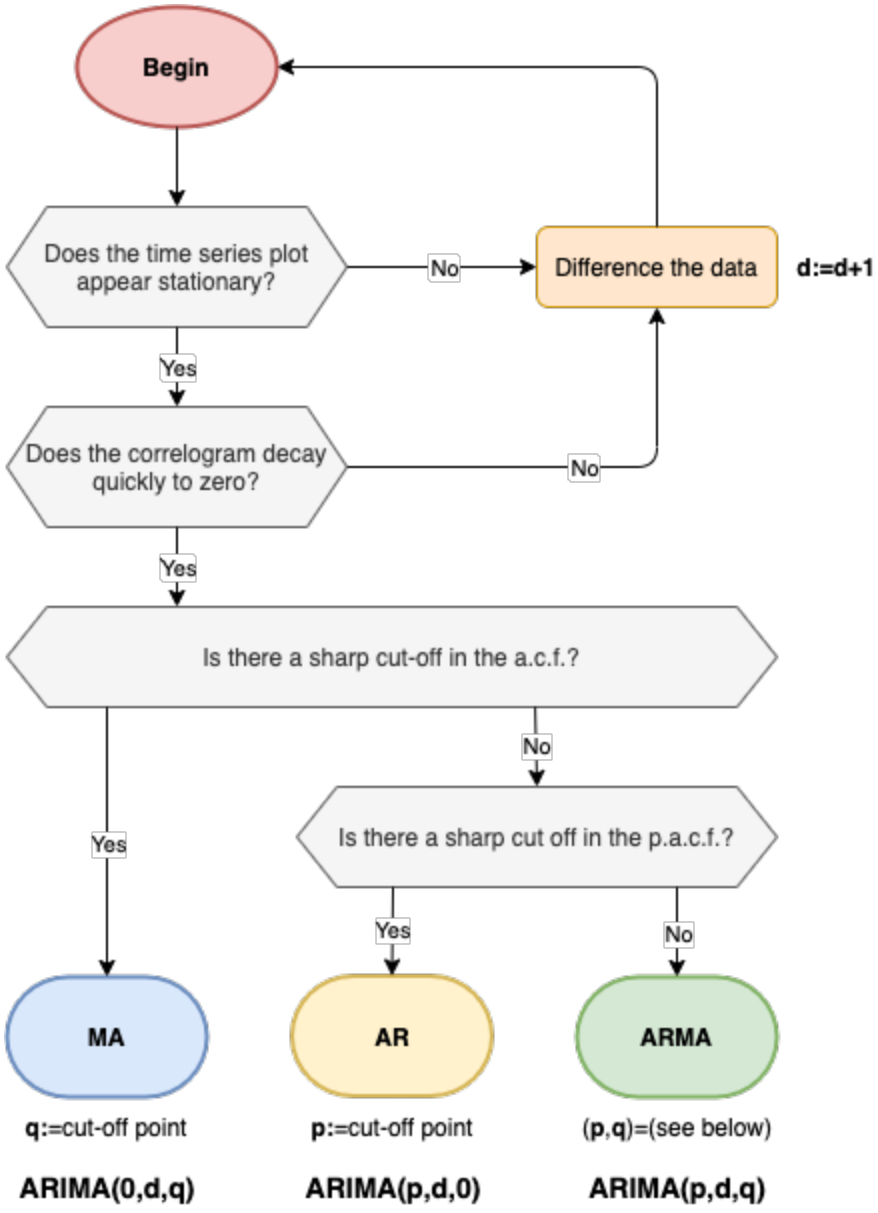
Thus, by assuming $Z_0 = 0$ we can compute $S(\mu, \beta)$ for any values of (μ, β). Computing values of $S(\mu, \beta)$ on a grid of points in the (μ, β)-plane we can then minimise the residual sum of squares numerically in order to determine the estimates $\hat{\mu}$ and $\hat{\beta}$. This is hard work, so it is tempting to employ a strategy analogous to the Yule-Walker estimates we found for an autoregressive process. That is, to equate the sample autocorrelation coefficient at lag 1 to its theoretical counterpart and determine $\hat{\beta}$ as the solution of

$$r_1 = \frac{\hat{\beta}}{1 + \hat{\beta}^2}$$

which satisfies $|\hat{\beta}| < 1$. Unfortunately, even for large N it can be shown that this gives rise to an inefficient estimator. However, it can provide a useful starting point for the grid search we outlined above.

4.3 Modelling an Autoregressive Integrated Moving Average Process

The Box-Jenkins procedure combines what we already know about the behaviour of the autocorrelation coefficient and the partial autocorrelation coefficient to give a simple step by step process for identifying a suitable ARIMA(p, d, q) process to model an observed time series:



For an ARIMA(p, d, q) process we choose p and q that minimise Akaike's Information Criterion

$$\text{AIC}(p, q) := N \log(\hat{\sigma}_Z^2) + 2(p + q).$$

Once the model orders (p, d, q) are identified the unknown parameters $\{\alpha_1, ..., \alpha_p, \beta_1, ..., \beta_q\}$ can be estimated computationally in **R**.

4.4 Fitting an ARIMA Model in R

The data set which we will explore in this section is held in a matrix called `wool`. This is a matrix containing 310 separate observations on 5 variables. The variables are described in table 4.1.

Variable No.	Description
1	Index number
2	Floor price (cents per kg)
3	Actual Price (cents per kg)
4	Ratio of Actual to Floor price
5	Log(Actual to Floor price)

Table 4.1: Variables in the wool data set

The wool data set contains prices monitored by the Australian Wool Corporation starting from June 1976. The prices are monitored weekly. Before the start of each week the Corporation sets a floor price for the week. The Corporation guarantees that it will pay this price for the wool during the week. The actual price of the wool is an average taken over the following week and is never less than the floor price (otherwise they could have sold it to the Corporation and made more money). A plot of these two variables appears in figure 4.1.

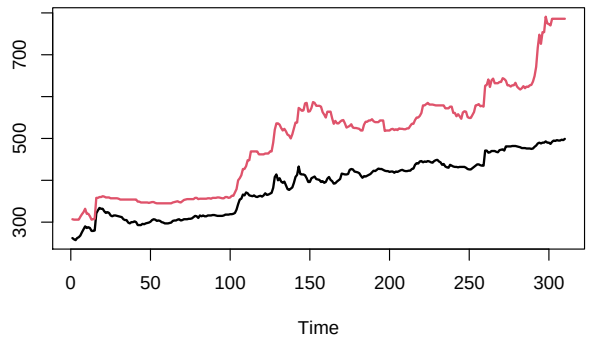


Figure 4.1: Weekly wool prices. The upper red series represents the average weekly price set by the market, the lower black series is a floor price set by the Australian Wool Corporation.

We will denote the actual price by A_t, and the floor price by F_t, in both cases $t = 1, 2, \ldots, 310$. The interest in this series centres around the Corporation's intervention policy and so it is the relative price movements that we are interested in. Therefore we can look at the ratio of actual to floor price as this will compensate for trends such as those caused by currency fluctuations and inflation. We denote the ratio by

$$R_t = \frac{A_t}{F_t} \qquad \text{for } t = 1, 2, ..., 310.$$

Another point to consider is that price movements are often multiplicative in nature (so price increases/decreases tend to be discussed in percentage terms rather than absolute terms). We can then feel justified in working with the log of the series, which we denote

$$L_t = \log(R_t) \qquad \text{for } t = 1, 2, ..., 310,$$

rather than the series itself.

Now that we have decided which time series we want to model we will investigate the possibility of fitting an ARIMA model to the series, $\{L_t\}$. To do this we follow our flowchart from the previous section:

- The first step is to see whether $\{L_t\}$ looks stationary. The time series of $\{L_t\}$ may be plotted in **R** using the command

```
ts.plot(wool[,5])
```

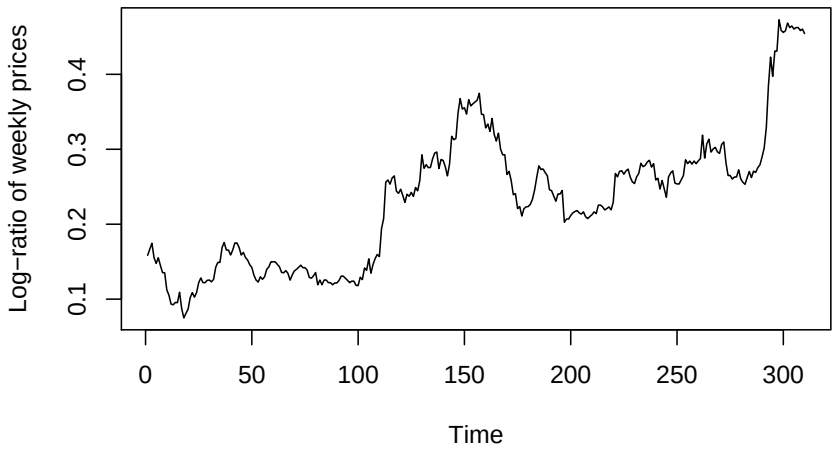


Figure 4.2: A time series plot of the log-ratio, L_t, of weekly wool prices.

The resulting plot is shown in figure 4.2, from which we can see that $\{L_t\}$ is clearly not stationary. The mean appears to change over time. More precisely, you might guess that the mean is piecewise constant. For example, you might guess that

$$\mu(t) = \begin{cases} \mu_1 = 0.15 & \text{for } 1 \leq t \leq 110 \\ \mu_2 = 0.25 & \text{for } 111 \leq t \leq 285 \\ \mu_3 = 0.46 & \text{for the rest of the series} \end{cases}$$

- Following the flowchart we difference the series. This is achieved in **R** by the command

```
ts.plot(diff(wool[,5]))
```

and produces the series in figure 4.3. The striking feature in figure 4.3 is that it appears to have a systematic change in variance of the data after about 100 observations. If we were seriously interested in modeling the series we would have to model the first 100 observations differently to the rest. For this example, we are interested in forecasting future behavior so we will simply discard the first 100 observations and model the remaining 209. To simplify matters we will create a new vector called `woolly` that contains the last 209 differenced observations. This is done by

```
woolly <- diff(wool[,5])[101:309]
```

in which the `:` operator constructs a vector so that `a:b` returns the vector $(a, a+1, a+2, \ldots, b-1, b)$. Note also that although `wool` is a matrix `woolly` is a vector because the `[,5]` extracts a column from the matrix.

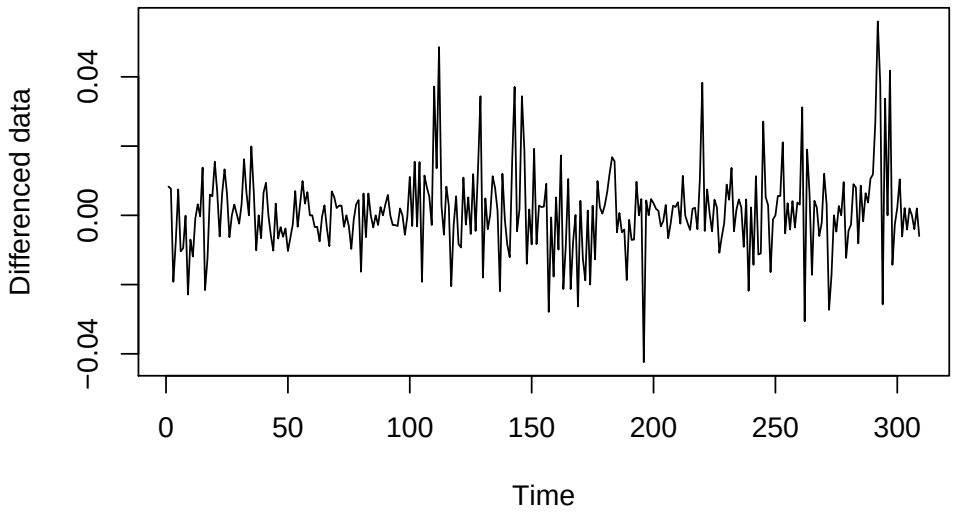


Figure 4.3: A time series plot of the log-ratio, L_t, of weekly wool prices.

- The next stage of the flowchart procedure was to produce a correlogram of the differenced data. This is done by

```
acf(woolly)
```

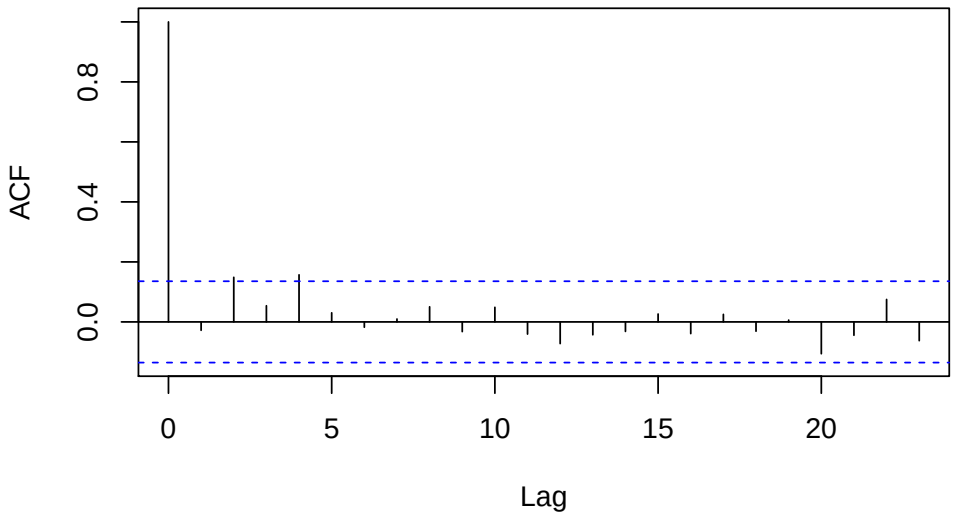


Figure 4.4: Correlogram of the last 209 first differences of L_t.

and the correlogram is shown in Figure 4.4. From this it can be seen that only the second and the fourth autocorrelation coefficients might possibly be significant. Later we might find it useful to take account of these autocorrelations and fit more complicated models, but in the interests of parsimony we want to fit the simplest model possible. Also there is no other pattern in the autocorrelations so we conclude that the differenced data is consistent with white noise (a purely random process).

4.4.1 Suggested Model for the Wool Data

Therefore our first model is an ARIMA$(0, 1, 0)$ process given by

$$L_t - L_{t-1} = \mu + Z_t,$$

where $\{Z_t\}$ is a purely random process with mean zero and variance σ_Z^2, and μ is the mean of the differenced series.

So, we have assumed a non-zero mean for this model. Can we do this? Well, the **R** command

```
mean(woolly)
```

```
## [1] 0.001555981
```

suggests that the differences have a non-zero mean, but is this significantly different from zero? Since we have assumed that the differences are uncorrelated we are justified in using a one-sample t-test to test the hypothesis

$$H_0 : \mu = 0 \qquad \text{against the alternative} \qquad H_A : \mu \neq 0.$$

This too can be carried out in **R** by

```
t.test(woolly)
```

```
##
##  One Sample t-test
##
## data:  woolly
## t = 1.6318, df = 208, p-value = 0.1042
## alternative hypothesis: true mean is not equal to 0
## 95 percent confidence interval:
##  -0.0003238799  0.0034358416
## sample estimates:
##   mean of x
## 0.001555981
```

The p-value suggests that we cannot reject the null hypothesis at the 10% level. So, we will assume $\mu = 0$. Finally, the **R** command `var()` allows us to find an estimate of σ_Z^2 (the variance of Z_t)

```
var(woolly)
```

```
## [1] 0.0001900347
```

Therefore our second model is

$$L_t = L_{t-1} + Z_t$$

where $\{Z_t\}$ is a purely random process with mean zero and variance $\hat{\sigma}_Z^2 = 0.00019$.

Chapter 5

Forecasting

Forecasting the future values of an observed time series is often our most important goal when analysing time series. We want to understand what has happened so that we can predict what will happen next and act accordingly. This is important in many areas, including climate change, economics, weather and sales forecasting. In order to predict what will happen next we need to use what we have observed in the past. This is a particularly difficult task, and different from the prediction problems seen in most other areas of statistics for one key reason: it relies on *extrapolation* rather than *interpolation*.

Say, for example, you are trying to predict ice cream sales based on the air temperature and you have a set of paired observations of these variables. Then, you would generally aim to fit a regression line and use this to predict ice cream sales for air temperatures within the range previously observed (interpolation). Meanwhile, you would generally avoid making predictions for temperatures outside the range previously observed (extrapolation) as your predictions will be subject to much more uncertainty.

In time series then you have no real choice but to make predictions based on extrapolation since interpolation would essentially involve predicting past values of your time series and is generally not so interesting. Returning to our example, if you were trying to predict ice cream sales next month and you had a set of ordered observations of ice cream sales in each month over the past few years then you could again fit a regression line, but you would have to extrapolate to make the prediction since you are outside your range of previously observed times. You would only be able to use interpolation if you were trying to 'predict' how many ice creams were sold in the past, say last July, which is probably not so interesting to an ice cream vendor when they're deciding how much stock to buy or whether they can afford to buy a new van.

Suppose the time series $x_1, ..., x_N$ of a process $\{X_t\}_{t\in\mathbb{Z}}$ has been *observed*. Forecasting refers to the prediction of the *unobserved* (future) values of this process, X_{N+h} for $h \in \mathbb{N}$, based on the observed values $x_1, ..., x_N$.

The natural number h is called the **lead time**.

The h**-step ahead** forecast of X_{N+h} made at time N is denoted by $\hat{x}(N, h)$.

Note: $\hat{x}(N, h)$ is a function of the observed values $\{x_1, ..., x_N\}$.

There is usually no single best procedure for forecasting. Different procedures will be more successful than others depending on the application.

5.1 Use of Trend Curves

The first option is to fit a trend curve using standard regression techniques (as in §2.3.1) and then extrapolate into the future. This method is simple and easy to communicate, but should only be used if there is an obvious reason for the choice of trend curve (e.g. linear, cubic,...).

Disadvantages:

- All observations are given equal weight, but it is often more natural to a greater weight to more recent observations.
- It is often possible to find several different curves that fit the observed data almost equally well, but which diverge in the future leading to very different forecasts.

5.2 Exponential smoothing

In principle, the 1-step ahead forecast $\hat{x}(N,1)$ can be any function of $x_1, ..., x_N$. However, it is simpler to only consider functions of the form

$$\hat{x}(N,1) = c_0 x_N + c_1 x_{N-1} + c_2 x_{N-2} + \cdots + c_{N-1} x_1$$

where the constants $c_0, c_1, ..., c_{N-1}$ are the weights. In this case $\hat{x}(N,1)$ is called a **linear predictor**.

It is usually natural to give more weight to recent observations and less weight to those in the past. A popular set of weights that achieve this are given by

$$c_i = \alpha(1-\alpha)^i \qquad \text{for } i = 0, 1, 2... \tag{5.1}$$

where α is some constant such that $0 < \alpha < 1$. Then, we have

$$\begin{aligned}
\hat{x}(N,1) &= c_0 x_N + c_1 x_{N-1} + c_2 x_{N-2} + \cdots + c_{N-1} x_1 \\
&= \alpha x_N + \alpha(1-\alpha) x_{N-1} + \alpha(1-\alpha)^2 x_{N-2} + \cdots + \alpha(1-\alpha)^{N-1} x_1 \\
&= \alpha x_N + (1-\alpha)[\alpha x_{N-1} + \alpha(1-\alpha) x_{N-2} + \cdots + \alpha(1-\alpha)^{N-2} x_1 \\
&= \alpha x_N + (1-\alpha)\hat{x}(N-1,1).
\end{aligned}$$

The recursion

$$\begin{aligned}
\hat{x}(1,1) &= x_1 \\
\hat{x}(N,1) &= \alpha x_N + (1-\alpha)\hat{x}(N-1,1)
\end{aligned} \tag{5.2}$$

is called **exponential smoothing**.

Notes:

- The recurrence equation (5.2) can be rewritten in error-correction form as

$$\begin{aligned}\hat{x}(N,1) &= \alpha(x_N - \hat{x}(N-1,1)) + \hat{x}(N-1,1) \\ &= \alpha e_N + \hat{x}(N-1,1)\end{aligned}$$

  where $e_N := x_N - \hat{x}(N-1,1)$ is the **prediction error** at time N *i.e.* it is the error that would be made at time N if x_N were predicted by $\hat{x}(N-1,1)$.

- If $\alpha = 1$ then $\hat{x}(N,1) = x_N$ and we simply predict what will happen at the next time step by what it is happening now - not an unreasonable suggestion. However, we typically choose $0.1 < \alpha < 0.3$ so that the weighting decreases more gradually as you move further into the past.

- The forecasting procedure we have described is often known as simple exponential smoothing as it is only applicable to observed time series that are constant in the mean, *i.e.* non-seasonal and with no systematic trend. It is possible to generalise this method to deal with trend and seasonality (the generalised procedures are usually called Holt's exponential smoothing and the Holt-Winter's procedure). But as we have seen it is also possible to estimate, record and remove systematic features such as trend and seasonality leaving a series for which simple exponential smoothing is appropriate.

5.3 The Box-Jenkins Approach

The Box-Jenkins Approach is a forecasting approach based on using autoregressive integrated moving average (ARIMA) models for the observed data as in the previous chapter. ARIMA models assume that, after differencing, the observed time series is generated by an ARMA(p,q) model

$$X_t = \sum_{i=1}^{p} \alpha_i X_{t-i} + Z_t + \sum_{i=1}^{q} \beta_i Z_{t-i}$$

where $\{Z_t\}$ is a purely random process with zero mean and variance σ_Z^2.

In operator notation this is written as

$$\phi_\alpha(B) X_t = \theta_\beta(B) Z_t.$$

If the process is **causal** then we can write it in the form

$$\begin{aligned}X_t &= \phi_\alpha^{-1}(B)\theta_\beta(B) Z_t \\ &= \sum_{i=0}^{\infty} \nu_i Z_{t-i} \qquad (5.3)\end{aligned}$$

where the $\{\nu_i\}$ are coefficients in the polynomial

$$\nu(z) = \phi_\alpha^{-1}(z)\theta_\beta(z) = \sum_{i=0}^{\infty} \nu_i z^i.$$

Our aim is to construct an h**-step ahead** forecast of the form

$$\begin{aligned}\hat{x}(N,h) &= c_0 X_N + c_1 X_{N-1} + \cdots + c_{N-1} X_1 \\ &= \sum_{j=0}^{N-1} c_j X_{N-j} \\ &= \sum_{j=0}^{N-1} c_j \sum_{i=0}^{\infty} \nu_i Z_{N-j-i} \\ &= \sum_{i=0}^{\infty} w_i Z_{N-i} \end{aligned} \tag{5.4}$$

where the coefficients $\{w_i\}$ can be obtained from the coefficients $\{c_j\}$ and $\{\nu_i\}$. We are not concerned with the exact form of the relationship between these coefficients. What *is* important here is that it is possible to choose weights such that our linear predictor, $\hat{x}(N,h)$, is causal *i.e.* it can be written as a convergent sum of current and past values of $\{Z_t\}$.

If this is the case then it is possible to derive the weights $\{c_j\}$ such that they minimise the mean squared prediction error

$$\begin{aligned}\mathbb{E}\left[(X_{N+h} - \hat{x}(N,h))^2\right] &= \mathbb{E}\left[\left(\sum_{i=0}^{\infty} \nu_i Z_{N+h-i} - \sum_{i=0}^{\infty} w_i Z_{N-i}\right)^2\right] \\ &= \mathbb{E}\left[\left(\sum_{i=0}^{h-1} \nu_i Z_{N+h-i} + \sum_{i=h}^{\infty} \nu_i Z_{N+h-i} - \sum_{i=h}^{\infty} w_{i-h} Z_{N+h-i}\right)^2\right] \\ &= \mathbb{E}\left[\left(\sum_{i=0}^{h-1} \nu_i Z_{N+h-i} + \sum_{i=h}^{\infty} (\nu_i - w_{i-h}) Z_{N+h-i}\right)^2\right] \\ &= \sigma_Z^2 \left[\sum_{i=0}^{h-1} \nu_i^2 + \sum_{i=h}^{\infty} (\nu_i - w_{i-h})^2\right]\end{aligned}$$

where the final line results since the $\{Z_t\}$ are independent and identically distributed with zero mean and variance σ_Z^2. The coefficients $\{\nu_i\}$ are specified by the ARMA(p,q) model whilst the coefficients $\{w_i\}$ may be chosen to minimise the mean squared prediction error. Thus we choose the weights $\{c_j\}$ such that $w_i = \nu_{i+h}$ for $i = 0, 1, 2, ...$

The h**-step ahead** Box-Jenkins forecast (5.4) is then

$$\begin{aligned}\hat{x}(N,h) &= \sum_{i=0}^{\infty} w_i Z_{N-i} \\ &= \sum_{i=0}^{\infty} \nu_{i+h} Z_{N-i} \\ &= \sum_{i=h}^{\infty} \nu_i Z_{N+h-i}. \end{aligned} \tag{5.5}$$

Comparing this with the 'true' value of X_{N+h}, as determined by (5.3), we see that

$$\begin{aligned} X_{N+h} &= \sum_{i=0}^{\infty} \nu_i Z_{N+h-i} \\ &= \sum_{i=0}^{h-1} \nu_i Z_{N+h-i} + \hat{x}(N,h) \end{aligned}$$

which leads to a simple interpretation of our optimal forecast.

The h**-step ahead Box-Jenkins forecast** is given by

$$\hat{x}(N,h) = \sum_{i=h}^{\infty} \nu_i Z_{N+h-i}$$

and may be calculated in the same way as the 'true' value X_{N+h}, but future values of Z_t (which have not yet been observed) are set equal to zero *i.e.* we set $Z_{N+1} = Z_{N+2} = \cdots = Z_{N+h} = 0$.

The prediction errors also have a simple interpretation in terms of the purely random process $\{Z_t\}$. The prediction error at time N is given by

$$\begin{aligned} e(N,h) &= X_{N+h} - \hat{x}(N,h) \\ &= \sum_{i=0}^{h-1} \nu_i Z_{N+h-i} \end{aligned} \tag{5.6}$$

i.e. it is made up of the unobserved, future elements of our ARMA process. In particular, for the 1-step ahead forecast $\hat{x}(N,1)$ we have the forecasting error $e(N,1) = \nu_0 Z_{N+1} = Z_{N+1}$ since $\nu_0 = 1$.

Example 5.1. Consider the AR(1) process

$$X_t = \alpha X_{t-1} + Z_t \tag{5.7}$$

where $\{Z_t\}$ is a purely random process with zero mean and variance σ_Z^2. Find the h-step ahead Box-Jenkins forecast, $\hat{x}(N,1)$.

Using our simple interpretation of the Box-Jenkins forecast we know that the 1-step ahead forecast $\hat{x}(N,1)$ is obtained from $\{Z_t\}$ in the same way as X_{N+1}, but with $Z_{N+1} = 0$. Using the defining equation (5.7) we have that the 'true' value is given by

$$X_{N+1} = \alpha X_N + Z_{N+1}.$$

Therefore the 1-step ahead forecast is given by $\hat{x}(N,1) = \alpha x_N$.

More generally, there are two possible ways to calculate the h-step ahead forecast:
Method 1:
Calculate forecasts recursively by using the appropriate forecast in place of future values of X_t to give

$$\hat{x}(N,h) = \alpha \hat{x}(N,h-1) = \alpha^2 \hat{x}(N,h-2) = \cdots = \alpha^{h-1} \hat{x}(N,1) = \alpha^h x_N$$

Method 2:
Rewrite X_{N+h} by successive substitution until you have eliminated all other future values of $\{X_t\}$ and then set all future values of Z_t equal to zero in your resulting equation. Thus, we find

$$\begin{aligned} X_{N+h} &= \alpha X_{N+h-1} + Z_{N+h} \\ &= \alpha(\alpha X_{N+h-2} + Z_{N+h-1}) + Z_{N+h} \\ &= \alpha^2 X_{N+h-2} + \alpha Z_{N+h-1} + Z_{N+h} \\ &= \dots \\ &= \alpha^h X_N + \alpha^{h-1} Z_{N+1} + \cdots + \alpha^2 Z_{N+h-2} + \alpha Z_{N+h-1} + Z_{N+h} \end{aligned}$$

and setting $Z_{N+1} = Z_{N+2} = ... = Z_{N+h} = 0$ we again find that the h-step ahead forecast is given by

$$\hat{x}(N,h) = \alpha^h x_N.$$

5.4 Assessing Forecast Uncertainty

The key to understanding uncertainty is the prediction error (5.6),

$$e(N,h) = X_{N+h} - \hat{x}(N,h) = \sum_{i=0}^{h-1} \nu_i Z_{N+h-i},$$

which has mean $\mathbb{E}[e(N,h)] = 0$ and variance $\text{Var}[e(N,h)] =: V(h) = \sigma_Z^2 \sum_{i=0}^{h-1} \nu_i^2$.

We estimate the variance of our prediction error using the 'plug-in' estimate

$$\hat{V}(h) = \hat{\sigma}_Z^2 \sum_{i=0}^{h-1} \hat{\nu}_i^2$$

in which $\hat{\sigma}_Z^2$ and $\hat{\nu}_i^2$ are estimates for σ_Z^2 and ν_i^2 respectively.

In order to quantify our uncertainty we want to construct a prediction interval that we are $100(1-\alpha)\%$ certain will contain the 'true' value, X_{N+h}. Noticing that $X_{N+h} = \hat{x}(N,h) + e(N,h)$ we write our **prediction interval** in the form

$$\left[\hat{x}(N,h) - C_\alpha \sqrt{\hat{V}(h)}, \hat{x}(N,h) + C_\alpha \sqrt{\hat{V}(h)}\right]$$

for an appropriately chosen constant, C_α. In practice, C_α is usually chosen by making the assumption that the prediction errors, $e(N,h)$, are approximately Normally distributed with zero mean and variance $\hat{V}(h)$. In this case then C_α is given by $z_{\alpha/2}$, namely the percentage point of the standard normal distribution satisfying $\mathbb{P}(Z \geq z_{\alpha/2}) = \alpha/2$ where $Z \sim N(0,1)$.

Example 5.2. Recall that in Chapter 4 we fitted an ARIMA model to the `wool` dataset. More specifically, we suggested the ARIMA(0,1,0) model

$$L_t = L_{t-1} + Z_t,$$

where $\{Z_t\}$ is a purely random process with zero mean and variance $\hat{\sigma}_Z^2 = 0.00019$, would be appropriate. We would like to determine the h-step ahead forecast and associated prediction interval for this model.

As we have seen, there are two different methods for computing the h-step ahead forecast:

1. Recursively, using the appropriate forecast in place of future values. This gives

$$\hat{l}(N,h) = \hat{l}(N,h-1) = \hat{l}(N,h-2) = \cdots = \hat{l}(N,1) = L_N$$

in which the final equality results since $L_{N+1} = L_N + Z_{N+1}$ in which we set the future value of the random process, Z_{N+1}, equal to zero.

2. By eliminating all other future values of $\{L_t\}$ from your expression for L_{N+h} using successive substitution. This gives

$$\begin{aligned} L_{N+h} &= L_{N+h-1} + Z_{N+h} \\ &= L_{N+h-2} + Z_{N+h-1} + Z_{N+h} \\ &= \ldots \\ &= L_N + Z_{N+1} + Z_{N+2} + \cdots + Z_{N+h} \end{aligned}$$

and setting $Z_{N+1} = Z_{N+2} = \cdots = Z_{N+h} = 0$ we again find that $\hat{l}(N,h) = L_N$.

In this case then we are also interested in determining the prediction interval for our forecast and this requires us to estimate the variance of the prediction error. Thus, we favour the second method as it allows us to write

$$e(N,h) := L_{N+h} - \hat{l}(N,h) = L_{N+h} - L_N = \sum_{i=0}^{h-1} Z_{N+h-i}$$

and so we find

$$V(h) = \mathrm{Var}[e(N,h)] = \sum_{i=0}^{h-1} \sigma_Z^2 = h\sigma_Z^2.$$

Assuming $e(N,h) \approx N(0, \hat{V}(h))$ then the $100(1-\alpha)\%$ prediction interval for L_{N+h} is given by

$$\left[L_N - z_{\alpha/2}\sqrt{\hat{V}(h)}, L_N + z_{\alpha/2}\sqrt{\hat{V}(h)}\right]$$

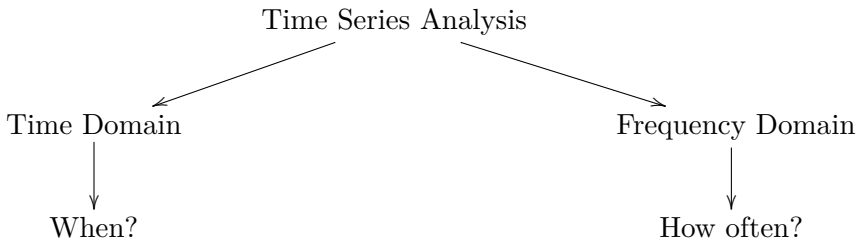
in which we use the 'plug-in' estimate $\hat{V}(h) = h\hat{\sigma}_Z^2 = 0.00019h$. Choosing a particular value for α we find that the limits of a 90% prediction interval for L_{N+h} are given by

$$L_N \pm 0.0227\sqrt{h}$$

where we have also used that $z_{0.05} = 1.645$.

Chapter 6

Stationary processes in the frequency domain



In the previous chapters, we thought about a time series as a process evolving in time. In the next chapters, we will instead think about a time series as the aggregation of different phenomena evolving at different frequencies. The time and frequency domains are two different perspectives for analysing the same system. In fact, in later chapters we will show that the two methods are equivalent, in that they carry the same amount of information. Furthermore we will see how the two perspectives are related and how we can move from one to the other.

6.1 Modelling periodic time series

If a time series is periodic, then a reasonable model might be

$$X_t = R\cos(\omega t + \theta) + Z_t, \tag{6.1}$$

where

- ω is the frequency of the periodic variation
- R is the amplitude of the variation at frequency ω
- θ is the phase and
- $\{Z_t\}$ is a stationary random process.

The frequency, ω measures the number of radians covered by the process in one unit of time. The period is then given by $T = 2\pi/\omega$ and measures the number of units of time spanned by a complete cycle, see figure 6.1.

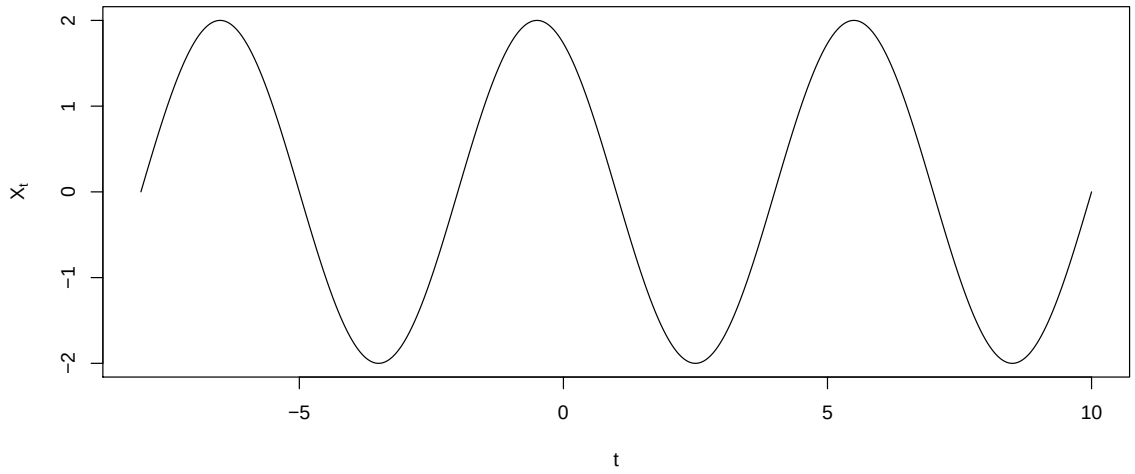


Figure 6.1: A graph of $R\cos(\omega t + \theta)$ where $R = 2$, $\omega = \pi/3$ and $\theta = \pi/6$.

The process $\{X_t\}$, given in (6.1), is not stationary if R, ω and θ are fixed (constants). Indeed, if we calculate the expectation of X_t for R, ω and θ constant, we get

$$\mathbb{E}\left[X_t\right] = \mathbb{E}\left[R\cos(\omega t + \theta)\right] = R\cos(\omega t + \theta),$$

which depends on t.

We can solve this problem by making appropriate assumptions. For example, it is often assumed that R and θ are independent random variables, with

$$\mathbb{E}(R) = 0, \qquad \text{Var}(R) < \infty \qquad \text{and} \qquad \theta \sim U[0, 2\pi].$$

In this case then

$$\mathbb{E}[X_t] = \mathbb{E}[R] \cdot \mathbb{E}[\cos(\omega t + \theta)] = 0$$

and

$$\begin{aligned}
\text{Cov}\left(X_t, X_{t+k}\right) &= \mathbb{E}[X_t X_{t+k}] \\
&= \mathbb{E}\left[R^2\right] \cdot \mathbb{E}\left[\cos\left(\omega t + \theta\right) \cdot \cos\left(\omega(t+k) + \theta\right)\right] + \mathbb{E}[Z_t Z_{t+k}] \\
&= \mathbb{E}\left[R^2\right] \mathbb{E}\left[\cos(\omega t + \theta) \cdot \cos(\omega k + \omega t + \theta)\right] + \mathbb{E}[Z_t Z_{t+k}] \\
&= \mathbb{E}\left[R^2\right] \cdot \mathbb{E}\left[\cos(\theta') \cdot \cos(\omega k + \theta')\right] + \mathbb{E}[Z_t Z_{t+k}] \\
&= \mathbb{E}\left[R^2 \cos(\theta') \cos(\omega k + \theta')\right] + \mathbb{E}[Z_0 Z_k] \\
&= \text{Cov}(X_0, X_k)
\end{aligned}$$

where $\theta' := \theta + \omega t \mod 2\pi \sim U[0, 2\pi]$. And so $\{X_t\}$ is second-order stationary.

If a time series appears to have more than one periodic oscillation (e.g. weekly, monthly, yearly etc), then model (6.1) could be generalised to

$$X_t = \sum_{j=1}^{k} R_j \cos(\omega_j t + \theta_j) + Z_t. \tag{6.2}$$

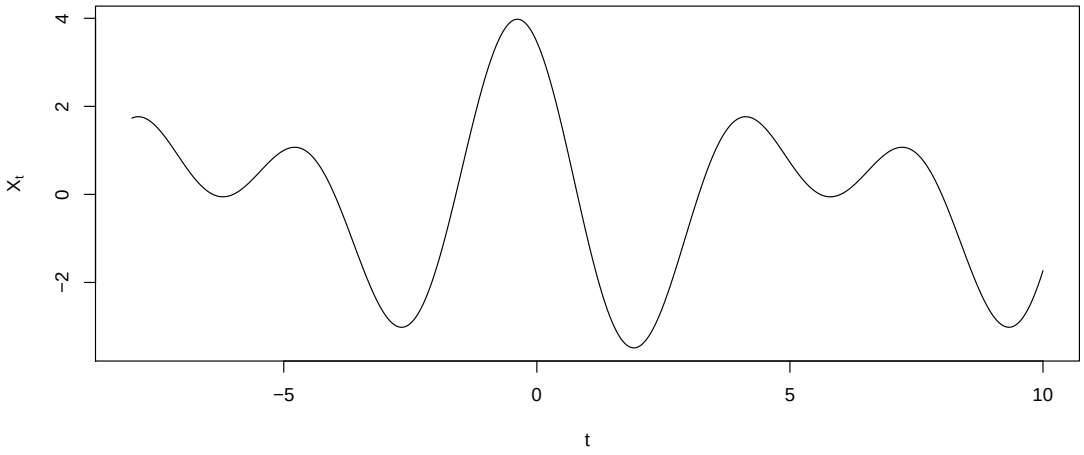


Figure 6.2: A graph of $R\cos(\omega t+\theta)+R\cos(\omega' t+\theta)$ where $R=2$, $\omega=\pi/3$, $\theta=\pi/6$, and $\omega'=\pi/2$.

Again, this process is not stationary if $\{R_j\}$, $\{\omega_j\}$ and $\{\theta_j\}$ are fixed constants. Similar assumptions to those made above for a single periodic component are usually made to ensure second-order stationarity. An example of a time series with two periodic components is shown in figure 6.2.

Expanding the cosine term in (6.2) we gain an alternative representation for X_t as a superposition of cosines and sines,

$$X_t = \sum_{j=1}^{k} R_j\Big\{\cos(\omega_j t)\cos(\theta_j) - \sin(\omega_j t)\sin(\theta_j)\Big\} + Z_t$$

$$= \sum_{j=1}^{k} a_j\cos(\omega_j t) + b_j\sin(\omega_j t) + Z_t, \tag{6.3}$$

where, $a_j = R_j\cos(\theta_j)$ and $b_j = -R_j\sin(\theta_j)$.

6.2 Spectral Representation

So far we have considered only a finite number, k, of frequencies. But there is no reason we should not consider a process made up of infinitely many different periodic oscillations. This idea is the basis of the **spectral representation theorem**, which says that any discrete stationary process measured at unit intervals can be represented in the form

$$X_t = \int_0^{\pi}\cos(\omega t)du(\omega) + \int_0^{\pi}\sin(\omega t)dv(\omega), \tag{6.4}$$

where the integral is a continuous stochastic integral in which $u(\omega)$ and $v(\omega)$ are uncorrelated processes with orthogonal increments. The derivation of (6.4) is beyond the scope of this course. In loose terms, this means that $\{X_t\}$ can be represented as a sum of randomly weighted sines and cosines over a continuous range of frequencies.

6.2.1 The Nyquist Frequency

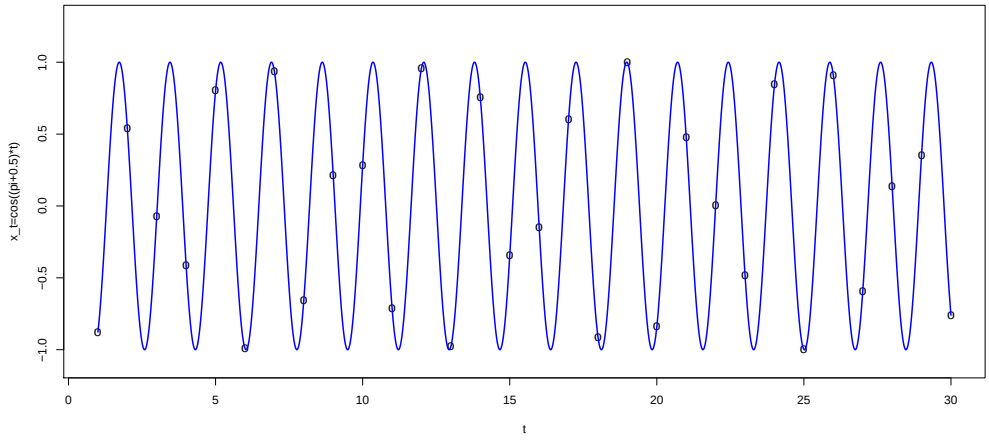
Why the upper limit of π? To understand this we consider a frequency $a > \pi$, this may be written in the form $a = \omega + k\pi$ where k is the integer part of a/π and $\omega \in [0, \pi]$ is the fractional part of a/π.

Now consider that

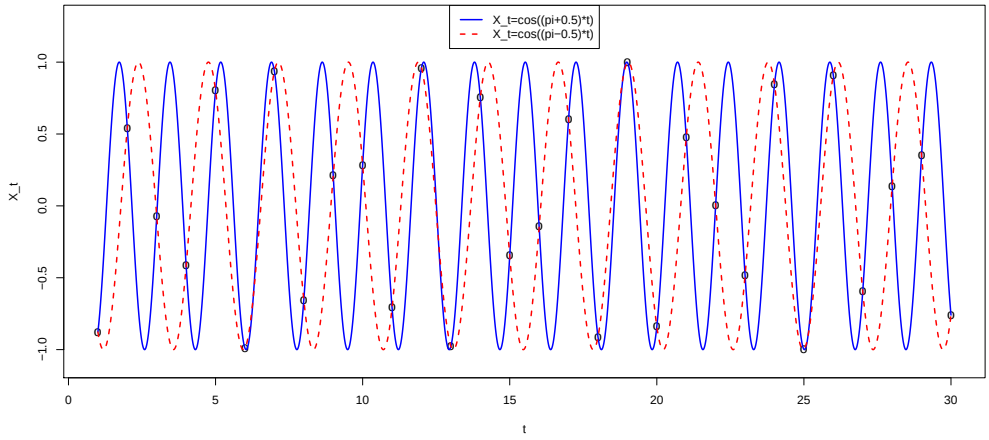
$$\cos(at) = \cos(\omega t + k\pi t) = \begin{cases} \cos \omega t & \text{if } k, t \in \mathbb{Z} \text{ and } k \text{ is even,} \\ \cos(\pi - \omega)t & \text{if } k, t \in \mathbb{Z} \text{ and } k \text{ is odd,} \end{cases} \tag{6.5}$$

in which both ω and $\pi - \omega$ are within the range $[0, \pi]$ and a similar formula holds for $\sin(at)$.

This means that the oscillations for frequencies greater than π cannot be distinguished from frequencies within $[0, \pi]$. Figure 6.3 gives a graphical explanation of this phenomenon (*aliasing*), by fitting two cosines on the same sequence of points. This corresponds to substituting $k = 1$ and $\omega = 0.5$ into (6.5).



(a) Points sampled from a process $X_t = \cos[(\pi + 0.5)t]$



(b) $\cos[(\pi - 0.5)t]$ (in red)

Figure 6.3: Graphical illustration of the Nyquist frequency.

Therefore, for discrete processes measured at unit time intervals ($t \in \mathbb{Z}^+$), the highest observable frequency is π. This frequency is called the **Nyquist frequency** (or Nyquist rate). In general, for a process measured at time intervals of Δt, the Nyquist frequency is $\pi/\Delta t$.

6.2.2 The Spectral Distribution Function

Our aim in this chapter is to develop an understanding of the frequency properties of time series. We've made a good start in establishing that every frequency in the range $[0, \pi]$ contributes to the variability of a process. However, the processes $u(\omega)$ and $v(\omega)$, introduced in the spectral representation (6.4), lack an intuitive interpretation. Of more interest is the **spectral distribution function**, $F(\omega)$. This is related to the autocovariance function and allows us to readily determine the contribution to the variability of the process that is accounted for by frequencies in a particular range. Much like the autocovariance function allowed us to determine the time lags at which the correlations played a significant role in the evolution of the process the spectral distribution allows us to determine the frequencies which play a significant role in the variability of the process.

The Wiener–Khintchine Theorem (non-examinable)
Let $\{X_t\}$ be a stationary discrete real-valued process with auto-covariance function $\gamma(k)$. Then, there exists a non-decreasing function $F(\omega)$ such that

$$\gamma(k) = \int_0^\pi \cos(\omega k) dF(\omega). \tag{6.6}$$

The function F is called the **spectral distribution function** and (6.6) is known as the spectral representation of the auto-covariance function.

Interpretation of the Wiener–Khintchine Theorem
$F(\omega)$ is the contribution to the variance of the time series which is accounted for by frequencies in the range $(0, \omega)$.

Since $\omega \in (0, \pi)$, we can set

$$F(0) = 0 \qquad \text{for} \qquad \omega < 0.$$

Moreover, for a discrete time process with $t \in \mathbb{Z}$, we have seen that the highest distinguishable frequency is π. In line with this,

$$F(\pi) = \sigma_X^2 \quad \text{(variance of } X_t\text{)}.$$

This is also consistent with (6.6) since

$$\sigma_X^2 = \text{Var}(X_t) = \gamma(0) = \int_0^\pi \cos 0 \; dF(\omega) = F(\pi) - F(0) = F(\pi).$$

Notes:

- Between 0 and π, F is monotonically increasing and is analogous to the cumulative distribution function of ordinary statistics, except that $F(\pi) = \sigma_X^2$ ($\neq 1$ in general).
- Sometimes a normalised form of $F(\omega)$ given by

$$F^*(\omega) = F(\omega)/\sigma_X^2$$

is used. Thus, $F^*(\omega)$ is the proportion of variance accounted for by frequencies in the range $(0, \omega)$.

Analogously to the cumulative distribution function we will also define a spectral density function, but before we do it will be useful to define the semi-discrete Fourier transform, which will be a very useful tool throughout the rest of this course.

6.2.3 Semi-Discrete Fourier Transform

Suppose $\{h(t)\}_{t\in\mathbb{Z}}$ is an infinite sequence. Then the **semi-discrete Fourier transform** of $\{h(t)\}$ is defined as the function

$$H(\omega) = \sum_{t=-\infty}^{\infty} h(t)e^{-i\omega t}, \qquad -\pi \leq \omega \leq \pi. \tag{6.7}$$

One surprising and fundamental point about Fourier analysis is that we can recover the original sequence $h(t)$ from $H(\omega)$ using the **inverse semi-discrete Fourier Transform**

$$h(t) = \frac{1}{2\pi}\int_{-\pi}^{\pi} H(\omega)e^{i\omega t}d\omega, \qquad t \in \mathbb{Z}. \tag{6.8}$$

Notes:

- The semi-discrete Fourier transform is so-called because it deals with discrete time – useful for the discrete stochastic processes considered in this course – and continuous frequency.
- If $\{h(t)\}_{t\in\mathbb{Z}}$ is an even function (i.e $h(-t) = h(t)$ for $t \in \mathbb{Z}$), then
  1. We can rewrite the semi-discrete Fourier transform as

$$H(\omega) = \Big\{h(0) + 2\sum_{t=1}^{\infty} h(t)\cos(\omega t)\Big\}, \qquad \text{for } 0 \leq \omega \leq \pi, \tag{6.9}$$

Proof:

$$H(\omega) = \sum_{t=-\infty}^{\infty} h(t)e^{-i\omega t} = \sum_{t=1}^{\infty} h(-t)e^{i\omega t} + h(0) + \sum_{t=1}^{\infty} h(t)e^{-i\omega t} = h(0) + 2\sum_{t=1}^{\infty} h(t)\cos(\omega t)$$

since $h(t) = h(-t)$ and $e^{i\omega t} + e^{-i\omega t} = 2\cos(\omega t)$.

  2. We can rewrite the inverse semi-discrete Fourier transform as

$$h(t) = \frac{1}{\pi}\int_{0}^{\pi} H(\omega)\cos(\omega t)d\omega, \qquad \text{for } t \in \mathbb{Z}, \tag{6.10}$$

Proof:

$$\begin{aligned} h(t) &= \frac{1}{2\pi}\int_{-\pi}^{\pi} H(\omega)e^{i\omega t}\,d\omega \\ &= \frac{1}{2\pi}\left[\int_{-\pi}^{0} H(\omega)e^{i\omega t}\,d\omega + \int_{0}^{\pi} H(\omega)e^{i\omega t}\,d\omega\right] \\ &= \frac{1}{2\pi}\left[\int_{0}^{\pi} H(-\omega)e^{-i\omega t}\,d\omega + \int_{0}^{\pi} H(\omega)e^{i\omega t}\,d\omega\right] \\ &= \frac{1}{\pi}\int_{0}^{\pi} H(\omega)\cos(\omega t)\,d\omega \end{aligned}$$

since $H(\omega) = H(-\omega)$ from (6.9) and $e^{i\omega t} + e^{-i\omega t} = 2\cos(\omega t)$.

6.2.4 The Spectral Density Function

For a purely indeterministic process the spectral distribution function, F, is continuous on $(0, \pi)$ and can therefore be differentiated. This results in the spectral density function.

The spectral density function, f, is defined to be the derivative of the spectral distribution function, F,

$$f(\omega) = \frac{dF(\omega)}{d\omega}.$$

In particular, (6.6) becomes

$$\gamma(k) = \int_{0}^{\pi} \cos(\omega k) f(\omega)\ d\omega, \tag{6.11}$$

This is an ordinary (Riemann) integral and therefore much easier to handle.

Physical meaning of the spectrum
The quantity $\int_{\omega}^{\omega+\delta\omega} f(w)dw$ represents the contribution to the variance of components with frequencies in the interval $(\omega, \omega + \delta\omega)$. Therefore, by graphing the spectrum, we could derive the following information:

1. The total variance of $\{X_t\}$ is the area under the graph of $f(\omega)$ for $\omega \in (0, \pi)$.
2. Peaks indicate the frequencies which have a large contribution to the total variance.

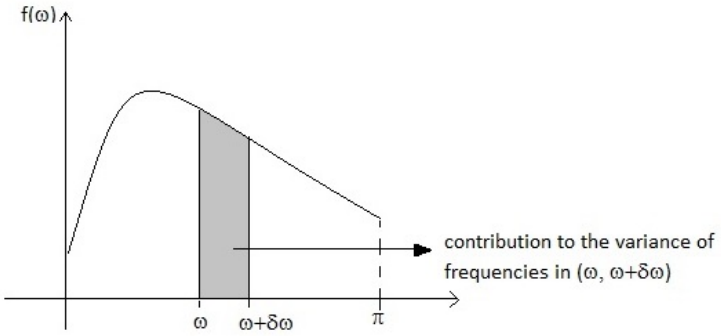


Figure 6.4: Graphical interpretation of the spectrum

Now, remembering our Fourier transform results, in particular (6.10), we see that (6.11) takes the same form as the inverse Fourier transform of $\pi f(\omega)$.

So, it follows that the spectrum can be written as

$$f(\omega) = \frac{1}{\pi}\left\{\gamma(0) + 2\sum_{k=1}^{\infty}\gamma(k)\cos(\omega k)\right\}. \tag{6.12}$$

Note: Expressions (6.11) and (6.12) suggest that the auto-covariance function, $\gamma(k)$, and the spectral density function, $f(\omega)$, are interchangeable and therefore they carry the same amount of information about the process.

6.3 Examples

6.3.1 A purely random process

Suppose that $\{Z_t\}$ is a purely random process with mean 0 and variance σ_Z^2.

We know that the auto-covariance function of $\{Z_t\}$ is given by

$$\gamma(k) = \begin{cases} \sigma_Z^2, & \text{for } k = 0 \\ 0, & \text{for } k \neq 0 \end{cases}$$

Hence, by (6.12) we find that

$$f(\omega) = \frac{1}{\pi}\left\{\gamma(0) + 0\right\} = \frac{\sigma_Z^2}{\pi}, \text{ for } \omega \in [0, \pi]$$

Notes:

- The variance is given by

$$\text{Var}(Z_t) = \int_0^{\pi} f(\omega)\, d\omega = \pi \cdot \frac{\sigma_Z^2}{\pi} = \sigma_Z^2,$$

  which is consistent with the definition of $\{Z_t\}$.

- The above result explains the name 'white noise', as the spectrum of such a process is constant for all values of the frequency domain.

6.3.2 Moving Average processes of order 1

Suppose that $\{X_t\}$ is a moving average process of order 1 with defining equation

$$X_t = Z_t + \beta Z_{t-1},$$

where $\{Z_t\}$ is a purely random process with mean 0 and variance σ_Z^2.

We know that the autocovariance function of $\{X_t\}$ is given by

$$\gamma(k) = \begin{cases} (1+\beta^2)\sigma_Z^2 & k = 0, \\ \beta\sigma_Z^2 & k = \pm 1, \\ 0 & \text{otherwise.} \end{cases}$$

Thus, using (6.12), the spectrum of $\{X_t\}$ is

$$f(\omega) = \frac{1}{\pi}\left\{(1+\beta^2)\sigma_Z^2 + 2\beta\sigma_Z^2\cos\omega\right\} = \frac{\sigma_Z^2}{\pi}\left(1+\beta^2+2\beta\cos\omega\right).$$

Having calculated the spectrum for a MA(1) process, let's have a look at the graphs of the spectrum for $\beta = \pm 1$:

- If $\beta = 1$ the process is $X_t = Z_t + Z_{t-1}$ and the spectrum is given by

  $$f(\omega) = \frac{2\sigma_Z^2}{\pi}\,(1+\cos\omega),$$

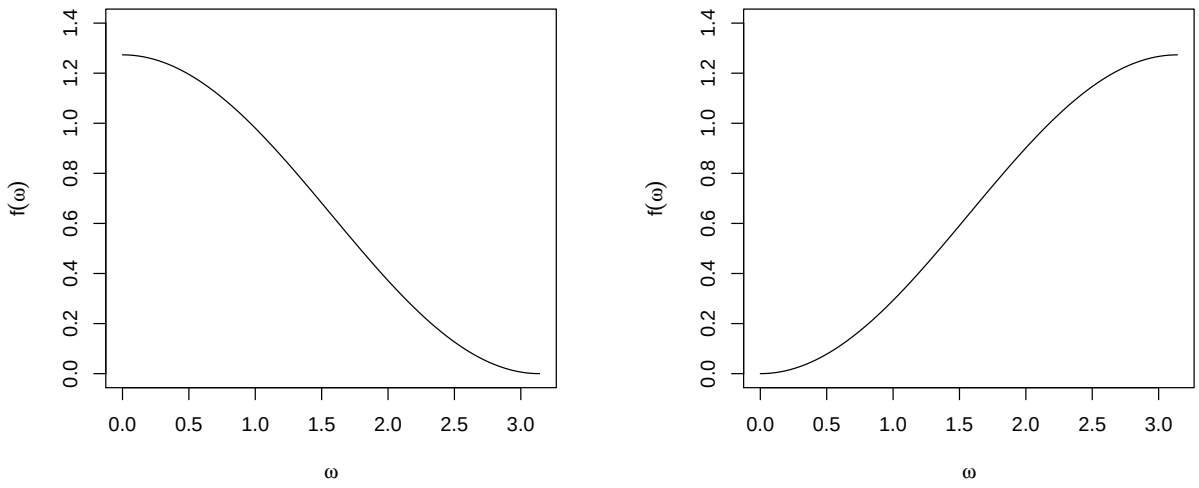
  which is decreasing in the interval $(0, \pi)$ and has power at low frequencies.

- If $\beta = -1$ the process is $X_t = Z_t - Z_{t-1}$ and the spectrum is given by

  $$f(\omega) = \frac{2\sigma_Z^2}{\pi}\,(1-\cos\omega),$$

  which is increasing in the interval $(0, \pi)$ and has power at high frequencies.

The graphs of the two cases are shown in figure 6.5 (for $\sigma_Z^2 = 1$).



(a) $X_t = Z_t + Z_{t-1}$ (b) $X_t = Z_t - Z_{t-1}$

Figure 6.5: Spectra for MA(1)

Note: Operations such as $X_t = Z_t + Z_{t-1}$ that concentrate power in the low frequency region are referred to as **low-pass**. Similarly, operations such as $X_t = Z_t - Z_{t-1}$ that concentrate power in the high frequencies are referred to as **high-pass**.

6.3.3 Autoregressive processes of order 1

Suppose that $\{X_t\}$ is a stationary autoregressive process of order 1 with defining equation

$$X_t = \alpha X_{t-1} + Z_t$$

for $|\alpha| < 1$, in which $\{Z_t\}$ is a purely random process with mean 0 and variance σ_Z^2.

We know that the autocovariance function of $\{X_t\}$ is given by

$$\gamma(k) = \sigma_X^2 \alpha^{|k|}, \ k \in \mathbb{Z}.$$

Using (6.7) we will calculate the spectral density function for this process.

$$\begin{aligned}
f(\omega) &= \frac{1}{\pi} \sum_{k=-\infty}^{\infty} \gamma(k) e^{-i\omega k} \\
&= \frac{1}{\pi} \Big\{ \gamma(0) + \sum_{k=-\infty}^{-1} \gamma(k) e^{-i\omega k} + \sum_{k=1}^{\infty} \gamma(k) e^{-i\omega k} \Big\} \\
&= \frac{1}{\pi} \Big\{ \gamma(0) + \sum_{k=1}^{\infty} \gamma(k) e^{i\omega k} + \sum_{k=1}^{\infty} \gamma(k) e^{-i\omega k} \Big\} \\
&= \frac{1}{\pi} \Big\{ \sigma_X^2 + \sum_{k=1}^{\infty} \sigma_X^2 \alpha^k e^{i\omega k} + \sum_{k=1}^{\infty} \sigma_X^2 \alpha^k e^{-i\omega k} \Big\} \\
&= \frac{\sigma_X^2}{\pi} \Big\{ 1 + \sum_{k=1}^{\infty} \left(\alpha e^{i\omega} \right)^k + \sum_{k=1}^{\infty} \left(\alpha e^{-i\omega} \right)^k \Big\}.
\end{aligned}$$

Since $|\alpha| < 1$, we know that $|\alpha e^{\pm i\omega}| < 1$ and therefore the two series of the last expression converge. Hence,

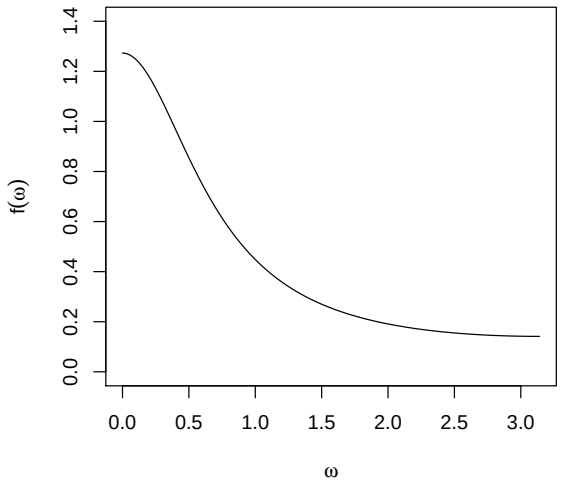
$$\begin{aligned}
f(\omega) &= \frac{\sigma_X^2}{\pi} \Big\{ 1 + \frac{\alpha e^{i\omega}}{1 - \alpha e^{i\omega}} + \frac{\alpha e^{-i\omega}}{1 - \alpha e^{-i\omega}} \Big\} \\
&= \frac{\sigma_X^2}{\pi} \Big\{ 1 + \frac{\alpha e^{i\omega} - \alpha^2 + \alpha e^{-i\omega} - \alpha^2}{1 - \alpha \left(e^{i\omega} + e^{-i\omega} \right) + \alpha^2} \Big\} \\
&= \frac{\sigma_X^2}{\pi} \left(\frac{1 - \alpha^2}{1 - 2\alpha \cos \omega + \alpha^2} \right)
\end{aligned}$$

and since

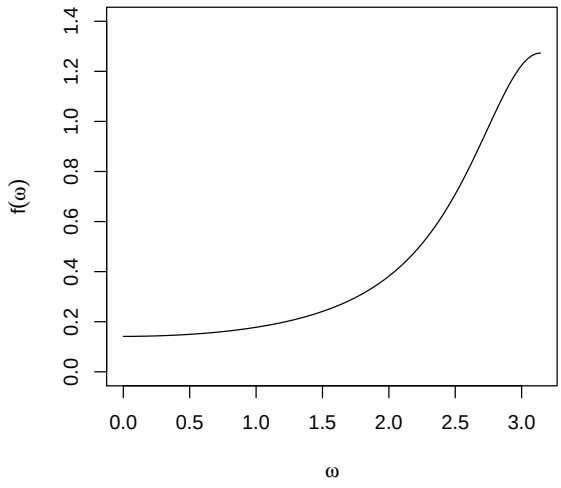
$$\sigma_X^2 = \frac{\sigma_Z^2}{1 - \alpha^2}$$

we finally get that

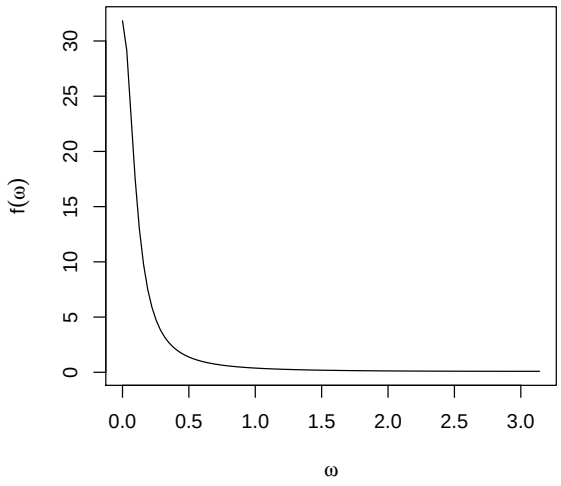
$$f(\omega) = \frac{\sigma_Z^2}{\pi(1 - 2\alpha \cos \omega + \alpha^2)}.$$



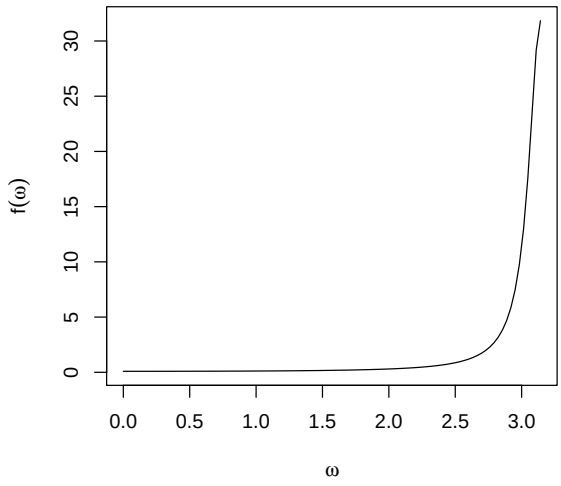
(a) $X_t = Z_t + 0.5X_{t-1}$



(b) $X_t = Z_t - 0.5X_{t-1}$



(c) $X_t = Z_t + 0.9X_{t-1}$



(d) $X_t = Z_t - 0.9X_{t-1}$

Figure 6.6: Spectra for AR(1)

Figure 6.6 shows the graphs of the spectra for the AR(1) model for $\sigma_Z^2 = 1$ and various values of α. We can see that when $\alpha > 0$ we get a low-frequency spectrum and the closer α is to 1, the more the power of the process is concentrated on lower frequencies. This happens since, in this case, successive values of $\{X_t\}$ are similar, meaning that there is no high frequency oscillation. On the other hand, if $\alpha < 0$, we get a high-frequency spectrum, since $\{X_t\}$ oscillates in this case. The spectrum becomes more powerful at high frequencies as α gets closer to -1.

6.3.4 Higher order autoregressive processes

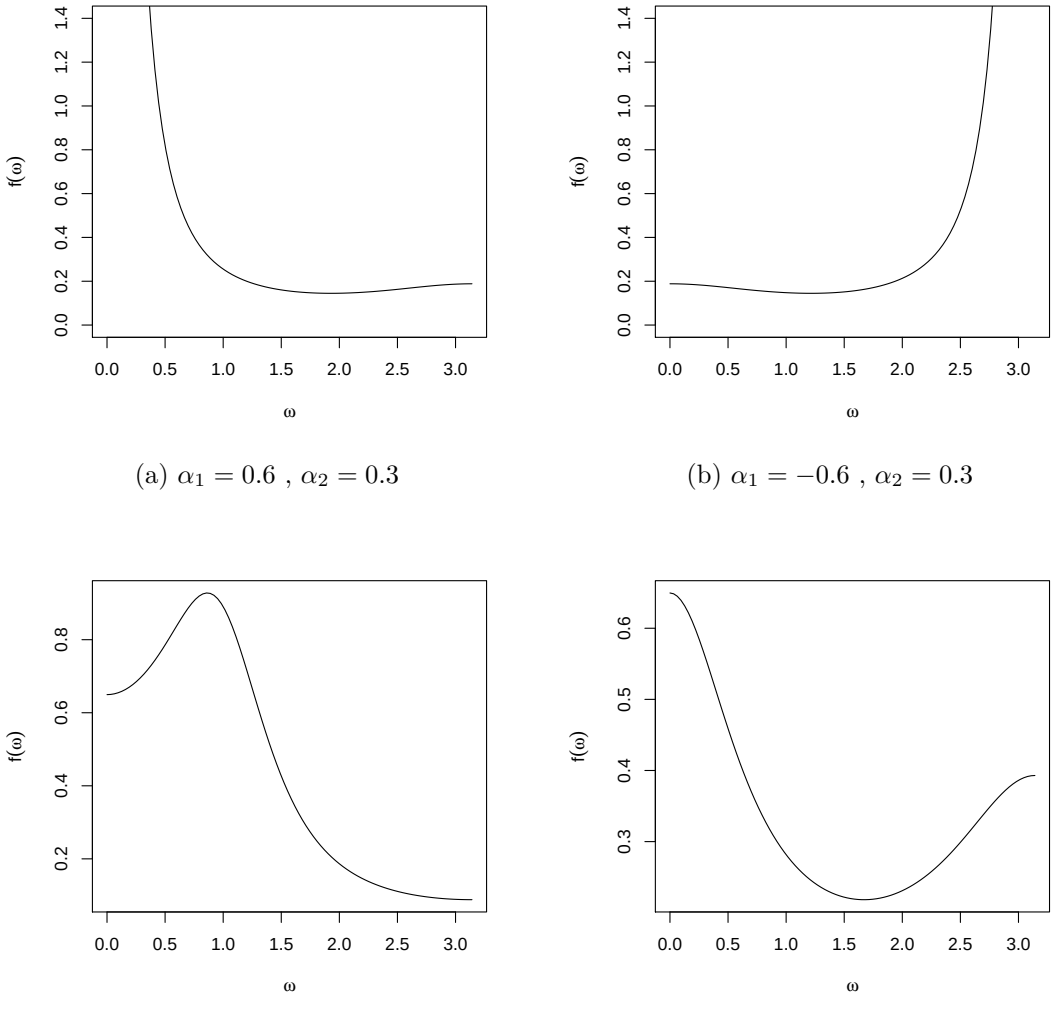
The spectrum of a stationary, second-order AR process $\{X_t\}$ with coefficients α_1 and α_2, i.e.

$$X_t = \alpha_1 X_{t-1} + \alpha_2 X_{t-2} + Z_t,$$

is given by

$$f(\omega) = \frac{\sigma_Z^2}{\pi\left\{1 + \alpha_1^2 + \alpha_2^2 - 2\alpha_1(1-\alpha_2)\cos\omega - 2\alpha_2\cos(2\omega)\right\}},\ \omega \in [0, \pi].$$

Clearly the shape of this spectrum depends on α_1 and α_2. It is possible to obtain spectra with power at either low or high frequencies or, indeed, peaks/troughs between the two, as shown in figure 6.7.



(a) $\alpha_1 = 0.6$, $\alpha_2 = 0.3$

(b) $\alpha_1 = -0.6$, $\alpha_2 = 0.3$

(c) $\alpha_1 = 0.6$, $\alpha_2 = -0.3$

(d) $\alpha_1 = 0.1$, $\alpha_2 = 0.2$

Figure 6.7: Spectra for AR(2) for various values of α_1, α_2.

6.3.5 A deterministic sinusoidal model

Suppose that $\{X_t\}$ is a deterministic process defined by

$$X_t = \cos(\omega_0 t + \theta),\ \omega_0 \in (0, \pi),\ \theta \sim U(0, 2\pi).$$

The autocovariance function of the process is given by

$$\gamma(k) = \frac{1}{2}\cos(\omega_0 k).$$

(Hint: $2\cos(a)\cos(b) = \cos(a-b) + \cos(a+b)$). First, let's attempt to apply (6.12), i.e.,

$$\begin{aligned} f(\omega) &= \frac{1}{\pi}\Big\{\gamma(0) + 2\sum_{k=1}^{\infty}\frac{1}{2}\cos(\omega_0 k)\cos(\omega k)\Big\} \\ &= \frac{1}{\pi}\Big[\frac{1}{2} + \frac{1}{2}\sum_{k=1}^{\infty}\cos\{(\omega_0+\omega)k\} + \cos\{(\omega_0-\omega)k\}\Big] \end{aligned}$$

Then, at $\omega = \omega_0$,

$$f(\omega_0) = \frac{1}{2\pi}\Big[1 + \sum_{k=1}^{\infty}\{\cos(2\omega_0 k) + 1\}\Big],$$

which does not converge.

However, recall from the Wiener–Khintchine Theorem that there must exist a non-decreasing F such that

$$\gamma(k) = \frac{1}{2}\cos(\omega_0 k) = \int_0^{\pi}\cos(\omega k)dF(\omega).$$

We find that one function that works is

$$F(\omega) = \begin{cases} 0 & \text{for } \omega < \omega_0, \\ \frac{1}{2} & \text{for } \omega \geq \omega_0. \end{cases}$$

Then

$$\frac{dF(\omega)}{d\omega} = \delta(\omega - \omega_0)$$

and so the spectral density function is 0 everywhere except at ω_0, where it is infinite.

Chapter 7

Spectral Analysis

In the previous chapter we saw that the spectral density function, f, provides useful information about a stochastic process $\{X_t\}$. In this chapter we'll look at how we can analyse the frequency properties of an observed time series, rather than an idealised model. Spectral analysis is the name given to methods of estimating the spectral density function based on observed time series data, $x_1, x_2, ..., x_N$, and deriving such methods is the ultimate goal of this chapter. Our first step is to search for the periodicities in the data — some of these might be obvious, such as the annual period of temperature data, but others might be harder to spot by eye. To do this we estimate, record and remove the various periodic effects using a simple sinusoidal model and standard linear regression techniques.

7.1 A Simple Sinusoidal Model

Consider a process for which we have reason to believe that a sinusoidal component at frequency ω is present. Then, assuming that observations are collected at unit time intervals, a reasonable model could be:

$$X_t = \mu + \alpha\cos(\omega t) + \beta\sin(\omega t) + Z_t,\ t = 1, 2, \ldots N, \tag{7.1}$$

where $\{Z_t\}$ is a purely random process with mean zero and variance σ_Z^2 whilst μ, α and β are the parameters that we will estimate using the data, $x_1, x_2, \ldots, x_N$.
Model (7.1) can be equivalently represented in matrix form as

$$\mathbf{X} = A\boldsymbol{\theta} + \mathbf{Z},$$

where $\mathbf{X} = (X_1, X_2, \ldots, X_N)^T$, $\boldsymbol{\theta} = (\mu, \alpha, \beta)^T$, and $\mathbf{Z} = (Z_1, Z_2, \ldots, Z_N)^T$ whilst

$$A = \begin{pmatrix} 1 & \cos(\omega) & \sin(\omega) \\ 1 & \cos(2\omega) & \sin(2\omega) \\ \vdots & \vdots & \vdots \\ 1 & \cos(N\omega) & \sin(N\omega) \end{pmatrix}. \tag{7.2}$$

Then, the least squares estimator of $\boldsymbol{\theta}$ is that which minimises $\sum_{t=1}^{N}\{x_t - \mu - \alpha\cos(\omega t) - \beta\sin(\omega t)\}^2$ and is given by

$$\hat{\boldsymbol{\theta}} = \left(A^T A\right)^{-1} A^T \mathbf{x} \tag{7.3}$$

in which $\mathbf{x} = (x_1, \ldots, x_N)^T$.

This is in no way different from the usual least-squares estimator for a linear regression, it is just in higher dimension and the most efficient calculation of the result requires us to compute some matrix derivatives - you don't need to be able to do this!

To find a more explicit form of $\hat{\boldsymbol{\theta}}$, we first calculate A^TA :

$$\begin{aligned} A^TA &= \begin{pmatrix} 1 & 1 & \dots & 1 \\ \cos(\omega) & \cos(2\omega) & \dots & \cos(N\omega) \\ \sin(\omega) & \sin(2\omega) & \dots & \sin(N\omega) \end{pmatrix} \begin{pmatrix} 1 & \cos(\omega) & \sin(\omega) \\ 1 & \cos(2\omega) & \sin(2\omega) \\ \vdots & \vdots & \vdots \\ 1 & \cos(N\omega) & \sin(N\omega) \end{pmatrix} \\ &= \begin{pmatrix} N & \sum_{t=1}^{N}\cos(\omega t) & \sum_{t=1}^{N}\sin(\omega t) \\ \sum_{t=1}^{N}\cos(\omega t) & \sum_{t=1}^{N}\cos^2(\omega t) & \sum_{t=1}^{N}\cos(\omega t)\sin(\omega t) \\ \sum_{t=1}^{N}\sin(\omega t) & \sum_{t=1}^{N}\cos(\omega t)\sin(\omega t) & \sum_{t=1}^{N}\sin^2(\omega t) \end{pmatrix} \end{aligned} \tag{7.4}$$

Interestingly, if N is even and ω is restricted to take a value among $\omega_p = \frac{2\pi p}{N},\ p = 1, 2, \dots, N/2$ (called the Fourier frequencies) then A^TA is a diagonal matrix (and hence easy to invert). This is due to the following trigonometric results:

(i) $$\sum_{t=1}^{N}\cos(\omega_p t) = \sum_{t=1}^{N}\sin(\omega_p t) = 0$$

(ii) $$\sum_{t=1}^{N}\cos(\omega_p t)\cos(\omega_q t) = \begin{cases} 0, & p \neq q \\ N, & p = q = \frac{N}{2} \\ \frac{N}{2}, & p = q \neq \frac{N}{2} \end{cases}$$

(iii) $$\sum_{t=1}^{N}\sin(\omega_p t)\sin(\omega_q t) = \begin{cases} 0, & p \neq q \\ 0, & p = q = \frac{N}{2} \\ \frac{N}{2}, & p = q \neq \frac{N}{2} \end{cases}$$

(iv) $$\sum_{t=1}^{N}\cos(\omega_p t)\sin(\omega_q t) = 0,$$

for $p, q \in \left\{1, 2, \dots, \frac{N}{2}\right\}$. The proof of these results is left as an exercise (Problem Sheet 8, Q1).

But we don't only restrict ourselves to these values of ω because it makes things easier. The highest and lowest values are also chosen to be the highest and lowest frequencies we can observe. We already know that the **highest observable frequency** is π — the Nyquist frequency — and this corresponds to the highest Fourier frequency (the special case $p = N/2$). But what about the lowest observable frequency? To be able to observe an oscillation we need it to complete at least one entire cycle within the range of the series. The lowest frequency that satisfies this condition completes exactly one cycle within the range of the series and so has period equal to N. Thus, the
lowest observable frequency is given by

$$N = 2\pi/\omega \qquad \Longrightarrow \qquad \omega = 2\pi/N,$$

and this corresponds to the lowest Fourier frequency ($p = 1$).

When applying the trigonometric results for Fourier frequencies we first consider the special case where $p = N/2$. In this case then $\omega_p = \pi$ and $\sin(\omega_p t)$ vanishes so our model becomes

$$\begin{aligned} X_t &= \mu + \alpha\cos(\pi t) + \beta \underbrace{\sin(\pi t)}_{=0} + Z_t \\ &= \mu + \alpha(-1)^t + Z_t, \qquad t = 1, 2, \ldots, N. \end{aligned} \tag{7.5}$$

Thus, the value of β is immaterial. The problem reduces to estimating $\boldsymbol{\theta}^T = (\mu, \alpha)$, where A is now an $N \times 2$ matrix, with the third column in (7.2) removed. Consequently,

$$A^T A = \begin{pmatrix} N & 0 \\ 0 & N \end{pmatrix} \implies \left(A^T A\right)^{-1} = \begin{pmatrix} \frac{1}{N} & 0 \\ 0 & \frac{1}{N} \end{pmatrix}$$

and we find that

$$\hat{\boldsymbol{\theta}} = \begin{pmatrix} \frac{1}{N} & 0 \\ 0 & \frac{1}{N} \end{pmatrix} \begin{pmatrix} 1 & \ldots & 1 \\ \cos(\pi) & \ldots & \cos(N\pi) \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_N \end{pmatrix} = \begin{pmatrix} \frac{1}{N}\sum_{t=1}^{N} x_t \\ \\ \frac{1}{N}\sum_{t=1}^{N} x_t \cos(\pi t) \end{pmatrix}$$

from which it is immediate that

$$\hat{\mu} = \frac{1}{N}\sum_{t=1}^{N} x_t, \qquad \text{and} \qquad \hat{\alpha} = \frac{1}{N}\sum_{t=1}^{N} (-1)^t x_t.$$

Meanwhile, considering the more general case where $p \neq N/2$, then we obtain

$$A^T A = \begin{pmatrix} N & 0 & 0 \\ 0 & \frac{N}{2} & 0 \\ 0 & 0 & \frac{N}{2} \end{pmatrix} \implies \left(A^T A\right)^{-1} = \begin{pmatrix} \frac{1}{N} & 0 & 0 \\ 0 & \frac{2}{N} & 0 \\ 0 & 0 & \frac{2}{N} \end{pmatrix}.$$

Hence, the calculation of $\hat{\boldsymbol{\theta}}$ simplifies to

$$\begin{aligned} \hat{\boldsymbol{\theta}} &= \left(A^T A\right)^{-1} A^T \mathbf{x} \\ &= \begin{pmatrix} \frac{1}{N} & 0 & 0 \\ 0 & \frac{2}{N} & 0 \\ 0 & 0 & \frac{2}{N} \end{pmatrix} \begin{pmatrix} 1 & 1 & \ldots & 1 \\ \cos(\omega_p) & \cos(2\omega_p) & \ldots & \cos(N\omega_p) \\ \sin(\omega_p) & \sin(2\omega_p) & \ldots & \sin(N\omega_p) \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_N \end{pmatrix} \\ &= \begin{pmatrix} \frac{1}{N} & \frac{1}{N} & \ldots & \frac{1}{N} \\ \frac{2}{N}\cos(\omega_p) & \frac{2}{N}\cos(2\omega_p) & \ldots & \frac{2}{N}\cos(N\omega_p) \\ \frac{2}{N}\sin(\omega_p) & \frac{2}{N}\sin(2\omega_p) & \ldots & \frac{2}{N}\sin(N\omega_p) \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_N \end{pmatrix} \\ &= \begin{pmatrix} \frac{1}{N}\sum_{t=1}^{N} x_t \\ \\ \frac{2}{N}\sum_{t=1}^{N} x_t \cos(\omega_p t) \\ \\ \frac{2}{N}\sum_{t=1}^{N} x_t \sin(\omega_p t) \end{pmatrix} \end{aligned}$$

and so for $p \neq N/2$ our estimates of the parameters are given by

$$\hat{\mu} = \frac{1}{N}\sum_{t=1}^{N} x_t \qquad (= \hat{x}, \text{ as expected}),$$

$$\hat{\alpha} = \frac{2}{N}\sum_{t=1}^{N} x_t \cos(\omega_p t), \tag{7.6}$$

$$\hat{\beta} = \frac{2}{N}\sum_{t=1}^{N} x_t \sin(\omega_p t). \tag{7.7}$$

7.2 The Finite Fourier Series Representation

The methodology used in the previous section to estimate the parameters of model (7.1) holds for all frequencies in the form $\omega_p = \frac{2\pi}{N}, \frac{4\pi}{N}, \ldots, \pi$.

Thus, the simple sinusoidal model given in (7.1) generalises to give the so-called **finite Fourier series representation** of the observed time series $\{x_t\}$. That is,

$$x_t = \mu + \sum_{p=1}^{\frac{N}{2}-1} \{\alpha_p \cos(\omega_p t) + \beta_p \sin(\omega_p t)\} + \alpha_{\frac{N}{2}} \cos(\pi t), \quad \text{for } t = 1, 2, \ldots, N \tag{7.8}$$

in which

$$\begin{cases} \hat{\mu} &= \bar{x} \\ \hat{\alpha}_{\frac{N}{2}} &= \frac{1}{N}\sum_{t=1}^{N}(-1)^t x_t \\ \hat{\alpha}_p &= \frac{2}{N}\sum_{t=1}^{N} x_t \cos(\omega_p t), \qquad p = 1, 2, \ldots, \frac{N}{2} - 1 \\ \hat{\beta}_p &= \frac{2}{N}\sum_{t=1}^{N} x_t \sin(\omega_p t), \qquad p = 1, 2, \ldots, \frac{N}{2} - 1. \end{cases} \tag{7.9}$$

Notes:

- As we saw in the previous section, when $\pi = \frac{N}{2}$ (i.e. $\omega_p = \pi$), $\sin(\omega_p t)$ is 0 and this explains the absence of parameter $\beta_{\frac{N}{2}}$ from (7.8).
- The Fourier series representation in (7.8) uses N parameters to describe N observations and so can be made to fit the data exactly. This explains why there is no error term in (7.8).

Thus, N numbers become N Fourier coefficients. The effect of this Fourier Analysis is to partition the variability of the time series into components at frequencies $\omega_p = \frac{2\pi}{N}, \frac{4\pi}{N}, \ldots, \pi$.

The component at frequency $\omega_p = \frac{2\pi p}{N}$, is called the **p−th harmonic** and, for $p \neq \frac{N}{2}$, this can be written as

$$\alpha_p \cos(\omega_p t) + \beta_p \sin(\omega_p t) \equiv R_p \cos(\omega_p t + \phi_p),$$

where

$$\begin{aligned} R_p &= \sqrt{\alpha_p^2 + \beta_p^2} \qquad \text{is the } \textbf{amplitude} \text{ of the } p\text{-th harmonic and} \\ \phi_p &= \tan^{-1}\left(-\frac{\beta_p}{\alpha_p}\right) \qquad \text{is the } \textbf{phase} \text{ of the } p\text{-th harmonic.} \end{aligned}$$

Equation (7.8) therefore decomposes the time series into a sum of $N/2$ harmonics.

7.3 The Periodogram

In fact, we can be more precise about how each harmonic contributes to the variability of the process, relating the variance in time- and frequency-domains using Parseval's theorem:

Parseval's Theorem (discrete version)
If $x_1, x_2, \ldots, x_N$ is an observed time series, R_p is the amplitude of the $p-$th harmonic for $p = 1, 2, \ldots, \frac{N}{2} - 1$ and $\alpha_{\frac{N}{2}}$ is the amplitude of the $\frac{N}{2}$th harmonic, then

$$\frac{1}{N}\sum_{t=1}^{N}(x_t - \bar{x})^2 = \frac{1}{2}\sum_{p=1}^{\frac{N}{2}-1} R_p^2 + \alpha_{\frac{N}{2}}^2. \tag{7.10}$$

Proof: We know that $\mu = \bar{x}$ and by (7.8) we have

$$x_t - \bar{x} = x_t - \mu = \sum_{p=1}^{\frac{N}{2}-1} \{\alpha_p \cos(\omega_p t) + \beta_p \sin(\omega_p t)\} + \alpha_{\frac{N}{2}} \cos(\pi t). \tag{7.11}$$

Taking the sum of squares of both sides in (7.11) we get

$$\sum_{t=1}^{N}(x_t - \bar{x})^2 = \sum_{t=1}^{N}\left[\sum_{p=1}^{\frac{N}{2}-1} \{\alpha_p \cos(\omega_p t) + \beta_p \sin(\omega_p t)\} + \alpha_{\frac{N}{2}} \cos(\pi t)\right]^2 = A_1 + A_2 + A_3, \tag{7.12}$$

in which

$$A_1 = \sum_{t=1}^{N}\left[\sum_{p=1}^{\frac{N}{2}-1} \{\alpha_p \cos(\omega_p t) + \beta_p \sin(\omega_p t)\}\right]^2,$$

whilst

$$A_2 = \sum_{t=1}^{N} 2\alpha_{\frac{N}{2}} \cos(\pi t) \sum_{p=1}^{\frac{N}{2}-1}\Big[\{\alpha_p \cos(\omega_p t) + \beta_p \sin(\omega_p t)\}\Big]$$

and

$$A_3 = \sum_{t=1}^{N} \alpha^2_{\frac{N}{2}} \overbrace{\cos^2(\pi t)}^{=1}.$$

We will work with the three quantities A_1, A_2 and A_3 separately. Starting with A_3, which is the easiest, we have

$$A_3 = N\alpha^2_{\frac{N}{2}}.$$

Now we show that A_2 vanishes:

$$A_2 = 2\alpha_{\frac{N}{2}} \left\{ \sum_{p=1}^{\frac{N}{2}-1} \alpha_p \underbrace{\sum_{t=1}^{N} \cos(\omega_p t)\cos(\pi t)}_{=0} + \sum_{p=1}^{\frac{N}{2}-1} \beta_p \underbrace{\sum_{t=1}^{N} \sin(\omega_p t)\cos(\pi t)}_{=0} \right\} = 0$$

where we have used our trigonometric identities again.

Finally,

$$\begin{aligned}
A_1 &= \sum_{t=1}^{N} \sum_{p=1}^{\frac{N}{2}-1} \sum_{q=1}^{\frac{N}{2}-1} \{\alpha_p \cos(\omega_p t) + \beta_p \sin(\omega_p t)\} \{\alpha_q \cos(\omega_q t) + \beta_q \sin(\omega_q t)\} \\
&= \sum_{p=1}^{\frac{N}{2}-1} \sum_{q=1}^{\frac{N}{2}-1} \sum_{t=1}^{N} \{\alpha_p\alpha_q \cos(\omega_p t)\cos(\omega_q t) + \alpha_p\beta_q \cos(\omega_p t)\sin(\omega_q t) \\
&\qquad + \beta_p\alpha_q \sin(\omega_p t)\cos(\omega_q t) + \beta_p\beta_q \sin(\omega_p t)\sin(\omega_q t)\} \\
&= \sum_{p=1}^{\frac{N}{2}-1} \sum_{q=1}^{\frac{N}{2}-1} \alpha_p\alpha_q \sum_{t=1}^{N} \cos(\omega_p t)\cos(\omega_q t) + \sum_{p=1}^{\frac{N}{2}-1} \sum_{q=1}^{\frac{N}{2}-1} \alpha_p\beta_q \underbrace{\sum_{t=1}^{N} \cos(\omega_p t)\sin(\omega_q t)}_{=0} \\
&\quad + \sum_{p=1}^{\frac{N}{2}-1} \sum_{q=1}^{\frac{N}{2}-1} \beta_p\alpha_q \underbrace{\sum_{t=1}^{N} \sin(\omega_p t)\cos(\omega_q t)}_{=0} + \sum_{p=1}^{\frac{N}{2}-1} \sum_{q=1}^{\frac{N}{2}-1} \beta_p\beta_q \sum_{t=1}^{N} \sin(\omega_p t)\sin(\omega_q t) \\
&= \sum_{p=1}^{\frac{N}{2}-1} \alpha_p^2 \cdot \frac{N}{2} + \sum_{p=1}^{\frac{N}{2}-1} \beta_p^2 \cdot \frac{N}{2} \\
&= \frac{N}{2} \sum_{p=1}^{\frac{N}{2}-1} \left(\alpha_p^2 + \beta_p^2\right) \\
&= \frac{N}{2} \sum_{p=1}^{\frac{N}{2}-1} R_p^2
\end{aligned}$$

having again used the trigonometric identities to evaluate sums. Substituting A_1, A_2, and A_3 into (7.12) and dividing by N we get (7.10).

Parseval's theorem illustrates how the (total) variance of the time series can be decomposed into contributions made by fluctuations at certain frequencies. More precisely, we regard $R_p^2/2$ as the contribution to variance in the range $\omega_p \pm \pi/N$. A visual representation of this decomposition of variance is given by **the periodogram**, a histogram-like estimator of the spectrum. The idea behind the periodogram is that every bar of the histogram should reflect the contribution of the corresponding frequencies to the total variance of the process. So, the height in the range $\omega_p \pm \pi/N$ (denoted $I(\omega_p)$) is such that

$$\begin{aligned}\frac{R_p^2}{2} &= \text{area of histogram rectangle} \\ &= \text{height of histogram} \times \frac{2\pi}{N}\end{aligned}$$

and so is given by

$$I(\omega_p) = \frac{NR_p^2}{4\pi} \quad \text{for } p = 1, \ldots, \frac{N}{2} - 1.$$

As usual we have to consider $p = N/2$ separately. In this case then we regard $\alpha_{N/2}^2$ as the contribution to the variance in the range $[\pi - \pi/N, \pi]$ so that $I(\pi) = N\alpha_{N/2}^2/\pi$.

Combining these results we find that the heights of the bars in the periodogram are given by

$$I(\omega) = \begin{cases} \frac{NR_p^2}{4\pi} & \text{for } \omega \in [\omega_p - \pi/N, \omega_p + \pi/N], \text{where } p = 1, \ldots, \frac{N}{2} - 1 \\ \frac{N\alpha_{N/2}^2}{\pi} & \text{for } \omega \in [\pi - \pi/N, \pi]. \end{cases} \tag{7.13}$$

Notes:

- For $p = 1, 2, \ldots, \frac{N}{2} - 1$, $I(\omega_p)$ can be computed directly from the data as follows

$$I(\omega_p) = \frac{\left\{\sum_{t=1}^{N} x_t \cos(\omega_p t)\right\}^2 + \left\{\sum_{t=1}^{N} x_t \sin(\omega_p t)\right\}^2}{N\pi}. \tag{7.14}$$

  How to derive this relationship is left as an exercise.

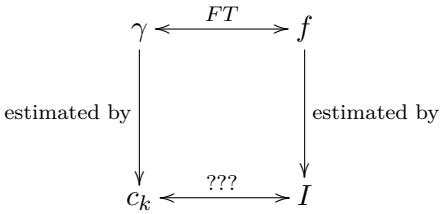
- Recall that the area under the spectral density function for a particular range of frequencies represents the contribution to variance of components with frequencies in that range. So, it is natural to think of the periodogram as an estimator of the spectral density. But how good is it? This fundamental question is our next subject.

7.4 The Periodogram and the Sample Auto-covariance Function

We have seen that $f(\omega)$ is related to $\gamma(k)$ by Fourier Transform, and we have proposed estimators for each: $I(\omega)$ and c_k respectively, where

$$c_k = \frac{1}{N}\sum_{t=1}^{N-k}(x_t - \bar{x})(x_{t+k} - \bar{x}).$$

It is then natural to ask: how are I and c_k related?



Recall that

$$\sum_{t=1}^{N}\cos(\omega_p t) = \sum_{t=1}^{N}\sin(\omega_p t) = 0 \quad \text{for } p = 1, \ldots, \frac{N}{2},$$

so we can write (7.14) as

$$\begin{aligned}
I(\omega_p) &= \left[\left\{\sum_{t=1}^{N}(x_t - \bar{x})\cos\omega_p t\right\}^2 + \left\{\sum_{t=1}^{N}(x_t - \bar{x})\sin\omega_p t\right\}^2\right]/N\pi \\
&= \sum_{s,t=1}^{N}(x_t - \bar{x})(x_s - \bar{x})(\cos\omega_p t\cos\omega_p s + \sin\omega_p t\sin\omega_p s)\ /N\pi \\
&= \Big\{\sum_{t=1}^{N}(x_t - \bar{x})(x_t - \bar{x})(\cos\omega_p t\cos\omega_p t + \sin\omega_p t\sin\omega_p t) \\
&\quad + 2\sum_{k=1}^{N-1}\sum_{t=1}^{N-k}(x_t - \bar{x})(x_{t+k} - \bar{x})(\cos\omega_p t\cos\omega_p(t+k) + \sin\omega_p t\sin\omega_p(t+k))\Big\}\ /N\pi,
\end{aligned}$$

so that:

$$\begin{aligned}
I(\omega_p) &= \left\{c_0 + 2\sum_{k=1}^{N-1}c_k\cos(\omega_p k)\right\}/\pi, \qquad (7.15)\\
&= \sum_{k=-(N-1)}^{N-1}c_k e^{-i\omega_p k}\ /\pi.
\end{aligned}$$

Compare this to the equation for the spectral density in terms of the autocovariance function in (6.12).

7.5 Statistical Properties of the Periodogram

Looking at (7.15) the periodogram appears to be a natural estimate for the spectral density

$$f(\omega) = \frac{1}{\pi}\left\{\gamma(0) + 2\sum_{k=1}^{\infty}\gamma(k)\cos(\omega k)\right\}.$$

In this section we will investigate how good it actually is. We start by defining a couple of familiar measures for the quality of an estimator.

Suppose $\hat{\theta}$ is an estimator of θ, then:

- We say that $\hat{\theta}$ is **unbiased** if $\mathbb{E}(\hat{\theta}) = \theta$.
- We say that $\hat{\theta}$ is **consistent** if, for any $\epsilon > 0$, $\lim_{N\to\infty} \mathbb{P}(|\hat{\theta} - \theta| > \epsilon) = 0$.

  This happens whenever $\lim_{N\to\infty} \mathbb{E}(\hat{\theta} - \theta) = 0$ and $\lim_{N\to\infty} \mathrm{Var}(\hat{\theta}) = 0$.

In general, the periodogram, $I(\omega)$, is neither unbiased nor consistent as an estimator of $f(\omega)$. In this section, we will focus on showing that the periodogram is not a consistent estimator of the spectrum in the simple case of a purely random Gaussian process

$$X_t = Z_t, \qquad Z_t \overset{i.i.d.}{\sim} \mathcal{N}(0, \sigma_Z^2).$$

If we set $A(\omega_p)$ and $B(\omega_p)$ to be

$$A(\omega_p) = \sum_{t=1}^{N} X_t \cos(\omega_p t) \quad \text{and} \quad B(\omega_p) = \sum_{t=1}^{N} X_t \sin(\omega_p t),$$

Then, from (7.14), the periodogram value at ω_p (when viewed as a random variable) is

$$I(\omega_p) = \frac{1}{N\pi}\left\{A^2(\omega_p) + B^2(\omega_p)\right\}, \qquad \text{for any } p \in \{1, \ldots, \frac{N}{2} - 1\}. \tag{7.16}$$

First, we calculate the expectations and variances of A and B.

$$\mathbb{E}\left\{A(\omega_p)\right\} = \mathbb{E}\left\{\sum_{t=1}^{N} X_t \cos(\omega_p t)\right\} = \sum_{t=1}^{N} \cos(\omega_p t)\mathbb{E}(X_t) = 0 \quad \text{and}$$

$$\mathbb{E}\left\{B(\omega_p)\right\} = \mathbb{E}\left\{\sum_{t=1}^{N} X_t \sin(\omega_p t)\right\} = \sum_{t=1}^{N} \sin(\omega_p t)\mathbb{E}(X_t) = 0.$$

Whilst

$$\begin{aligned}
\mathrm{Var}\left\{A(\omega_p)\right\} &= \mathrm{Var}\left\{\sum_{t=1}^{N} X_t \cos(\omega_p t)\right\} \\
&= \sum_{t=1}^{N} \cos^2(\omega_p t)\mathrm{Var}(X_t) \qquad \text{(because } X_t \text{ are i.i.d.)} \\
&= \sigma_Z^2 \sum_{t=1}^{N} \cos^2(\omega_p t) = \frac{\sigma_Z^2 N}{2},
\end{aligned}$$

and

$$\text{Var}\left\{B(\omega_p)\right\} = \frac{\sigma_Z^2 N}{2},$$

for $p \neq \frac{N}{2}$.

Since A and B are linear combinations of Gaussian random variables, they are also Gaussian, and therefore

$$A(\omega_p) \sim \mathcal{N}\left(0, \frac{\sigma_Z^2 N}{2}\right) \qquad \text{and} \qquad B(\omega_p) \sim \mathcal{N}\left(0, \frac{\sigma_Z^2 N}{2}\right).$$

Standardising both variables gives,

$$\sqrt{\frac{2}{N\sigma_Z^2}} A(\omega_p) \sim \mathcal{N}(0,1) \qquad \text{and} \qquad \sqrt{\frac{2}{N\sigma_Z^2}} B(\omega_p) \sim \mathcal{N}(0,1).$$

We also have that A and B are independent since they are multivariate normal (this means that any linear combination of them has a univariate normal distribution) and

$$\begin{aligned}
\text{Cov}\left\{A(\omega_p), B(\omega_p)\right\} &= \text{Cov}\left\{\sum_{t=1}^{N} X_t \cos(\omega_p t), \sum_{t=1}^{N} X_t \sin(\omega_p t)\right\} \\
&= \sum_{t=1}^{N}\sum_{s=1}^{N} \cos(\omega_p t)\sin(\omega_p s)\text{Cov}(X_t, X_s) \\
&= \sigma_Z^2 \sum_{t=1}^{N} \cos(\omega_p t)\sin(\omega_p t) = 0,
\end{aligned}$$

using our usual trigonometric results. So, it follows that

$$\frac{2\pi}{\sigma_Z^2} I(\omega_p) = \frac{2}{N\sigma_Z^2}\left\{A^2(\omega_p) + B^2(\omega_p)\right\} \sim \mathcal{X}_2^2.$$

We can now readily calculate the mean and variance,

$$\mathbb{E}\left\{I(\omega_p)\right\} = \frac{\sigma_Z^2}{2\pi} \cdot 2 = \frac{\sigma_Z^2}{\pi} = f_X(\omega_p)$$

and

$$\text{Var}\left\{I(\omega_p)\right\} = \left(\frac{\sigma_Z^2}{2\pi}\right)^2 \cdot 4 = \frac{\sigma_Z^4}{\pi^2}.$$

So $I(\omega_p)$ is an unbiased estimator for $f_X(\omega_p)$, but NOT a consistent one (the variance does not tend to 0 as $N \to \infty$).

7.6 Consistent Estimation of the Spectral Density

First, let us understand why the periodogram is not a consistent estimator.

For this argument, recall that

$$I(\omega_p) = \frac{1}{\pi}\left[c_0 + 2\sum_{k=1}^{N-1} c_k \cos(\omega_p k)\right] \tag{7.17}$$

and note that, in (7.17),

$$c_k = \frac{1}{N}\sum_{t=1}^{N-k}(X_t - \bar{X})(X_{t+k} - \bar{X})$$

is included for $k = 0, 1, \ldots, N-1$.

We see that, as N gets larger, c_k will provide a better estimate of $\gamma(k)$ for any fixed k, but if k is also allowed to vary then this is not necessarily the case.

In particular, as k increases the estimation of $\gamma(k)$ by c_k becomes less reliable, as the number of data points c_k is based upon decreases as k increases. This effect has a significant impact on the quality of the periodogram as an estimator of the spectrum as there are too many c_k with k large (with respect to N) that contribute to the non-vanishing variance of $I(\omega_p)$.

For example,

$$c_{N-1} = \frac{1}{N}(X_1 - \bar{X})(X_N - \bar{X})$$

has only one summand.

We discuss two methods for reducing the variance of the periodogram without sacrificing the asymptotic unbiasedness to obtain consistent estimates.

7.6.1 Lag-Window Estimator

As we had argued before, we don't have consistent estimation, because c_k with large k are unreliable. In the lag window estimator these c_k are given less weight;

$$\hat{f}(\omega) = \frac{1}{\pi}\left\{\lambda_0 c_0 + 2\sum_{k=1}^{M}\lambda_k c_k \cos(\omega_p k)\right\}, \tag{7.18}$$

where $\{\lambda_k\}$ are the weights, and $M < N$ is the truncation point. The estimation problem, then, is to choose appropriate values for $\{\lambda_k\}$ and M. For the lag window, there have been several forms suggested, some given in Table (7.1). The Bartlett window is no longer commonly used, as the other two (Tukey and Parzen) have superior properties. Figure (7.1) shows how the weights are distributed according to the lag, for each of the lag windows of Table (7.1).

Lag Window	λ_k
Unit	$\lambda_k = 1,\ 0 \le k \le M$
Bartlett	$\lambda_k = 1 - k/M,\ 0 \le k \le M$
Tukey	$\lambda_k = \frac{1}{2}\left(1 + \cos\frac{\pi k}{M}\right),\ 0 \le k \le M$
Parzen	$\lambda_k = \begin{cases} 1 - 6\left(\frac{k}{M}\right)^2 + 6\left(\frac{k}{M}\right)^3 & 0 \le k \le M/2 \\ 2\left(1 - \frac{k}{M}\right)^3 & M/2 \le k \le M \end{cases}$

Table 7.1: Different lag-windows

The choice of the truncation point is a complicated matter. Choosing a small M may cause us to miss significant features. On the other hand, if we choose a large M, the estimator will act as the periodogram and show too much variability. A rule-of-thumb is to choose M to be approximately equal to $2\sqrt{N}$.

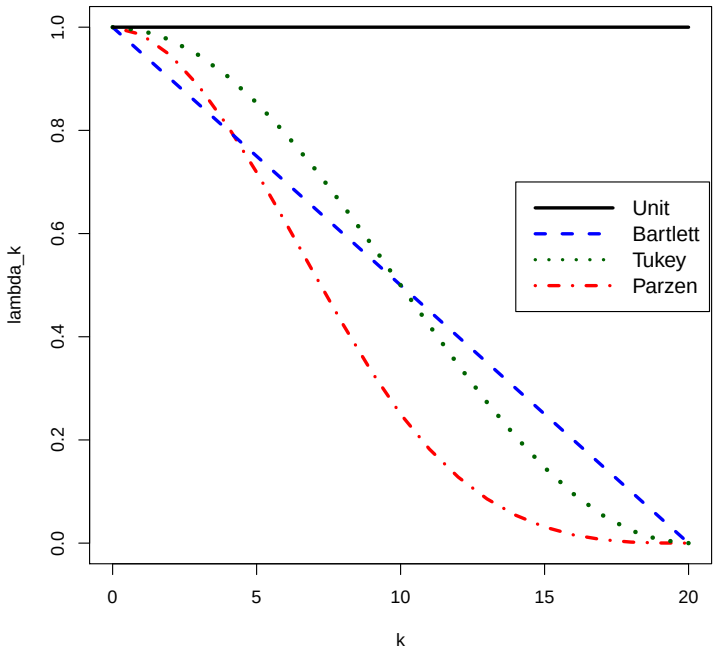


Figure 7.1: Different lag windows for $M = 20$.

7.6.2 Smoothing the Periodogram

The second method for obtaining consistent estimators for the spectral density function is to smooth the periodogram, by grouping the frequencies in sets. This works, under smoothness assumptions, because periodogram estimates are asymptotically unbiased and independent at the Fourier frequencies (not proved here).

The smoothed periodogram near ω is therefore given by

$$\hat{f}(\omega) = \sum_{j=-m}^{m} \mu_j I\left(\frac{2\pi(s+j)}{N}\right) \quad \text{for } \omega \in (0, \pi)$$

where $\{\mu_j\}$ are weights satisfying $\sum_{j=-m}^{m} \mu_j = 1$ and s is an integer such that $2\pi s/N$ is near ω.

Chapter 8

Bivariate Processes

Up until now our focus has been on understanding how a single time series evolves, treating it singly and in isolation. However, in practice you might often have data from more than one different, but related, process. This is common, for example, within meteorology where a number of different measurements (rain, temperature, wind speed etc) are made periodically at various sites across the country. In this chapter we will consider the situation in which we have data from two different processes and we are interested in the relationship between them. These kinds of processes are called bivariate, *bi* for two and *variate* for varying *i.e.* they vary in two ways.

Returning to our meteorology example then there are two distinct situations that might arise and the approach we take to understanding each of these is different. Say, for example, you were to have observations of rainfall from two different sites and one was upwind of the other. In this case we might think of one time series as an input and the other as an output, with the rainfall at the upwind site being a good predictor of the incoming weather front and thus what we would expect to see after a delay at the downwind site. This approach to modelling bivariate time series, in which we introduce the notion of a causal relationship between the two sets of data, will be discussed in chapter 9 and can be thought of as roughly analogous to regression in 'ordinary' statistics. Another situation could see us considering the relationship between measurements made on two different variables at the same site, for example rain and air temperature. In this case it is not necessarily so relevant to assume a directionality to the relationship, and so we might instead set the two series on an equal footing in order to study the interrelationship of the two series. This is the approach we will be concerned with in chapter 8 and can be thought of as roughly analogous to correlation in 'ordinary' statistics.

8.1 Time Domain Quantities

We consider the bivariate process $\{(X_t, Y_t)\}_{t\in\mathbb{Z}}$. An observed time series is regarded as a finite realisation of this process and is denoted by $(x_1, y_1), (x_2, y_2), \ldots, (x_N, y_N)$.

As in chapter 3, we describe the time series in terms of the first- and second-order moments. These include not only the mean, variance and autocovariance of each component process as before, but also a new function, called the cross-covariance function.

The **cross-covariance function** of the bivariate process $\{(X_t, Y_t)\}$, γ_{XY}, is defined as

$$\gamma_{XY}(t, \tau) = \text{Cov}\left(X_t, Y_{t+\tau}\right).$$

Henceforth, we will assume that $\{(X_t, Y_t)\}$ is stationary, meaning that the first and second order moments do not change over time, and the notation is simplified to:

$$\gamma_{XY}(\tau) = \text{Cov}\left(X_t, Y_{t+\tau}\right). \tag{8.1}$$

Notes:

- $\gamma_{XX}(\tau) = \text{Cov}\left(X_t, X_{t+\tau}\right)$ and $\gamma_{YY}(\tau) = \text{Cov}\left(Y_t, Y_{t+\tau}\right)$ give the auto-covariance functions of the component processes X_t and Y_t, respectively.
- If $\{(X_t, Y_t)\}$ is stationary, then

  $$\gamma_{XY}(-\tau) = \gamma_{YX}(\tau).$$

  Proof:

  $$\gamma_{XY}(-\tau) = \text{Cov}(X_t, Y_{t-\tau}) = \text{Cov}(X_{t+\tau}, Y_t) = \text{Cov}(Y_t, X_{t+\tau}) = \gamma_{YX}(\tau).$$

- In contrast with the auto-covariance function, the cross-covariance function is not even.

The **cross-correlation function** of the bivariate process $\{(X_t, Y_t)\}$, ρ_{XY}, is defined as

$$\rho_{XY}(\tau) = \frac{\gamma_{XY}(\tau)}{\sqrt{\gamma_{XX}(0)\gamma_{YY}(0)}}.$$

Notes:

- $\rho_{XY}(0)$ is not necessarily 1.
- $\rho_{XY}(-\tau) = \rho_{YX}(\tau)$
- $|\rho_{XY}(\tau)| \leq 1$ (see problem sheet 9, Q2).
- For cases where $\{X_t\}$ and $\{Y_t\}$ are measured on different scales, the cross-correlation function is more appropriate to use than the cross-covariance function.

Example 8.1. Find the cross-covariance and cross-correlation function of the bivariate process $\{(X_t, Y_t)\}$, where

$$X_t = Z_t \text{ and } Y_t = \frac{1}{2}Z_{t-1} + \frac{1}{2}Z_{t-2},$$

in which $\{Z_t\}$ is a purely random process with mean 0 and variance σ_Z^2.

First we calculate the cross-covariance function:

$$\begin{aligned}
\gamma_{XY}(\tau) &= \mathrm{Cov}(X_t, Y_{t+\tau}) \\
&= \mathrm{Cov}\left(Z_t, \frac{1}{2}Z_{t+\tau-1} + \frac{1}{2}Z_{t+\tau-2}\right) \\
&= \frac{1}{2}\mathrm{Cov}\left(Z_t, Z_{t+\tau-1}\right) + \frac{1}{2}\mathrm{Cov}\left(Z_t, Z_{t+\tau-2}\right) \\
&= \begin{cases} \frac{1}{2}\sigma_Z^2 & \text{for } \tau = 1, 2, \\ 0 & \text{otherwise.} \end{cases}
\end{aligned}$$

Meanwhile, $\gamma_{XX}(0) = \mathrm{Var}(X_t) = \sigma_Z^2$ and $\gamma_{YY}(0) = \mathrm{Var}(Y_t) = \frac{1}{2}\sigma_Z^2$.

The cross-correlation function is then

$$\rho_{XY}(\tau) = \frac{\gamma_{XY}(\tau)}{\sqrt{\gamma_{XX}(0)\gamma_{YY}(0)}} = \frac{\gamma_{XY}(\tau)}{\sqrt{\sigma_Z^2 \cdot \frac{1}{2}\sigma_Z^2}} = \begin{cases} \frac{1}{\sqrt{2}} & \tau = 1, 2, \\ 0 & \text{otherwise.} \end{cases}$$

To estimate the time domain quantities we follow the same arguments as those made for a univariate process in Section 2.6. Thus, we consider pairs of the form $(x_t, y_{t+\tau})$ from our observations.

The **sample cross-covariance function** is thus defined by

$$c_{XY}(\tau) = \begin{cases} \frac{1}{N}\sum_{t=1}^{N-\tau}(x_t - \bar{x})(y_{t+\tau} - \bar{y}) & \text{for } \tau = 0, 1, \ldots, N-1 \\ \frac{1}{N}\sum_{t=1-\tau}^{N}(x_t - \bar{x})(y_{t+\tau} - \bar{y}) & \text{for } \tau = -(N-1), \ldots, -2, -1, \end{cases}$$

and the sample cross-correlation function is defined by

$$r_{XY}(\tau) = \frac{c_{XY}(\tau)}{\sqrt{c_{XX}(0)c_{YY}(0)}}.$$

8.2 Frequency Domain Quantities

The basic quantity used to study a bivariate process in the frequency domain is the **cross-spectrum**. Remember our definition for the spectral density $f(\omega)$. Starting from (6.11), we have

$$f(\omega) = \frac{1}{\pi}\left\{\gamma(0) + 2\sum_{k=1}^{\infty}\gamma(k)\cos(\omega k)\right\} = \frac{1}{\pi}\sum_{k=-\infty}^{\infty}\gamma(k)e^{-i\omega k} \qquad \text{for } 0 < \omega < \pi.$$

The cross-spectrum is defined analogously:

Let $\{(X_t, Y_t)\}$ be a discrete time bivariate process with cross-covariance function γ_{XY}. Then, the **cross-spectrum** is defined as

$$f_{XY}(\omega) = \frac{1}{\pi}\sum_{k=-\infty}^{\infty}\gamma_{XY}(k)e^{-i\omega k}, \qquad 0 < \omega < \pi. \tag{8.2}$$

Note: The cross-spectrum is in general a complex-valued function since γ_{XY} is not in general even. This is usually decomposed into two real-valued components as follows:

The **co-spectrum**, $c(\omega)$, is defined as the real part of the cross-spectrum and is given by

$$c(\omega) = \mathrm{Re}\{f_{XY}(\omega)\} = \frac{1}{\pi}\sum_{k=-\infty}^{\infty}\gamma_{XY}(k)\cos(\omega k), \qquad 0 < \omega < \pi. \tag{8.3}$$

The **quadrature-spectrum**, $q(\omega)$, is defined as the negative of the imaginary part of the cross-spectrum and is given by

$$q(\omega) = -\,\mathrm{Im}\{f_{XY}(\omega)\} = \frac{1}{\pi}\sum_{k=-\infty}^{\infty}\gamma_{XY}(k)\sin(\omega k), \qquad 0 < \omega < \pi. \tag{8.4}$$

The cross-spectrum may then be written as

$$f_{XY}(\omega) = c(\omega) - iq(\omega).$$

As with any complex number, we can also write f_{XY} in polar form. We do so as follows:

The **cross-amplitude spectrum**, α_{XY}, is defined as

$$\alpha_{XY}(\omega) = \sqrt{c^2(\omega) + q^2(\omega)}, \quad 0 < \omega < \pi.$$

Meanwhile, the **phase spectrum**, ϕ_{XY}, satisfies

$$\tan\left(\phi_{XY}(\omega)\right) = -\frac{q(\omega)}{c(\omega)}, \quad 0 < \omega < \pi$$

and is usually constrained to lie within the range $(-\pi, \pi]$.

The cross-spectrum can then be written in the form:

$$f_{XY}(\omega) = \alpha_{XY}(\omega)e^{i\phi_{XY}(\omega)}.$$

A useful standardised version of the cross-spectrum is the coherency

$$C(\omega) = \frac{f_{XY}(\omega)}{\sqrt{f_X(\omega)f_Y(\omega)}}, \quad 0 < \omega < \pi,$$

and the coherence (also known as the squared coherency), $|C(\omega)|^2$.

Notes:

- $0 \leq |C(\omega)| \leq 1$
- The coherence measures the square of the linear correlation between the two components of the bivariate process at frequency ω. The closer $|C(\omega)|^2$ is to one the more closely related the processes are at frequency ω.
- The coherence is perhaps the most useful, and most readily understood, of the functions introduced in this section. The other functions are more readily understood in the context of linear systems so we will discuss their interpretation in chapter 9.

Example 8.2. Calculate the cross-amplitude, phase spectrum and coherence of the following bivariate process:

$$\begin{aligned} X_t &= Z_t \\ Y_t &= \frac{1}{2}Z_{t-1} + \frac{1}{2}Z_{t-2}, \end{aligned}$$

where $\{Z_t\}$ is a purely random process with mean 0 and variance σ_Z^2.

In section 8.1 we calculated the cross-covariance function for this process:

$$\gamma_{XY}(\tau) = \begin{cases} \frac{1}{2}\sigma_Z^2 & \tau = 1, 2, \\ 0 & \text{otherwise.} \end{cases}$$

Hence, we can calculate the cross-spectrum:

$$\begin{aligned} f_{XY}(\omega) &= \frac{1}{\pi}\sum_{k=-\infty}^{\infty}\gamma_{XY}(k)e^{-i\omega k} \\ &= \frac{1}{\pi}\left\{\frac{1}{2}\sigma_Z^2 e^{-i\omega} + \frac{1}{2}\sigma_Z^2 e^{-2i\omega}\right\} \\ &= \frac{\sigma_Z^2}{2\pi}\left\{\cos(\omega) - i\sin(\omega) + \cos(2\omega) - i\sin(2\omega)\right\} \\ &= \frac{\sigma_Z^2}{2\pi}\Big[\left\{\cos(\omega) + \cos(2\omega)\right\} - i\left\{\sin(\omega) + \sin(2\omega)\right\}\Big] \end{aligned}$$

So, the co-spectrum is given by

$$c(\omega) = \frac{\sigma_Z^2}{2\pi}\left\{\cos(\omega) + \cos(2\omega)\right\}$$

whilst the quadrature spectrum is

$$q(\omega) = \frac{\sigma_Z^2}{2\pi}\left\{\sin(\omega) + \sin(2\omega)\right\}.$$

The cross-amplitude is then

$$\begin{aligned}
\alpha_{XY}(\omega) &= \sqrt{\left[\frac{\sigma_Z^2}{2\pi}\left\{\cos(\omega)+\cos(2\omega)\right\}\right]^2+\left[\frac{\sigma_Z^2}{2\pi}\left\{\sin(\omega)+\sin(2\omega)\right\}\right]^2} \\
&= \frac{\sigma_Z^2}{2\pi}\sqrt{\cos^2(\omega)+2\cos(\omega)\cos(2\omega)+\cos^2(2\omega)+\sin^2(\omega)+2\sin(\omega)\sin(2\omega)+\sin^2(2\omega)} \\
&= \frac{\sigma_Z^2}{2\pi}\sqrt{2+2\cos(2\omega-\omega)} \\
&= \frac{\sigma_Z^2}{2\pi}\sqrt{2(1+\cos\omega)} \\
&= \frac{\sigma_Z^2}{\pi}\cos\left(\frac{\omega}{2}\right),
\end{aligned}$$

and the phase spectrum is

$$\tan\left\{\phi_{XY}(\omega)\right\} = -\frac{q(\omega)}{c(\omega)} = -\frac{\sin\omega+\sin(2\omega)}{\cos\omega+\cos(2\omega)}.$$

Finally, the coherence is given by

$$\begin{aligned}
|C(\omega)|^2 &= \left|\frac{f_{XY}(\omega)}{\sqrt{f_X(\omega)f_Y(\omega)}}\right|^2 \\
&= \left|\frac{\sigma_Z^2\cos(\omega/2)e^{i\phi_{XY}(\omega)}/\pi}{\sqrt{\left(\sigma_Z^2\cos^2(\omega/2)/\pi\right)\left(\sigma_Z^2/\pi\right)}}\right|^2 \\
&= \left|e^{i\phi_{XY}(\omega)}\right|^2 = 1,
\end{aligned}$$

in which we have used that

$$f_X(\omega) = \frac{\sigma_Z^2}{\pi}$$

as usual for a purely random process and

$$f_Y(\omega) = \frac{\sigma_Z^2(1+\cos(\omega))}{2\pi} = \frac{\sigma_Z^2\cos^2(\omega/2)}{\pi}$$

from the examples of the previous chapter.

This result may appear surprising at first, but it is plausible, because $\{X_t\}$ and $\{Y_t\}$ are both generated from the same white noise process $\{Z_t\}$.

To estimate the frequency domain quantities, we can use the weighted and truncated cross-covariance function, as we did in Section 7.6.1 in the univariate process case, by appropriately choosing the truncation point, M, and the lag-window weights $\{\lambda_k\}$.

The estimators of the co-spectrum and the quadrature spectrum are then given by

$$\begin{aligned}\hat{c}(\omega) &= \frac{1}{\pi}\sum_{k=-M}^{M}\lambda_k c_{XY}(k)\cos(\omega k)\\ &= \frac{1}{\pi}\left[\lambda_0 c_{XY}(0)+\sum_{k=1}^{M}\lambda_k\left\{c_{XY}(k)+c_{XY}(-k)\right\}\cos(\omega k)\right]\end{aligned}$$

and

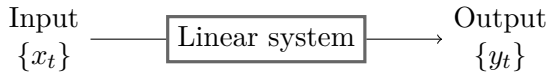
$$\begin{aligned}\hat{q}(\omega) &= \frac{1}{\pi}\sum_{k=-M}^{M}\lambda_k c_{XY}(k)\sin(\omega k)\\ &= \frac{1}{\pi}\left[\sum_{k=1}^{M}\lambda_k\left\{c_{XY}(k)-c_{XY}(-k)\right\}\sin(\omega k)\right].\end{aligned}$$

Chapter 9

Linear Systems

In the previous chapter we considered bivariate processes in which both components were put on an equal footing and we were interested their interrelationship. In this chapter we instead think of the first process as an input to a system and the second as an output thereby introducing a hierarchy. This is often relevant in physical situations where two time series are causally linked, for example daily rainfall and reservoir levels, with rainfall being thought of as the input and reservoir levels as the output. But studying linear systems is not only useful for examining the relationship between different time series, it's also useful for examining the properties of linear filtering procedures such as those introduced in chapter 2 for removing trend and seasonality. In this case we think of the original time series as our input and its filtered or detrended counterpart as the output.

For the purposes of this course we will confine our attention to time-invariant, linear systems that transform an input $\{x_t\}$ to an output $\{y_t\}$,



and satisfy the following conditions:

- A system is **linear** if the input $\left\{\lambda_1 x_t^{(1)} + \lambda_2 x_t^{(2)}\right\}$ gives rise to the output $\left\{\lambda_1 y_t^{(1)} + \lambda_2 y_t^{(2)}\right\}$, where $\left\{y_t^{(1)}\right\}$ and $\left\{y_t^{(2)}\right\}$ are the outputs for $\left\{x_t^{(1)}\right\}$ and $\left\{x_t^{(2)}\right\}$, respectively.
- A linear system is **time invariant** if a delay in the input, $\{x_{t-\tau}\}$, produces the same delay in the output, $\{y_{t-\tau}\}$.

Note: Time invariance means that the relationship between x_t and y_t does not change with time.

9.1 Linear Systems in the Time Domain

A system is time-invariant and linear if and only if it can be written in the form:

$$y_t = \sum_{k=-\infty}^{\infty} h_k x_{t-k}. \tag{9.1}$$

The function $\{h_k\}$, which maps k to h_k, is called the **impulse response function**.

Notes:

- The impulse response function h_k does not depend on time, ensuring that the systems are time invariant.
- If $h_k = 0$ for $k < 0$, the system is called *causal*, i.e. the output depends only on present and past values of the input.

Example 9.1. Consider the following moving average filter

$$y_t = \frac{1}{3}x_{t-1} + \frac{1}{3}x_t + \frac{1}{3}x_{t+1}.$$

The impulse response function is

$$h_k = \begin{cases} \frac{1}{3} & k = -1, 0, 1, \\ 0 & \text{otherwise.} \end{cases}$$

The system is not causal.

Example 9.2. Consider the difference equation

$$y_t = \frac{1}{2}y_{t-1} + x_t.$$

To determine h_k we first need to write the system in the form of (9.1).

$$\begin{aligned} y_t &= x_t + \frac{1}{2}y_{t-1} \\ &= x_t + \frac{1}{2}\left(x_{t-1} + \frac{1}{2}y_{t-2}\right) \\ &= x_t + \frac{1}{2}x_{t-1} + \frac{1}{4}x_{t-2} + \frac{1}{8}y_{t-3} \\ &= \dots \\ &= x_t + \frac{1}{2}x_{t-1} + \frac{1}{4}x_{t-2} + \frac{1}{8}x_{t-3} + \frac{1}{16}x_{t-4} + \dots \\ &= \sum_{k=0}^{\infty} \frac{1}{2^k} x_{t-k}. \end{aligned}$$

Hence the system is causal and its impulse response function is

$$h_k = \begin{cases} \left(\frac{1}{2}\right)^k & k = 0, 1, 2, \ldots, \\ 0 & \text{otherwise.} \end{cases}$$

Example 9.3. Consider the simple delay system

$$y_t = x_{t-d}, \qquad d \geq 0.$$

The system is causal with impulse response function

$$h_k = \begin{cases} 1 & k = d, \\ 0 & \text{otherwise.} \end{cases}$$

Example 9.4. Consider the simple gain system

$$y_t = g x_t.$$

In this case

$$h_k = \begin{cases} g & k = 0, \\ 0 & k \neq 0. \end{cases}$$

Why is it called an impulse response function?
The name comes from the effect of inputting a unit impulse at time 0, i.e. the series

$$x_t = \begin{cases} 1 & \text{if } t = 0, \\ 0 & \text{otherwise.} \end{cases}$$

The output is then

$$y_t = \sum_{k=-\infty}^{\infty} h_k x_{t-k} = h_t,$$

i.e. the output is the impulse response function.

9.2 Linear systems in the Frequency Domain

As we have seen from previous chapters, the quantities in the time domain have their equivalent quantities in the frequency domain, usually related through a Fourier transform. Following this reasoning we define the frequency response function.

The **frequency response function**, $H(\omega)$, of a stable linear system is defined as the Fourier transform of the impulse response function

$$H(\omega) = \sum_{k=-\infty}^{\infty} h_k e^{-i\omega k}, \quad \text{for } 0 < \omega < \pi. \tag{9.2}$$

Notes:

- Linear systems are called **stable** if they have a summable impulse response function, i.e. $\sum_{k=-\infty}^{\infty} |h_k| < \infty$.
- The frequency response function and impulse response function are equivalent ways of describing stable, time-invariant linear systems.
- We will see that when we deal with linear systems, the frequency response function, and the frequency quantities in general, could make things a lot simpler compared to the time domain quantities, where calculations are often really hard, or in some cases impossible.

9.2.1 Gain & Phase Diagrams

The frequency response function, $H(\omega)$, of a linear system is a complex function unless $h_k = h_{-k}$. It is often written in polar form as

$$H(\omega) = G(\omega)e^{i\phi(\omega)}, \tag{9.3}$$

where $G(\omega)$ and $\phi(\omega)$ are called the **gain** and **phase**, respectively. The gain is determined by the modulus of the frequency response function

$$G(\omega) = |H(\omega)|, \tag{9.4}$$

whilst the phase is determined by the argument of the frequency response function, which satisfies

$$\tan(\phi(\omega)) = \frac{\mathrm{Im}\{H(\omega)\}}{\mathrm{Re}\{H(\omega)\}}$$

and is usually constrained to lie within the range $(-\pi, \pi]$.

Why are they called gain and phase?
The names come from inputting a purely sinusoidal signal at frequency ω, i.e. the series

$$x_t = \cos(\omega t)$$

to a stable, time-invariant, linear system.

The output is then

$$\begin{aligned} y_t &= \sum_{k=-\infty}^{\infty} h_k \cos(\omega(t-k)) \\ &= \text{Re}\left\{ \sum_{k=-\infty}^{\infty} h_k e^{i\omega(t-k)} \right\} \\ &= \text{Re}\left\{ e^{i\omega t} H(\omega) \right\} \\ &= \text{Re}\left\{ e^{i\omega t} G(\omega) e^{i\phi(\omega)} \right\} \\ &= G(\omega) \cos\left(\omega t + \phi(\omega)\right). \end{aligned}$$

Similarly, the input

$$x_t = \sin(\omega t)$$

corresponds to the output

$$y_t = G(\omega) \sin\left(\omega t + \phi(\omega)\right).$$

So an input at frequency ω is amplified by a factor of $G(\omega)$, called the gain, and translated by an angle $\phi(\omega)$, called the phase.

Gain and phase diagrams give us information about the frequency properties of the system:

- **Gain Diagram** - A plot of $G(\omega)$ against ω is called a gain diagram. If G is relatively large in the low frequencies then h is called a low-pass filter since it disproportionately amplifies low frequencies. Similarly, when G is larger at the higher frequencies, the system is considered to be a high-pass filter since it disproportionately amplifies high frequencies.
- **Phase Diagram** - A plot of $\phi(\omega)$ against ω is called a phase diagram and demonstrates the phase shift. The phase diagram, unlike its counterpart, is not always easy to interpret (or to plot).

9.3 Design of Linear Filters

Gain diagrams are useful not only in understanding the frequency properties of a linear system, but also in designing linear filters. The filters we considered in chapter 2 constitute linear systems since they are of the form

$$y_t = \sum_{k=-q}^{s} a_k x_{t-k}$$

and the impulse response function is thus given by

$$h_k = \begin{cases} a_k & \text{for } k = -q, ..., s \\ 0 & \text{otherwise.} \end{cases}$$

We used these filters to either estimate or eliminate trends from time series data. The design of a filter involves choosing the weightings, and hence h_k, to achieve the desired effect.

When we are estimating trends our goal is to filter out the high frequency oscillations, or 'noise', to leave only low frequency oscillations indicative of longer term, systematic features. Thus we want a low-pass filter. Conversely, when we are trying to eliminate a trend (to detrend the data) then we want to remove the low frequency variations indicative of longer term, systematic features leaving only the high frequency residuals. Thus we want a high-pass filter. Two types of 'ideal' filters, aimed at achieving these goals are shown in figure 9.1.

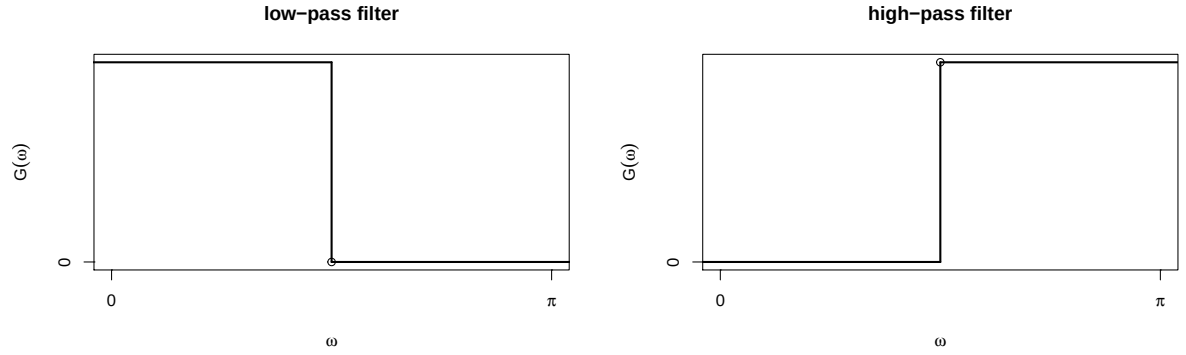


Figure 9.1: Two types of ideal filter: (a) a low-pass filter or trend estimator, (b) a high-pass filter or trend eliminator.

Both have sharp cut-offs, the low-pass filter completely eliminating high-frequency variation and the high-pass filter completely eliminating low-frequency variation. In reality 'ideal' filters of this type are impossible to achieve with a finite set of weightings, but we can try to design filters that approximate them.

Trend Estimation

Moving average filters are an example of trend estimators, since they are smoothing filters designed to 'smooth' out high frequency oscillating leaving only low frequency oscillations indicative of long term variation. An example is the simple moving average filter from example 9.1,

$$y_t = \frac{1}{3}x_{t-1} + \frac{1}{3}x_t + \frac{1}{3}x_{t+1}.$$

The impulse response function is

$$h_k = \begin{cases} \frac{1}{3} & k = -1, 0, 1, \\ 0 & \text{otherwise.} \end{cases}$$

The frequency response function is then

$$\begin{aligned} H(\omega) &= \sum_{k=-\infty}^{\infty} h_k e^{-i\omega k} \\ &= \frac{1}{3}e^{-i\omega} + \frac{1}{3}e^{0} + \frac{1}{3}e^{i\omega} \\ &= \frac{1}{3} + \frac{2}{3}\cos\omega, \qquad \text{for } 0 < \omega < \pi. \end{aligned}$$

This means that the frequency response function is real and

$$H(\omega) > 0 \qquad \text{for} \quad 0 < \omega < \frac{2\pi}{3} \qquad \text{whilst}$$

$$H(\omega) < 0 \qquad \text{for} \quad \frac{2\pi}{3} < \omega < \pi.$$

Therefore, the gain is given by

$$G(\omega) = |H(\omega)| = \left|\frac{1}{3} + \frac{2}{3}\cos\omega\right| = \begin{cases} \frac{1}{3} + \frac{2}{3}\cos\omega & 0 < \omega < \frac{2\pi}{3}, \\ -\frac{1}{3} - \frac{2}{3}\cos\omega & \frac{2\pi}{3} < \omega < \pi. \end{cases}$$

Meanwhile, for the phase, we have

$$\tan\{\phi(\omega)\} = \frac{\text{Im}\{H(\omega)\}}{\text{Re}\{H(\omega)\}} = \frac{0}{\frac{1}{3} + \frac{2}{3}\cos\omega} = 0,$$

implying that $\phi(\omega) = 0$ or $\phi(\omega) = \pi$ (assuming that $\phi(\omega) \in (-\pi, \pi]$). To figure out which, note that $H(\omega) = G(\omega)e^{i\phi(\omega)}$, and therefore to recover the original sign of $H(\omega)$ we must have

$$\phi(\omega) = \begin{cases} 0 & 0 < \omega < \frac{2\pi}{3}, \\ \pi & \frac{2\pi}{3} < \omega < \pi. \end{cases}$$

The gain and phase diagrams for this system are shown in Figure 9.2.

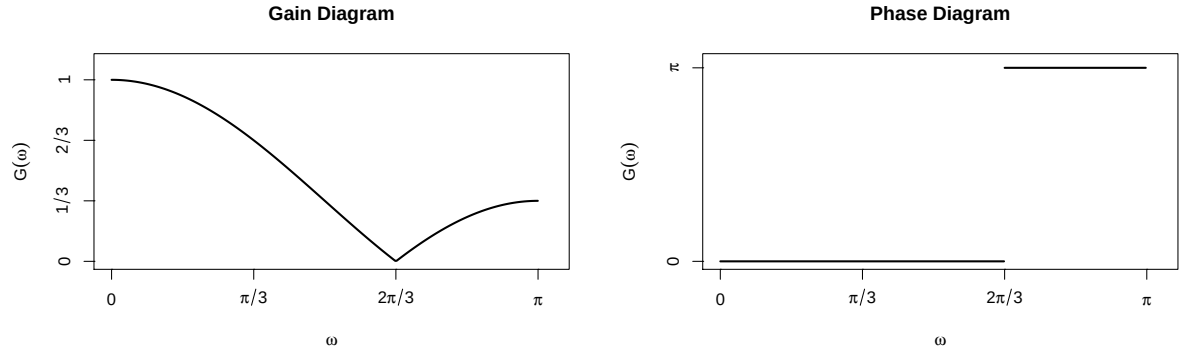


Figure 9.2: Gain and phase diagrams for $y_t = \frac{1}{3}x_{t-1} + \frac{1}{3}x_t + \frac{1}{3}x_{t+1}$.

So, the gain diagram of a simple moving average of three successive observations is of low-pass type, but has a much less sharp cut-off than the ideal low-pass filter shown in figure 9.1.

More generally, for a symmetric moving average filter,

$$y_t = \sum_{k=-q}^{q} a_k x_{t-k}$$

in which $a_{-k} = a_k$, then we have

$$h_k = \begin{cases} a_k & \text{for } |k| = 0, ..., q \\ 0 & \text{otherwise} \end{cases}$$

and so

$$G(\omega) = |H(\omega)| = \left| \sum_{k=-q}^{q} a_k e^{-i\omega k} \right| = \left| a_0 + 2 \sum_{k=1}^{q} a_k \cos(\omega k) \right|.$$

The properties of the filter improve as you include more terms even for equal weightings and with a more sophisticated choice of weights, such as Spencer's 15-point moving average, then the filter has much better cut-off properties as shown in figure 9.3.

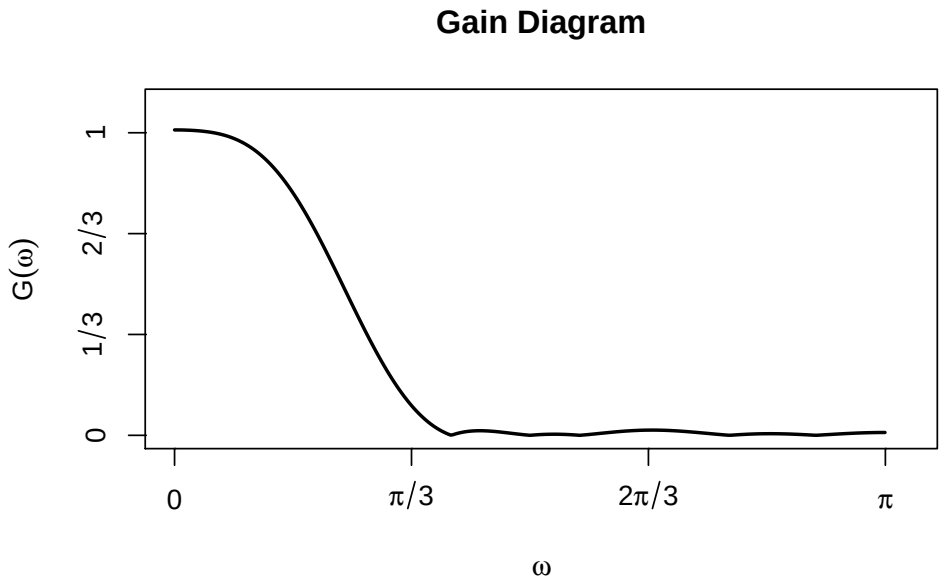


Figure 9.3: Gain diagram for Spencer's 15-point moving average.

Trend Elimination

To remove a trend using a linear filter we usually employ lag 1 differencing. For example, the first order differencing given by

$$y_t = x_t - x_{t-1},$$

which corresponds to the impulse response function

$$h_k = \begin{cases} 1 & \text{if } k = 0 \\ -1 & \text{if } k = 1 \\ 0 & \text{otherwise.} \end{cases}$$

This corresponds to the impulse response function

$$H(\omega) = 1 - e^{-i\omega}$$

and gain

$$G(\omega) = \sqrt{2(1 - \cos\omega)} = 2\sin(\omega/2),$$

which is plotted in figure 9.4.

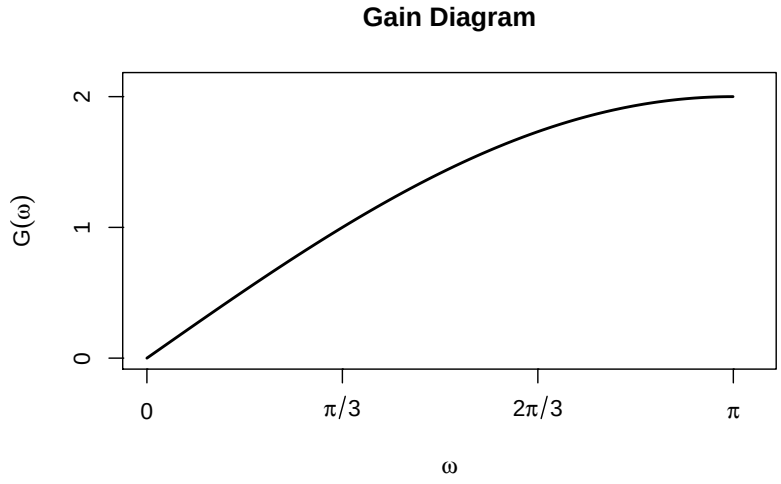


Figure 9.4: Gain diagram for lag 1 differencing of order 1.

This is indeed of high-pass type, as required, but the shape is a long way from the ideal filter in figure 9.1. We can improve this by increasing the order of the differencing. If we apply lag 1 differencing n times then the overall gain function is given by the product of the individual gains (as we will see in section 9.5) and we find that for the linear filter

$$y_t = \nabla^n x_t$$

then the gain is given by

$$G(\omega) = 2^n \sin^n(\omega/2),$$

which is plotted in figure 9.5 for $n = 1, 2, 3$ and shows some improvement.

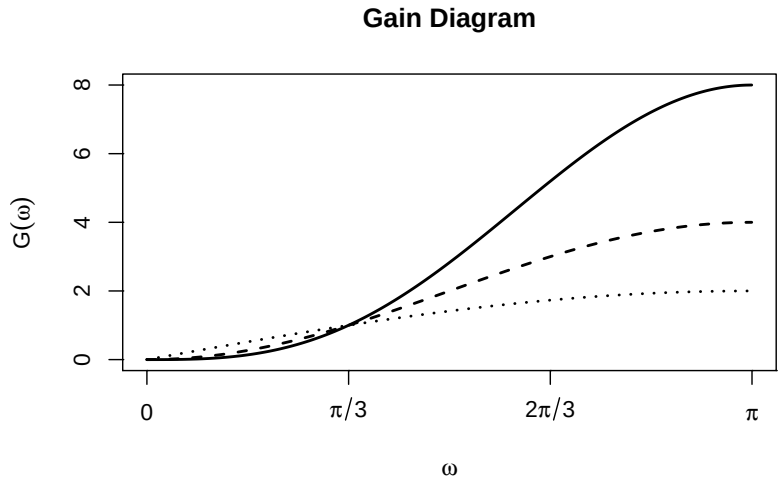


Figure 9.5: Gain diagram for differencing of orders 1, 2, 3 (dotted, dashed and solid lines respectively).

9.4 General Relationship Between Input and Output

In this section we will show that, for a stable time-invariant linear system, working in the frequency domain can be much more convenient than working in the time domain. This claim is supported by the following result, which provides an alternative (and much easier) way to calculate the spectrum of an output process, without using its auto-covariance function.

Let $\{X_t\}$ and $\{Y_t\}$ be the input and the output of a stable, time-invariant linear system respectively, with gain $G(\omega)$. Suppose also that the input process $\{X_t\}$ is stationary, with spectral density function $f_X(\omega)$. Then, the output process $\{Y_t\}$ is also stationary with spectral density function

$$f_Y(\omega) = G^2(\omega) f_X(\omega). \tag{9.5}$$

Proof: Since we are dealing with a stable, time-invariant linear system we can write

$$Y_t = \sum_{k=-\infty}^{\infty} h_k X_{t-k}.$$

where the impulse response function $\{h_k\}$ is summable, i.e. $\sum |h_k| < \infty$.

We first show that $\{Y_t\}$ is stationary. We have the following three results:

(i) the mean is constant and finite,

$$\mathbb{E}(Y_t) = \sum_{k=-\infty}^{\infty} h_k \mathbb{E}(X_{t-k}) = \mu_X \sum_{k=-\infty}^{\infty} h_k < \infty,$$

(ii) the autocovariance depends only on lag, τ,

$$\begin{aligned} \mathrm{Cov}(Y_t, Y_{t+\tau}) &= \sum_{j,k=-\infty}^{\infty} h_j h_k \mathrm{Cov}(X_{t-j}, X_{t+\tau-k}) \\ &= \sum_{j,k=-\infty}^{\infty} h_j h_k \gamma_X(\tau - k + j), \end{aligned}$$

(iii) the variance is constant and finite,

$$\begin{aligned} \mathrm{Var}(Y_t) &= \mathrm{Cov}(Y_t, Y_{t+0}) \\ &= \sum_{j,k=-\infty}^{\infty} h_j h_k \gamma_X(j-k) \\ &\leq \sum_{j,k=-\infty}^{\infty} |h_j||h_k| \gamma_X(0) \\ &< \infty, \end{aligned}$$

and so the process $\{Y_t\}$ is stationary.

Now, from (ii) and the definition of $f_Y(\omega)$:

$$\begin{aligned}
f_Y(\omega) &= \frac{1}{\pi}\sum_{\tau=-\infty}^{\infty}\gamma_Y(\tau)e^{-i\tau\omega} \\
&= \frac{1}{\pi}\sum_{j,k,\tau=-\infty}^{\infty}\gamma_X(\tau-k+j)h_jh_ke^{-i\tau\omega} \\
&= \sum_{j,k=-\infty}^{\infty}h_jh_k\frac{1}{\pi}\sum_{\ell=-\infty}^{\infty}\gamma_X(\ell)e^{-i(\ell+k-j)\omega} \\
&= \left(\sum_{j=-\infty}^{\infty}h_je^{ij\omega}\right)\cdot\left(\sum_{k=-\infty}^{\infty}h_ke^{-ik\omega}\right)\cdot\left(\frac{1}{\pi}\sum_{\ell=-\infty}^{\infty}\gamma_X(\ell)e^{-i\ell\omega}\right) \\
&= \overline{H(\omega)}\cdot H(\omega)\cdot f_X(\omega) \\
&= G^2(\omega)f_X(\omega)
\end{aligned}$$

since $G(\omega) = |H(\omega)|$.

This relationship between the spectra of the input and the output of a linear system, as shown in (9.5), is remarkably simple in its form, but very important in its application. The corresponding result in the time domain, which relates the autocovariances of the input and the output of a linear system, is given from (ii) by

$$\gamma_Y(\tau) = \sum_{j,k=-\infty}^{\infty}h_jh_k\gamma_X(\tau-k+j),$$

which is much more complicated. In many cases the summations are very hard or even impossible to compute.

There are many different uses for (9.5), in particular it can be used as an alternative method to find the spectral density function of a stationary process. This method is usually less complicated than working with the time domain quantities as we will now see.

9.4.1 Moving Average Processes

Consider the MA(q) process defined by

$$X_t = \beta_0 Z_t + \beta_1 Z_{t-1} + \ldots \beta_q Z_{t-q},$$

in which $\{Z_t\}$ is a purely random process with mean 0 and variance σ_Z^2.

The MA(q) process can be regarded a linear system whose input is $\{Z_t\}$ and output is $\{X_t\}$, since

$$X_t = \sum_{k=0}^{q}\beta_k Z_{t-k}.$$

The impulse response function is thus

$$h_k = \begin{cases}\beta_k & k = 0,\ldots,q, \\ 0 & \text{otherwise,}\end{cases}$$

and the frequency response function is

$$H(\omega) = \sum_{k=-\infty}^{\infty} h_k e^{-i\omega k} = \sum_{k=0}^{q} \beta_k e^{-i\omega k}.$$

The gain is the modulus of $H(\omega)$, so

$$G(\omega) = |H(\omega)| = \left| \sum_{k=0}^{q} \beta_k e^{-i\omega k} \right| \tag{9.6}$$

and hence,

$$f_X(\omega) = G^2(\omega) \cdot f_Z(\omega) = \left| \sum_{k=0}^{q} \beta_k e^{-i\omega k} \right|^2 \cdot \frac{\sigma_Z^2}{\pi}.$$

For the MA(1) process

$$X_t = Z_t + \beta Z_{t-1}$$

$$\begin{aligned} G^2(\omega) &= \left|1 + \beta e^{-i\omega}\right|^2 \\ &= \left(1 + \beta e^{-i\omega}\right)\left(1 + \beta e^{i\omega}\right) \\ &= 1 + 2\beta \cos\omega + \beta^2 \end{aligned}$$

and so we find

$$f_X(\omega) = \left(1 + 2\beta \cos\omega + \beta^2\right) \frac{\sigma_Z^2}{\pi}$$

as in chapter 6.

9.4.2 Autoregressive Processes

Consider the AR(p) process defined by

$$X_t = \alpha_1 X_{t-1} + \ldots + \alpha_p X_{t-p} + Z_t$$

in which $\{Z_t\}$ is a purely random process with mean 0 and variance σ_Z^2. Then we can rewrite this as

$$Z_t = \sum_{k=-\infty}^{\infty} h_k X_{t-k}$$

where

$$h_k = \begin{cases} 1 & \text{for } k = 0 \\ -\alpha_k & \text{for } k = 1, \ldots, p \\ 0 & \text{otherwise} \end{cases} \tag{9.7}$$

and regard $\{Z_t\}$ as the output of a linear system with input $\{X_t\}$ and impulse response function $\{h_k\}$. The spectral densities of $\{Z_t\}$ and $\{X_t\}$ are thus related through

$$f_Z(\omega) = G^2(\omega) f_X(\omega),$$

where

$$H(\omega) = \sum_{k=-\infty}^{\infty} h_k e^{-i\omega k} = 1 - \sum_{k=1}^{p} \alpha_k e^{-i\omega k}.$$

Thus

$$G^2(\omega) = \left|1 - \sum_{k=1}^{p} \alpha_k e^{-i\omega k}\right|^2$$

and we find that the spectral density of $\{X_t\}$ is given by

$$f_X(\omega) = \frac{1}{\left|1 - \sum_{k=1}^{p} \alpha_k e^{-i\omega k}\right|^2} \cdot \frac{\sigma_Z^2}{\pi}.$$

For the AR(1) process

$$X_t = \alpha X_{t-1} + Z_t$$

this gives

$$f_X(\omega) = \frac{1}{|1 - \alpha e^{-i\omega}|^2} \cdot \frac{\sigma_Z^2}{\pi} = \frac{\sigma_Z^2}{\pi\left(1 - 2\alpha\cos\omega + \alpha^2\right)}.$$

9.4.3 Mixed Autoregressive Moving Average Processes

Consider the ARMA(p, q) process defined by

$$X_t = \sum_{j=1}^{p} \alpha_j X_{t-j} + Z_t + \sum_{i=1}^{q} \beta_i Z_{t-i}$$

where $\{Z_t\}$ is a purely random process with mean 0 and variance σ_Z^2.

We can rewrite this as

$$W_t = \sum_{k=-\infty}^{\infty} h_k X_{t-k}$$

where

$$W_t = \sum_{i=0}^{q} \beta_i Z_{t-i}$$

is an MA(q) process and $\{h_k\}$ is as in (9.7).

Thus $\{W_t\}$ is the output of a linear system with impulse response function $\{h_k\}$ and input $\{X_t\}$.

Again, using the theorem we have that

$$f_W(\omega) = G^2(\omega) f_X(\omega)$$

and so

$$f_W(\omega) = \left|1 - \sum_{k=1}^{p} \alpha_k e^{-i\omega k}\right|^2 f_X(\omega)$$

as in section 9.4.2. Meanwhile, we also know that the spectrum of an MA(q) process is given by

$$f_W(\omega) = \left|1 + \sum_{j=1}^{q} \beta_j e^{-i\omega j}\right|^2 \frac{\sigma_Z^2}{\pi}$$

and so combining these two results we find that

$$f_X(\omega) = \frac{\left|1 + \sum_{j=1}^{q} \beta_j e^{-i\omega j}\right|^2}{\left|1 - \sum_{k=1}^{p} \alpha_k e^{-i\omega k}\right|^2} \cdot \frac{\sigma_Z^2}{\pi}.$$

Recall that in operator notation we had

$$\phi_\alpha(B)X_t = \theta_\beta(B)Z_t,$$

where

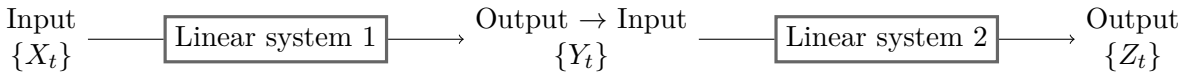
$$\phi_\alpha(z) = 1 - \sum_{k=1}^{p} \alpha_k z^k \qquad \text{and} \qquad \theta_\beta(z) = 1 + \sum_{j=1}^{q} \beta_j z^j.$$

With this notation, we have

$$f_X(\omega) = \frac{\left|\theta_\beta\left(e^{-i\omega}\right)\right|^2}{\left|\phi_\alpha\left(e^{-i\omega}\right)\right|^2} \cdot \frac{\sigma_Z^2}{2\pi}.$$

9.5 Cascades of Linear Systems

Frequency domain analysis is particularly useful when considering the effect of applying a number of linear systems in series, for example if we take $\{X_t\}$ to be the input of one linear system with output $\{Y_t\}$ and then take $\{Y_t\}$ in turn to be the input of a second linear system with output $\{Z_t\}$ (as shown below).



It is often of interest to evaluate the properties of the overall combined system, with input $\{X_t\}$ and output $\{Z_t\}$. In the time domain then the relationship between $\{X_t\}$ and $\{Z_t\}$ takes the form of a double sum,

$$\begin{aligned} Z_t &= \sum_{i=-\infty}^{\infty} h_i^{(2)} Y_{t-i} \\ &= \sum_{i=-\infty}^{\infty} \sum_{j=-\infty}^{\infty} h_j^{(1)} h_i^{(2)} X_{t-i-j} \\ &= \sum_{k=-\infty}^{\infty} \left[\sum_{j=-\infty}^{\infty} h_j^{(1)} h_{k-j}^{(2)} \right] X_{t-k} \qquad \text{using the change of variables } k = i + j, \end{aligned}$$

in which $\{h_j^{(1)}\}$ and $\{h_i^{(2)}\}$ are the impulse response functions associated with linear system 1 and 2 respectively and the combined impulse response function is given by

$$h_k = \sum_{j=-\infty}^{\infty} h_j^{(1)} h_{k-j}^{(2)}.$$

This is called a convolution and is rather complicated even for a simple cascade of two linear systems. It turns out that in the frequency domain the combined effect of applying multiple linear systems reduces to simple multiplication.

Suppose we have two stable, time-invariant linear systems in series, with respective frequency response functions $H_1(\omega)$ and $H_2(\omega)$, then the overall frequency response function associated with the combined system is given by

$$H(\omega) = H_1(\omega)H_2(\omega).$$

Notes:

- The combined frequency response function is given by the Fourier transform of the combined impulse response function (non-examinable):

$$\begin{aligned}
H(\omega) &= \sum_{k=-\infty}^{\infty} h_k e^{-i\omega k} \\
&= \sum_{k=-\infty}^{\infty} \sum_{j=-\infty}^{\infty} h_j^{(1)} h_{k-j}^{(2)} e^{-i\omega k} \\
&= \sum_{k=-\infty}^{\infty} \sum_{j=-\infty}^{\infty} h_j^{(1)} e^{-i\omega j} h_{k-j}^{(2)} e^{-i\omega(k-j)} \\
&= \sum_{j=-\infty}^{\infty} h_j^{(1)} e^{-i\omega j} \sum_{l=-\infty}^{\infty} h_l^{(2)} e^{-i\omega l} \qquad \text{using the change of variables } l = k - j \\
&= H_1(\omega)H_2(\omega).
\end{aligned}$$

- If the individual frequency response functions are written in polar coordinates as

$$H_1(\omega) = G_1(\omega)e^{i\phi_1(\omega)} \qquad \text{and} \qquad H_2(\omega) = G_2(\omega)e^{i\phi_2(\omega)}$$

  then the combined frequency response function is given by

$$H(\omega) = H_1(\omega)H_2(\omega) = G_1(\omega)G_2(\omega)e^{i(\phi_1(\omega)+\phi_2(\omega))}.$$

  So the overall gain is the **product** of the component gains and the overall phase is the **sum** of the component phases.

- The above result may be immediately applied to the general relationship between input and output to give

$$f_Z(\omega) = G_1^2(\omega)G_2^2(\omega)f_X(\omega).$$

- This can be easily extended to the situation where you have m linear systems in series, with component frequency response functions given by

$$H_j(\omega) = G_j(\omega)e^{i\phi_j(\omega)} \qquad \text{for } j = 1, ..., m.$$

  The combined frequency response function is then

$$H(\omega) = H_1(\omega)H_2(\omega)\dots H_m(\omega) = G_1(\omega)G_2(\omega)\dots G_m(\omega)e^{i(\phi_1(\omega)+\phi_2(\omega)+\cdots+\phi_m(\omega))}.$$

Chapter 10

Modelling Changes in Variance

So far our main tool for modelling stationary time series has been to use ARMA(p, q) models,

$$X_t = \alpha_1 X_{t-1} + \cdots + \alpha_p X_{t-p} + Z_t + \beta_1 Z_{t-1} + \ldots \beta_q Z_{t-q},$$

where $\{Z_t\}$ is a purely random process with mean 0 and variance σ_Z^2. Thus, we assume that what we observe now may be described as a (linear) combination of out past observations, X_{t-1}, ..., X_{t-p}, plus some shocks/error terms, Z_t, ..., Z_{t-q}. Such models are widely applicable and can be used to facilitate forecasting, a process that is closely related to the conditional expectation. Take, for example, the AR(1) model

$$X_t = \alpha X_{t-1} + Z_t$$

where $|\alpha| < 1$. In this case then the unconditional mean is given by $\mathbb{E}(X_t) = 0$, which is naturally independent of time as the process is stationary. However, we don't forecast based on the unconditional mean — this would mean forecasting all future values to be equal to the global mean regardless of any information we had about the local level — instead we use the conditional mean,

$$\mathbb{E}\left(X_t | X_{t-1} = x_{t-1}\right) = \mathbb{E}\left(\alpha X_{t-1} + Z_t | X_{t-1} = x_{t-1}\right) = \alpha x_{t-1}.$$

This depends on time and allows us to forecast what will happen next, X_{N+1}, based on what is happening now, x_N. Thus, we update our forecast based on the local level of the series.

Another important feature of X_{N+1} is its variability, since this decides our prediction interval and allows us to assess our uncertainty. Returning to our example, then the unconditional variance is naturally constant due to stationarity, but what about the conditional variance? This is given by

$$\begin{aligned}
\mathrm{Var}(X_t | X_{t-1} = x_{t-1}) &= \mathbb{E}(X_t^2 | X_{t-1} = x_{t-1}) - \left[\mathbb{E}(X_t | X_{t-1} = x_{t-1})\right]^2 \\
&= \mathbb{E}(\alpha^2 X_{t-1}^2 + 2\alpha X_{t-1} Z_t + Z_t^2 | X_{t-1} = x_{t-1}) - \alpha^2 x_{t-1}^2 \\
&= \alpha^2 x_{t-1}^2 + 2\alpha x_{t-1} \mathbb{E}(Z_t) + \mathbb{E}(Z_t^2) - \alpha^2 x_{t-1}^2 \\
&= \sigma_Z^2,
\end{aligned}$$

in which we have used causality of $\{X_t\}$ to say that Z_t is independent of X_{t-k} for $k > 0$. So, the conditional variance is also constant, and our assumption is thus that local variation in $\{X_t\}$ is constant. This is not always reasonable. In practice we often see that uncertainty today can lead to uncertainty tomorrow, and very variable observations can tend to cluster. This is particularly true for financial time series.

Changes in variation in a time series are referred to as its heteroskedasticity, from 'hetero' (which means different) and 'skedasis' (which means scattering or dispersion) so that the word literally means different amounts of scattering. Modelling these changes in variance can lead to better estimation of the local variation, which in turn allows us to compute more reliable prediction intervals. This allows us to make a better assessment of our uncertainty and the associated risk.

One way to obtain an appropriate model for data with changing local variation is therefore to reconsider the model used for $\{Z_t\}$ so that it satisfies the following two properties:

(i) the unconditional variance remains constant (so that the model is still stationary).

(ii) the conditional variance changes with time (so that the model is conditionally heteroskedastic).

10.1 An Alternative Model for $\{Z_t\}$

Instead of assuming $\{Z_t\}$ is purely random, we specify the following model,

$$Z_t = h_t V_t \tag{10.1}$$

where $\{V_t\}$ is a purely random process with mean 0 and variance $\sigma_V^2 = 1$. The term h_t thus scales the variability associated with the process and is usually referred to as the **volatility**.

But how do we specify h_t so that our model satisfies the properties given at the end of the previous section?

10.1.1 An ARCH(1) Model

One option is to use a model for which the conditional variance is of an AR(1) form so that knowledge of Z_{t-1} allows us to infer something about the variability of Z_t, that is

$$\text{Var}(Z_t|Z_{t-1}) = \nu + \phi_1 Z_{t-1}^2.$$

It turns out that the volatility given by

$$h_t^2 = \nu + \phi_1 Z_{t-1}^2, \tag{10.2}$$

in which $\nu > 0$ and $0 \leq \phi_1 < 1$, does exactly this whilst also maintaining stationarity as we will see.

A model for $\{Z_t\}$ that satisfies (10.1) and (10.2) is called an **autoregressive conditionally heteroskedastic model of order 1**, abbreviated as ARCH(1).

First we check that the model satisfies property (i), so that it is still stationary. The unconditional expectation and variance are given by

$$\mathbb{E}(Z_t) = \mathbb{E}(h_t V_t) = \mathbb{E}(\sqrt{\nu + \phi_1 Z_{t-1}^2})\mathbb{E}(V_t) = 0$$

and

$$\text{Var}(Z_t) = \mathbb{E}(Z_t^2) = \mathbb{E}(h_t^2 V_t^2) = \mathbb{E}(\nu + \phi_1 Z_{t-1}^2)\mathbb{E}(V_t^2) = \nu + \phi_1 \mathbb{E}(Z_{t-1}^2).$$

So we have

$$\mathbb{E}(Z_t^2) = \nu + \phi_1 \mathbb{E}(Z_{t-1}^2) = \nu + \phi_1(\nu + \phi_1 \mathbb{E}(Z_{t-2}^2)) = \cdots = \nu \sum_{n=0}^{\infty} \phi_1^n$$

and hence

$$\mathrm{Var}(Z_t) = \frac{\nu}{1 - \phi_1}$$

provided $|\phi_1| < 1$. So the unconditional mean and variance are constant, as required. It can also be shown that the $\mathrm{Cov}(Z_t, Z_{t-k}) = 0$ for all $k \neq 0$ and so only depends on lag. Hence the ARCH(1) process is stationary for $|\phi_1| < 1$ and the autocorrelation function behaves much like that of white noise, as shown in figure 10.1.

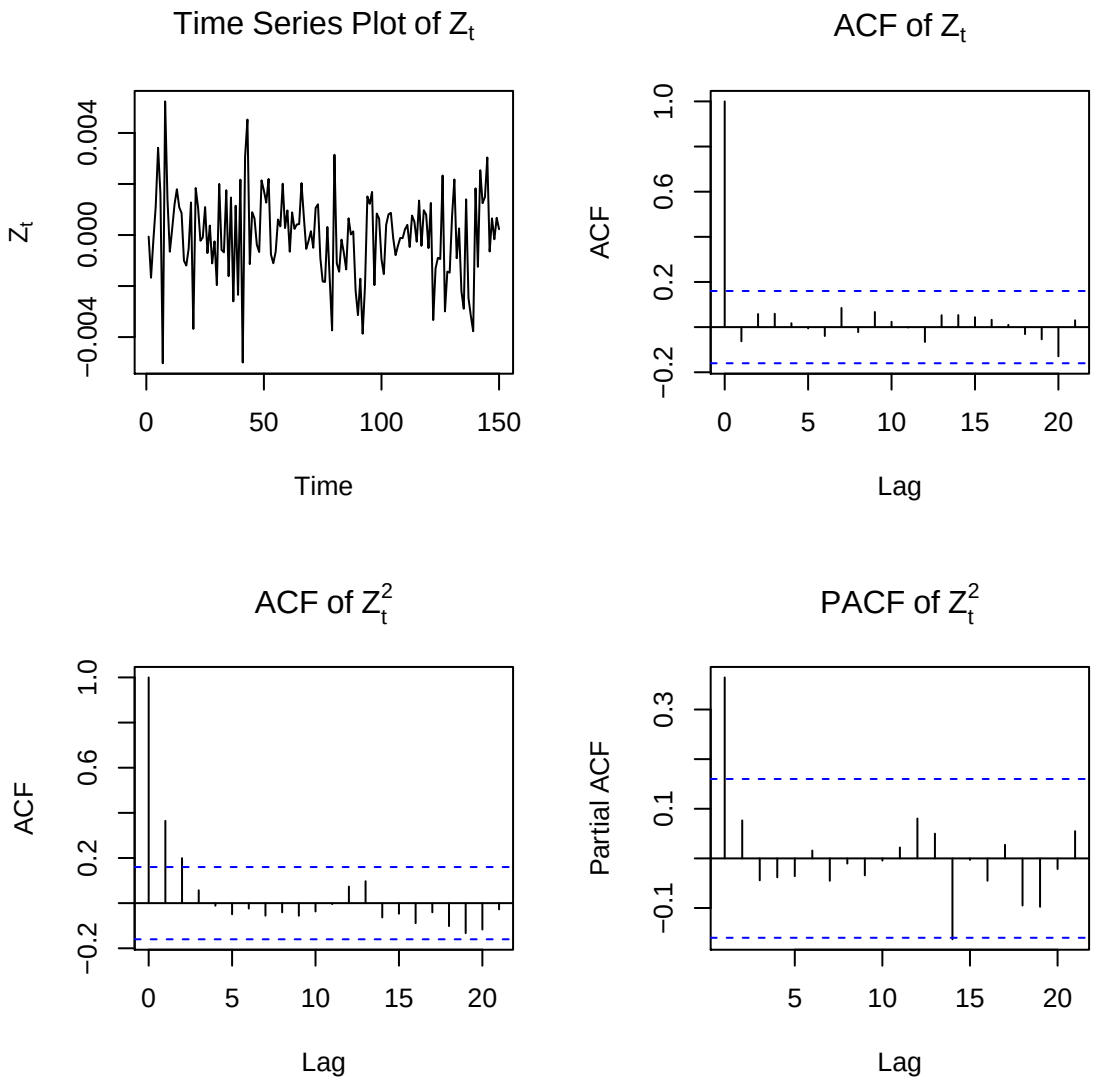


Figure 10.1: Simulated ARCH(1) data.

Next we check that our model satisfies property (ii), leading to heteroskedastic behaviour. The conditional expectation and variance are given by

$$\mathbb{E}(Z_t|Z_{t-1}) = \mathbb{E}(h_t V_t|Z_{t-1}) = \sqrt{\nu + \phi_1 Z_{t-1}^2}\mathbb{E}(V_t|Z_{t-1}) = 0$$

and

$$\begin{aligned}\mathrm{Var}(Z_t|Z_{t-1}) &= \mathbb{E}(Z_t^2|Z_{t-1}) - [\mathbb{E}(Z_t|Z_{t-1})]^2 \\ &= \mathbb{E}(h_t^2 V_t^2|Z_{t-1}) \\ &= (\nu + \phi_1 Z_{t-1}^2)\mathbb{E}(V_t^2|Z_{t-1}) \\ &= \nu + \phi_1 Z_{t-1}^2.\end{aligned}$$

Hence, the conditional variance now depends on time as required. This expression also explains the name, since the conditional variance depends on time (the process is conditionally heteroskedastic) and, more specifically, Z_t^2 is regressed on Z_{t-1}^2 (it is autoregressive of order 1). This means that the autocorrelation and partial autocorrelation of $\{Z_t^2\}$ will behave like an AR(1) process with exponential decay of the autocorrelation and a sharp cut-off at lag 1 of the partial autocorrelation as seen in figure 10.1. This behaviour is particularly useful when trying to fit a heteroskedastic model.

10.1.2 Extensions

There are many other ways to model the volatility, h_t, resulting in different behaviour of the conditional variation and thus the short term correlation in the variability. A few such models are given below:

- An autoregressive conditionally heteroskedastic model of order p, ARCH(p), has volatility given by

  $$h_t^2 = \nu + \phi_1 Z_{t-1}^2 + \cdots + \phi_p Z_{t-p}^2$$

  and results in the conditional variance

  $$\begin{aligned}\mathrm{Var}(Z_t|Z_{t-1}, Z_{t-2}, \ldots, Z_{t-p}) &= \mathbb{E}(Z_t^2|Z_{t-1}, Z_{t-2}, \ldots, Z_{t-p}) \\ &= \nu + \phi_1 Z_{t-1}^2 + \cdots + \phi_p Z_{t-p}^2\end{aligned}$$

  so that Z_t^2 is regressed on $Z_{t-1}^2, ..., Z_{t-p}^2$ and thus follows an AR(p) process.

- A generalised autoregressive conditionally heteroskedastic model of order (p, q), GARCH(p,q), has volatility given by

  $$h_t^2 = \nu + \phi_1 Z_{t-1}^2 + \cdots + \phi_p Z_{t-p}^2 + \psi_1 h_{t-1}^2 + \ldots \psi_q h_{t-q}^2$$

  and results in $\{Z_t^2\}$ behaving much like an ARMA(p, q) model.

10.1.3 Model Fitting

We use many of the same tools as in chapter 4 when fitting a conditionally heteroskedastic model to our residuals, $\{Z_t\}$:

- The best identification tool may be a **time series plot** of the series. It's usually easy to spot periods of increased variation strewn through the series and thus decide whether a conditionally heteroskedastic model would be appropriate.
- If we decide that a conditionally heteroskedastic model is appropriate then we turn our attention to the **ACF and PACF** of $\{Z_t\}$. For all of the models suggested we would expect the ACF and PACF of $\{Z_t\}$ to behave like white noise, as they should for residual data.
- The behaviour of the **ACF and PACF** of $\{Z_t^2\}$ then allows us to choose between the models suggested in this chapter:
  - (i) If $\{Z_t^2\}$ appears to follow an AR(1) model — the ACF shows exponential decay and the PACF cuts off at lag 1 — then an ARCH(1) model is suggested.
  - (ii) If $\{Z_t^2\}$ appears to follow an AR(p) model — the ACF shows exponential decay and the PACF cuts off at lag p — then an ARCH(p) model is suggested.
  - (iii) If $\{Z_t^2\}$ appears to follow an ARMA(p, q) model — there is no cut off in either the ACF or PACF — then a GARCH model may be suggested, though choosing appropriate model orders can be problematic.

Appendix A

Essential Concepts

This appendix reiterates important concepts that you should be aware of from the first year.

A.1 Random variables

Let X be a random variable (r.v.) taking values in $\mathbb{R}$. The distribution function F of X is given by

$$F(x) = \mathbb{P}(X \leq x)$$

where $\mathbb{P}(A)$ is the probability of event A occurring.

If X is a continuous r.v. its distribution function F may be written as

$$F(x) = \int_{-\infty}^{x} f(u)du$$

for some $x \in \mathbb{R}$ and integrable function $f : \mathbb{R} \to [0, \infty)$. The function f is called the density function of X.

A.2 Properties of density functions

A density function f always integrates to 1. In mathematical notation,

$$\int_{-\infty}^{\infty} f(x)dx = 1.$$

A.3 Expectation

Define the expectation or mean of X as

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} xf(x)dx.$$

Note that the expectation can be equal to $\pm\infty$ and in this case we say that the moment does not exist (the mean is the first moment).

Expectation is linear. That is, if $a, b \in \mathbb{R}$ are constants and X, Y are r.v.s then

$$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y].$$

A.4 Variance

The variance of a r.v. X is defined to be

$$\mathrm{Var}(X) = \mathbb{E}\{(X - \mathbb{E}X)^2\}$$

and describes the variation of X about its mean.

The following relation is always true: if $a \in \mathbb{R}$ is constant then

$$\mathrm{Var}(aX) = a^2 \mathrm{Var}(X).$$

If the collection $\{X_i\}_{i=1}^n$ of r.v.s is independent then

$$\mathrm{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \mathrm{Var}(X_i).$$

A.5 Covariance and correlation

The covariance between two r.v.s X and Y is given by

$$\mathrm{Cov}(X, Y) = \mathbb{E}\{(X - \mathbb{E}X)(Y - \mathbb{E}Y)\}$$

and describes the extent of the linear relationship between X and Y.

The correlation between X and Y is a normalized measure of the linear association:

$$\mathrm{Corr}(X, Y) = \frac{\mathrm{Cov}(X, Y)}{\mathrm{Var}(X)^{\frac{1}{2}} \mathrm{Var}(Y)^{\frac{1}{2}}}.$$

Always

$$|\mathrm{Corr}(X, Y)| \leq 1.$$

A.6 Multivariate distribution

If $X_1, \ldots, X_n$ are r.v.s then their joint distribution function is

$$F_{X_1,\ldots,X_n}(x_1, \ldots, x_n) = \mathbb{P}(X_1 \leq x_1, \ldots, X_n \leq x_n).$$

If the r.v.s are all mutually independent then

$$F_{\mathbf{X}}(\mathbf{x}) = \prod_{i=1}^n F_{X_i}(x_i).$$

A.7 Normal distribution

A r.v. X is said to have the normal distribution with mean μ and variance σ^2 (or standard deviation of σ) or in symbols $X \sim \mathrm{N}(\mu, \sigma^2)$ if

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}.$$

The density and distribution functions of a standard normal distribution $\mathrm{N}(0,1)$ are denoted $\phi(x)$ and $\Phi(x)$ respectively.

If $X \sim \mathrm{N}(\mu_1, \sigma_1)$ and $Y \sim \mathrm{N}(\mu_2, \sigma_2)$ and they are independent then

$$X + Y \sim \mathrm{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2).$$

A.8 Estimation

Suppose $X_1, \ldots, X_n$ are an independently distributed random sample with mean μ and variance σ^2 (not necessarily normal). We can estimate μ (population mean) by

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i,$$

the sample mean. Note the following properties of the sample mean:

$$\mathbb{E}[\bar{X}] = \frac{1}{n}\sum_{i=1}^{n} \mathbb{E}X_i = \frac{1}{n} n\mu = \mu.$$

That is the sample mean, which is unbiased for the population mean (i.e. "on average the sample mean hits the target"). Further

$$\mathrm{Var}(\bar{X}) = n^{-2}\sum_{i=1}^{n} \mathrm{Var}(X_i) = n^{-2} n\sigma^2 = \frac{\sigma^2}{n}.$$

That is the accuracy of the sample mean, which improves as more and more data are collected ($n \to \infty$; i.e. "the spread around the target gets smaller and smaller").

We can estimate σ^2 (population variance) with

$$s_{n-1}^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2.$$

Also, it can be shown that

$$\mathbb{E}[s_{n-1}^2] = \sigma^2.$$

The variance of s_{n-1}^2 can be worked out.

Note that the quantities $\bar{X}$ and s_{n-1}^2 are random. They depend on the data sample, $X_1, \ldots, X_n$ that you collected. If you take another random sample then the values of $\bar{X}$ and s_{n-1}^2 change.

A.9 Bias

Suppose that $X_1, \ldots, X_n$ is an independent and identically distributed (i.i.d.) random sample from a distribution dependent on a parameter θ (e.g., mean, variance). Suppose that $\hat{\theta}$ is an estimator for θ. Then the bias of $\hat{\theta}$ is defined by

$$\text{bias}(\hat{\theta}) = \mathbb{E}(\hat{\theta} - \theta) = \mathbb{E}\{\hat{\theta}(\mathbf{X})\} - \theta.$$

The notation $\hat{\theta}(\mathbf{X})$ is to suggest the dependence of the estimator $\hat{\theta}$ on the data sample $\mathbf{X}$ collected.

The variance of the estimator is given by

$$\text{Var}(\hat{\theta}) = \mathbb{E}\{(\hat{\theta} - \mathbb{E}\hat{\theta})^2\}.$$

Sometimes it is important to take bias and variance into account jointly. This can be done using the expected mean squared error of $\hat{\theta}$ which takes an overall view of how well the estimator does in estimating θ:

$$\text{MSE}(\hat{\theta}) = \mathbb{E}\{(\hat{\theta} - \theta)^2\} = \text{Var}(\hat{\theta}) + \text{bias}(\hat{\theta})^2.$$

A.10 Sample covariance

If $X_1, \ldots, X_n$ and $Y_1, \ldots, Y_n$ are both i.i.d. samples from (possibly) different distributions then their sample covariance, c, is given by

$$c = \frac{1}{n} \sum_{i=1}^{n} (X_i - \bar{X})(Y_i - \bar{Y})$$

and is an estimator of $\text{Cov}(X, Y)$ (where X has the same distribution as any of $X_1, \ldots, X_n$, they all have the same distribution as they are i.i.d.). The estimator c is not unbiased.

The sample correlation is given by

$$r = \frac{c}{s_X s_Y}$$

where $s_X = \sqrt{s_{n-1}^2}$ for the Xs and similarly for the Ys. Also $|r| \leq 1$ for any sample of observations.